



Federal-Mogul

Cojinetes para motores

Bronzinas

Engine Bearings

**Instrucciones para el uso
de este Manual**

How to use the Catalog

**Instruções para uso de
este Manual**

INSTRUCCIONES PARA EL USO DE ESTE MANUAL

Federal Mogul ha agrupado en este manual la información técnico-comercial que consideramos de uso fácil y práctico para rectificadores, vendedores y mecánicos. El lado izquierdo de la página se ha dedicado a datos de catálogo, mientras que el lado derecho contiene especificaciones técnicas.

INFORMACIÓN RELATIVA A LOS GRUPOS

Cada grupo se divide en 9 columnas. Cada columna de izquierda a derecha, tiene la siguiente explicación:

A.- CILINDROS, DIÁMETRO, CARRERA Y CILINDRADA

La primera cifra indica la cantidad de cilindros que tiene el motor. Inmediatamente debajo se encuentran las dimensiones del diámetro interior y de la carrera, en milímetros, seguidos del desplazamiento cúbico o cilindrada, en centímetros cúbicos y/o en pulgadas cúbicas, cuando también se conociera al motor por esta última referencia.

B.- NUMEROS DE PARTE Y POSICIÓN

Esta columna indica la aplicación de cada pieza o juego en la posición correspondiente ejemplo:

BIELAS Juego de Cojinetes de Bielas
BANCADAS Juego de Cojinetes de Bancada
SEMIARAND.
DE EMPUJE Juego de Semiarandela de Empuje
LEVAS Juego de Cojinetes del Árbol de Levas

Las posiciones de los muñones del cigüeñal y de las levas, están indicadas en el orden en que aparecen, a contar desde el frente del motor.

Nº 1 - Cojinete delantero
Nº 2-3 - Cojinetes intermedios
Nº 4 - Cojinete trasero

C.- NUMERO DE JUEGO Y DE LA POSICIÓN

Los números en esta columna son nuestros números de juegos o piezas.

Todos los cojinetes de bielas, bancadas, levas y semiarandelas de empuje se suministran en juegos.

En los cojinetes de bielas, el dígito delante del número indica la cantidad de pares de semicojinetes que compone cada juego.

Los números de pieza correspondientes a las posiciones están detallados con el propósito de identificación solamente, ya que éstas no se suministran sueltas, sino en el juego completo.

IDENTIFICACIÓN DE SUFIJOS

Los tipos y/o materiales de los cojinetes y otras piezas aquí catalogadas se identifican por letras de sufijos que siguen inmediatamente después de los números

de juegos o piezas. La interpretación de estas letras de sufijo es la siguiente

M Juego de cojinetes de bandadas o árbol de levas

Cojinetes de bielas y bancadas

AP, APA Respaldo de acero. Material de revestimiento: aleación sobre base de aluminio-cadmio. Película base plomo-estaño.

AT, ATA Cojinete de aleación de aluminio con 6 % de estaño.

B, AB, BA, BI Respaldo de bronce. Material de revestimiento: Babbit.

CA, CAA, CAB Respaldo de acero. Material de revestimiento: aleación de cobre-plomo.

CP, CPA, CPB Respaldo de acero. Material de revestimiento: aleación de cobre-plomo. Película base plomo-estaño.

RA, RAA, RAB Respaldo de acero. Material de revestimiento: aleación de aluminio con 20 % de estaño

SA, SB, SBI, SI Respaldo de acero. Material de revestimiento. Babbit.

Arandelas de empuje

AF, AFA, AFF Sin indicación específica del material.

BF, BFA, BFF Sin indicación específica del material.

Bujes de cojinetes de árbol de levas

DR, DRI, DRO Sin indicación específica del material.

Bujes en general

Y, W Buje bimetálico.

X Buje de bronce.

D.- MEDIDAS

Los cojinetes con bajo medidas están terminados a precisión, con diferentes espesores de respaldos de acero o de revestimiento. Los cojinetes semiterminados pueden suministrarse con distintos espesores, a elección, los cuales son rebajables a las medidas que en cada caso se indican.

E.-DIAMETRO ESTANDAR DEL EJE

En esta columna se indican las dimensiones de los límites mínimo y máximo correspondientes a los muñones del cigüeñal (bielas y bancadas) y del árbol de levas, en las medidas estándar.

F.- DIAMETRO DE ALOJAMIENTO

Esta columna muestra las dimensiones con los límites mínimo y máximo del alojamiento del cojinete.

G.- LUZ DE ACEITE

Las cifras bajo esta columna demuestran los límites mínimo y máximo sugeridos para el huelgo del aceite de los cojinetes.

H.- ESPESOR MAXIMO DEL COJINETE ESTANDAR

Esta columna contiene el espesor máximo de la pared del semicojinete, mediada al centro.

I.- ANCHO TOTAL

Es el ancho máximo del cojinete, incluyendo las pestañas, cuando son parte integrante del mismo.

DATOS PARA EL SERVICIO Y TALLER

Estas páginas, conjuntamente con las de la sección alfabética, están dedicadas a la información pertinente a los cojinetes del motor y piezas relacionadas.

En todo cambio de cojinetes, el resultado deseado es obtener nuevamente la luz de aceite original especificadas por el fabricante. La luz de aceite es la diferencia entre la medida del eje y el diámetro interior del cojinete después de instalado éste. Para obtener la luz de aceite apropiada, deben considerarse siempre tres elementos: el eje, el cojinete y el alojamiento, como los factores determinantes.

Se suministra también información adicional sobre las torsiones del aprieto recomendadas para la fijación de los bulones para los cojinetes de bielas, bancadas y de la tapa de cilindros.

EL CIGÜEÑAL

En este catálogo se indica la medida estándar original de los muñones del cigüeñal. Esta medida debe conocerse con exactitud cuando se efectúa la rectificación del eje a fin de poder emplearse cojinetes terminados a precisión en bajomedidas.

Por ejemplo, nominalmente el cigüeñal de una determinada marca de vehículo importado, es de 2-1/4" para los muñones de las bielas. El rectificador debe estimar necesariamente tamaño es de 2.250" en decimales, pero en realidad, este cigüeñal, según se produjo originalmente, es de 2.2488"/2.2498". El diámetro interior de un cojinete estándar para este muñón mide entre 2.2505/2.2510". Las bajomedidas que usualmente suministran las fábricas de automóviles o de cojinetes midan .010", .020", .030". El rectificador, a fin de adaptar correctamente estos cojinetes, debe rectificar el cigüeñal a 2.2388"/2.2398" para las medidas .010". 2.2188"/2.2198" para las medidas 0.030". Si se hubiera rectificado siguiendo la cifra nominal de 2-1/4 se habría eliminado el huelgo necesario para el aceite.

DIAMETRO DEL ALOJAMIENTO DE LOS COJINETES DE BIELAS Y BANCADAS

El diámetro de los alojamientos de los cojinetes de bielas y bancadas es, generalmente, igual al diámetro exterior de los cojinetes correspondientes.

DIMENSIONES

Todas las dimensiones están expresadas en milímetros. Cuando originalmente los valores del motor estuvieran dados en pulgadas, se incluye debajo la co-

rrespondiente equivalencia.

Los datos contenidos en este manual han sido cuidadosamente recopilados y revisados con el objeto de asegurar su exactitud no obstante, declinamos asumir responsabilidad por reclamos originados en posibles errores u omisiones.

Los alojamientos de los cojinetes, en el uso regular, algunas veces se deforman de sus dimensiones originales. Las bielas cuyos alojamientos se hubieran ovalizado más de 0.050 mm (.002") deben ser reemplazadas o reacondicionadas a la medida correcta. Los alojamientos ovalizados de las bancadas pueden ser reparados alesándolos en línea a los diámetros originales, e instalando cojinetes terminados. También pueden colocarse cojinetes semiterminados y alesarlos a la medida de cigüeñal dejando el espacio necesario para la correcta luz de aceite.

HUELGOS PARA EL ACEITE

Los huelgos correctos para el aceite son especificados por el fabricante del motor.

En los reemplazos de cojinetes, este huelgo de aceite debe mantenerse. Debe compensarse el desgaste del cigüeñal mediante el uso de cojinetes de bajomedidas. En ciertos casos puede haber disponible un material de cojinetes diferente M indicado por el número de pieza. Si la pieza a instalar en sustitución es un cojinete terminado a precisión que no requiere ser rebajado, cualquier variación en el huelgo requerido para el aceite por el nuevo material está ya calculado en la medida del cojinete.

Si se está usando un cojinete de sustitución semiterminados, al ser alesado debe hacerse el ajuste para el huelgo de aceite especificado.

Así como los cojinetes de bielas y bancadas son fabricados dentro de los límites de tolerancia de máxima precisión, también los cojinetes del árbol de levas (bujes) son producidos de manera que permitan mantener el huelgo correcto de aceite luego de instalados a presión en sus respectivos alojamientos.

INSTALACION DE COJINETES

Los orificios para el aceite de los cojinetes deben coincidir correctamente con los orificios en el block o en las bielas.

El cigüeñal debe estar perfectamente alineado, y los muñones, después de rectificados, deben ser bien pulidos. El alesado simultáneo de los cojinetes hace que los cojinetes de bancada queden alineados, pero debe verificarse también que el eje del cigüeñal esté en la misma condición de alineamiento.

Todo debe estar limpio, el interior del motor, el cárter de aceite, los tubos de aceite, los elementos del conjunto de las bielas, el cigüeñal, los alojamientos de los cojinetes, etc. Una pequeña partícula de metal, de carbón, de arena o de polvo, puede eliminar el huelgo para el aceite en un área determinada, lo cual ocasionará el comienzo de fallas en los cojinetes o ralladuras en el cigüeñal, provocando su desgaste prematuro.

Al instalar cojinetes de bielas nuevos, es altamente recomendable usar nuevos bulones y tuercas en las bie-

las, como también usar una llave de torsión al armar las tapas de los cojinetes de bielas y bancadas.

En determinadas reparaciones es conveniente que los cojinetes de árbol de levas sean alesados después de instalados.

Esto es especialmente aconsejable cuando los muñones han sido rectificados. Los cojinetes desgastados del árbol de levas, son con frecuencia la probable causa de la pérdida de la presión y de alto consumo de aceite.

Pueden ocurrir fugas excesivas de aceite en los cojinetes desgastados del árbol de levas, lo mismo que en los cojinetes de bielas y bancadas. Es muy fácil y práctico localizar cojinetes desgastados de bielas, bancadas y levas, o verificar su correcto montaje, usando el Prelubricador Federal-Mogul para motores. Además de observar el correcto huelgo para el aceite, la superficie antifricción del cojinete debe ser lubricada. El aceite debe llegar en suficiente cantidad y a la presión correcta.

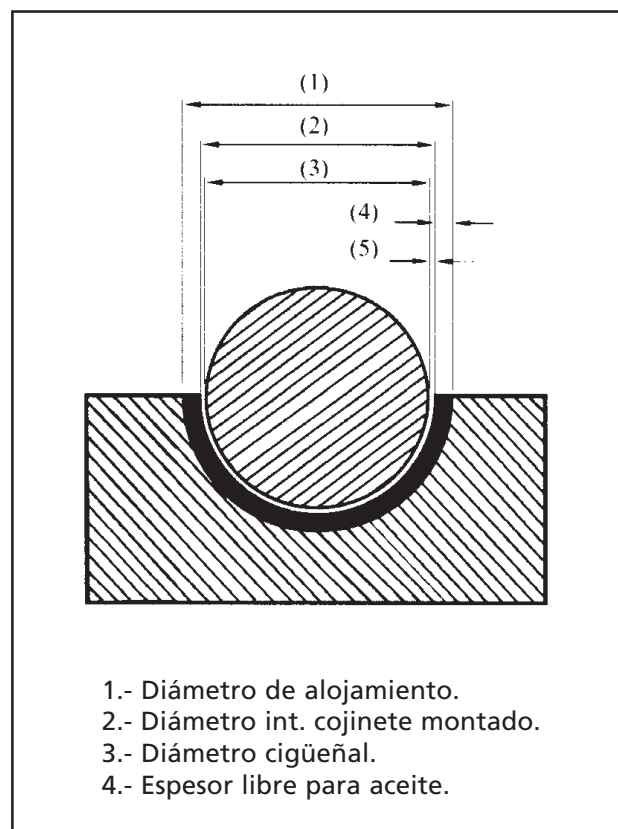
El sistema completo de lubricación, comenzando por el colador de aceite, debe ser revisado cuidadosamente. La bomba de aceite debe ser precargada. Los cojinetes de bancada deben revisarse para ver si hay fugas excesivas pues esta condición puede ser la causa de que los cojinetes no tengan suficiente lubricación, como también de excesivo desgaste de los aros y de los cilindros.

En caso de hallarse diferencias entre ambos cojinetes, conviene averiguar debidamente el motivo, ya que algunas veces suelen introducirse modificaciones para mejorar el rendimiento del cojinete. De persistir dudas, sírvase consultar al departamento de Asistencia Técnica de Federal -Mogul.

DETERMINACION DEL HUELGO DE ACEITE

	Milímetros	Pulgadas
Diámetro aloj. mín.	-67.183	-2.6450
Espesor máx. coj. (x2)	3.683	.1450
Diámetro int.mín. montado	-63.500	-2.5000
Diámetro máx. cigüeñal	63.475	2.4990
Mínimo espesor libre para aceite	0.025	.0010

	Milímetros	Pulgadas
Diámetro aloj. máx.	-67.196	-2.6455
Espesor mín. coj. (x2)	3.858	.1440
Diámetro int. máx. montado	-63.538	-2.5015
Diámetro máx. cigüeñal	63.462	2.4985
Máximo espesor libre para aceite	0,076	.0030



COJINETES DE EMPUJE DEL CIGÜEÑAL CON PESTAÑA

Los cojinetes de empuje con pestaña de bajomedida terminados, y los semiterminados, tiene exceso de materia en las caras de las pestañas para un ajuste axial del eje. Por consiguiente, es esencial medir con exactitud (en centésimas de milímetros o en milésimas de pulgadas) las distancias entre las superficies de empuje en el cigüeñal durante su rectificación. Los cojinetes de pestaña se entregan con las siguientes medidas en su ancho total:

Sobre medida	ancho
STD-010	STD
10 a 50	+0,30
SEMI	SEMI

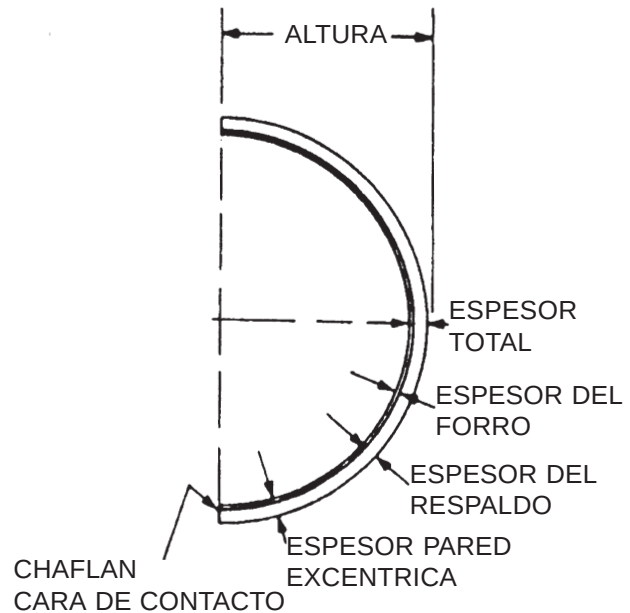
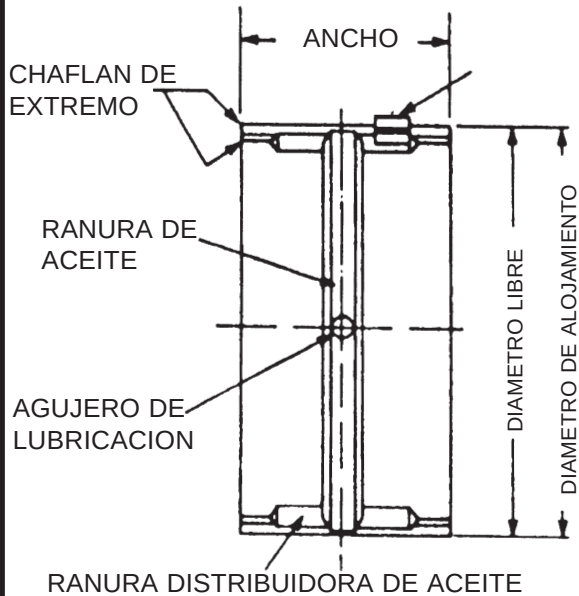
COJINETES DE EMPUJE JUEGO AXIAL RECOMENDADO

El juego (axial) recomendado por el fabricante del motor debe verificarse con láminas calibradas (sondas). En caso de no disponerse de estos valores, se pueden utilizar los indicados en el cuadro siguiente.

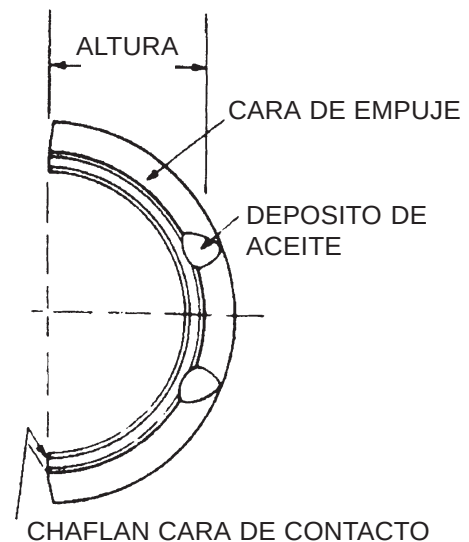
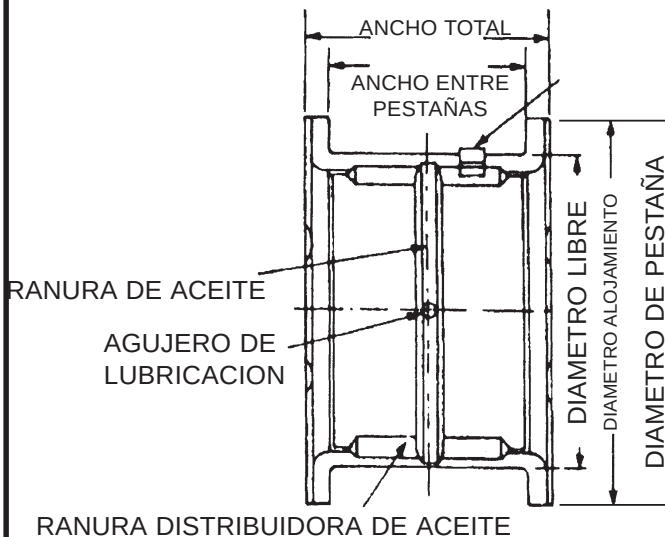
Diámetro de los muñones del cigüeñal	Juego axial
50 a 70 mm	0,10 a 0,15 mm
71 a 90 mm	0,15 a 0,20 mm
más de 90 mm	0,20 a 0,25 mm

NOMENCLATURA DE COJINETES

Cojinetes de bielas y bancadas



Cojinetes de empuje



Cojinete de pared concéntrica
(pared de espesor uniforme)



Cojinete de pared excéntrica
(pared más gruesa en la corona que
en las caras divisorias)

HOW TO USE THE CATALOG

The top of each page is divided into 9 main columns which continue downward. Each column heading is explained below.

A.- CYLINDER BORE & STROKE

Number of cylinders, bore and stroke in, and cc displacement appear in descending order for each new line

B.- PART NUMBERS AND POSITIONS

This illustration shows the placement of specific bearing types: and their relative positions numbered from the front or the engine to the rear (flywheel). The following terms are used to describe bearing types.

BIELAS Rod BRG Set

BANCADAS Main BRG Set

SEMI ARAND.

DE EMPUJE Thrust Washer

LEVAS Cam Bearing Set

BUJE DE BIELA Piston Pin Bushing

C.- SET NUMBERS AND POSITIONS

This column shows our set numbers and positions numbers.

BEARING MATERIALS

Con Rods-Main Bearings

AP, APA Steel back, lining material alloy on aluminum- cadmium base. Overplating on a lead-tin base.

AT, ATA Solid aluminum alloy bearing with 6% tin content.

B, AB, BA, BI Bronze back, lining material Babbitt.

CA, CAA, CAB Steel back, lining material copperlead alloy Overplating on a lead-tin base.

RA, RAA, RAB. Steel back, lining material aluminum alloy with 20% tin content.

SA, SB, SBI, SI Steel back, lining material babbitt.

Thrust Washers

AF, ADA, AFF,

BF, BFA, BIFF No specific material indication

Cambshaft bushings

DR, DRI, DRO No specific material indication

Bushings-generally

Y, W Bimetallic bushing

X Bronze bushing

D.- UNDERSIZES

This column shows the available bearing sizes.

Finished bearing inch undersizes are printed:

10-20-30-40-50-60 which are in fact: .010" - .020" etc.

Millimeter undersizes are printed as 0,25-0.50 mm. etc.

Resizeable semi finished bearings.

These bearings must be rebored after fitting into their housing.

E.- SHAFT DIAMETER DIMENSIONS.

As illustrated, are listed in this column. Minimum and maximum dimensions are listed.

F.- HOUSING BORE DIMENSIONS

As illustrated, are listed in this column. Minimum and maximum dimensions are listed.

G.- OIL CLEARANCE

This column shows the minimum and maximum oil clearance specified.

H.- WALL THICKNESS

Maximum wall thickness of the bearing at the crown of the bearing is listed in this column.

I.- OVERALL LENGTH

Maximum overall bearing length dimension is listed in this column.

The information contained in the manual has been compiled and edited with utmost care to insure accuracy, however, proof of the correctness of some of the data is beyond our control. We, therefore, assume no responsibility whatsoever for any errors that may appear herein.

MANUAL FOREWORD FOR MACHINISTS AND MECHANICS

These pages, together with the alphabetical section, are devoted to data pertaining to engine bearings and related parts.

In any bearing replacement the desired result is to again obtain the original manufacturers oil clearance. The oil clearance is the difference between shaft size and bearing inside diameter after bearing installation. To obtain the proper oil clearance, three members shaft, the bearing and the housing hole, must always be considered determining factors in the resulting oil clearance. Additionally, information is given concerning camshafts and cam bearings, recommended torque or tension wrench readings for bearing bolts and for cylinder head studs or cap screws.

THE CRANKSHAFT

In this Manual it is shown the original factory standard size of crankpins for the shaft. This exact original size must be known when regrinding the shaft for use with stock undersize replacement bearings. For example, the crankshaft of a given engine the diameter of the crankpin is given as 2 1/4". The crankshaft reginder must necessarily figure such a vein 2.250 " in decimals, but in reality this shaft as original/y produced is 2.2488"V2.2498". The inside diameter of a finished standard bearing for this crankpin is 2.2505"V2.251Y. The finished undersize bearings usually furnished by car factories or bearings manufacturers are .010", .020", .030" ' The reginder, in order to properly fit these bearings, should grind the shaft to 2.2388 " for .010 under, 2.2288"V2.2298"for .020" under and 2.2188" / 2.2198" for .030 undersize. To grind from the nominal figure of 2 1/4" would have eliminated the necessary oil clearance.

ROD OR MAIN BEARING HOUSING BORES

The maximum diameter of rod and main bearing housing bores is usually equal to the outside diameter of the respective bearings. Bearing housing bores in regular usage sometimes become distorted from their original dimensions. Connecting rods with bearing bores distorted more than 0, 050 mm, (00Z) should be replaced or reconditioned to proper size. Distorted main bearing bores can be restored by align-boring the bores and using precision bearings or by use of suitable undersize bearings and aligned bored to suit the crankshaft size allowing for proper oil clearance space.

OIL CLEARANCES

The correct oil clearances in the engine installation are provided by the engine manufacturer. In replacement

installation this oil clearance must be maintained. Any crankshaft wear encountered should be compensated for by use of undersize bearings.

In some instances, a bearing material other than the one indicated by the part number is also available. If the substitute part is a precision bearing and requires no resizing, any variation in the oil clearance required by the new material is built into the bearing.

If a substitute resizeable bearing is being used and it is being bored, adjustment of the oil clearance should be made to the manufacturer's specifications.

BEARING INSTALLATION

Bearing oil holes must match correctly with oil holes in crankcase or connecting rods.

Crankshaft must be straight, free from bend, and journals and crankpins polished smooth. Align borin produces alignment of finished main bearings, but conditions of shaft for alignment must be checked. Everything must be clean -interior of engine, oil pan, oil lines, members of rod assembly, crankshaft, bearing housing bores, etc. One small particle of metal, carbon, sand or dust, can eliminate the oil clearance in a given area which will start a bearing failure or score a crankshaft, producing early wear.

In the installation of new rod bearings, it is highly recommended that new connecting rod bolts and nuts be used; that in the assembling of rod and main bearing caps, a tension wrench be used

In certain installations, it is desirable that cam bearings be finish-machined in place.

This is especially desirable where the camshaft journals have been reground. Worn camshaft bearings, are often responsible for cases of lack of oil pressure and high oil consumption.

Excessive oil leakage can occur at worn camshaft bearings just the same as at worn mains and rods. Proper installation and worn cam bearings are easily located by the use of Federal-Mogul Engine Pre-Lubricator. Proper oil clearance alone is not enough, the bearing must be lubricated. Oil must reach the bearing in sufficient quantity at the right pressure.

The entire oiling system from pick-up screen on, must be checked. Oil pumps should be primed. Main bearings must be checked for excessive leakage as this condition may be the cause of starved crankpin bearings and excessive ring and cylinder wear.

In the installation of new bearings to replace the ones that have become worn, it is highly recommended that both should be compared, as the oil groove, hole and locating lip should be exactly the same. If the two bearings differ it is convenient to find the reason as sometimes, a variation is required to improve the bearing performance.

For further information call the Federal-Mogul Technical Assistance Department. Our staff can quickly handle your problem.

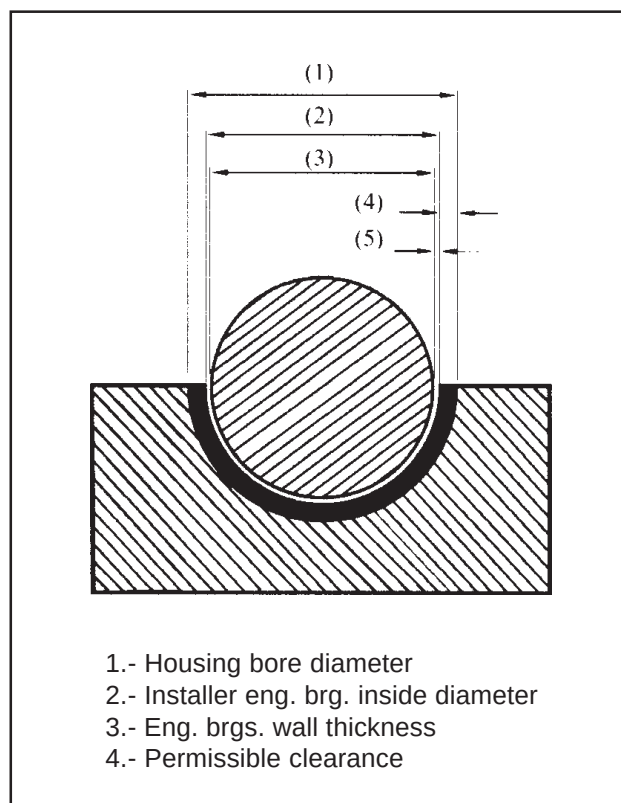
RECOMMENDED OIL CLEARANCE FOR THE ENGINE BEARINGS

	Milimeter	Inches
Minimum housing bore diameter	-67.183	-2.6450
Eng. Brgs. max. wall thickness (x 2)	3.683	.1450
Installed min. ins. diameter	-63.500	-2.5000
Crankshaft max. diameter	63.475	2.4990
Minimum permissible clearance	0.025	.0010
	Millimeter	Inches
Minimum housing bore diameter	-67.196	-2.6455
Eng. Brgs. max. wall thickness (x 2)	3.658	.1440
Installed eng. brg. max. ins. diameter	-63.538	-2.5015
Crankshaft max. diameter	63.462	2.4985
Minimum permissible clearance	0.076	.0030

CRANKSHAFT THRUST BEARINGS - FLANGED

Undersize and semi-finished undersize flange thrust bearings carry excessive stock on the flange so that they can be fitted lengthwise to take up wear on crankshaft thrust collars. It is essential, therefore, to accurately measure (in hundredths of a millimeter and thousandths of an inch) the distance between thrust surfaces on the crankshaft when regrinding. The flange bearings are supplied in the following total width sizes.

Undersizes	Width
STD-010	STD
10 a 50	+0.30
SEMI	SEMI



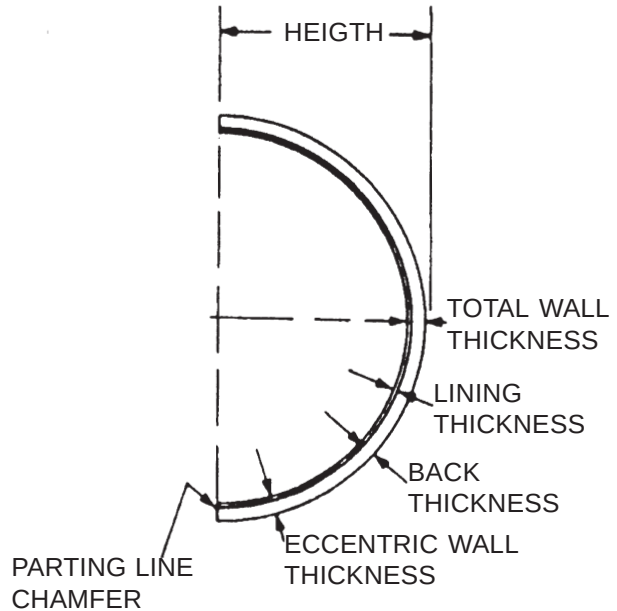
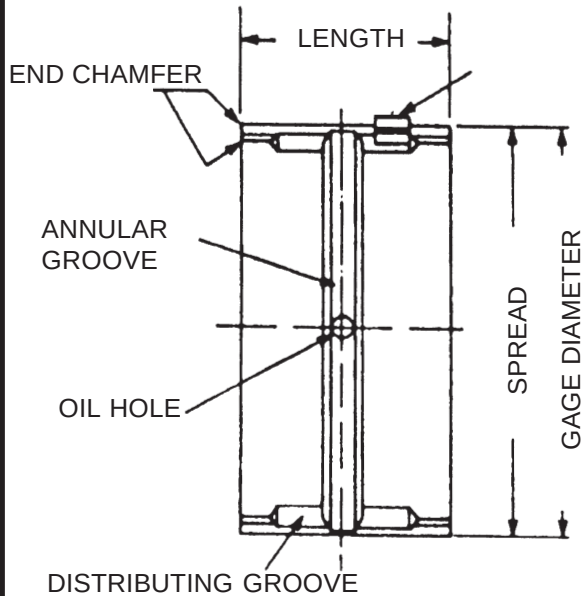
CRANKSHAFT THRUST BEARINGS - RECOMMENDED CRANKSHAFT END CLEARANCE

The crankshaft end clearance recommended by the engine manufacturer should be checked with gages. If these values are not available, the following chart provides dimensions of flange bearings that can be helpful.

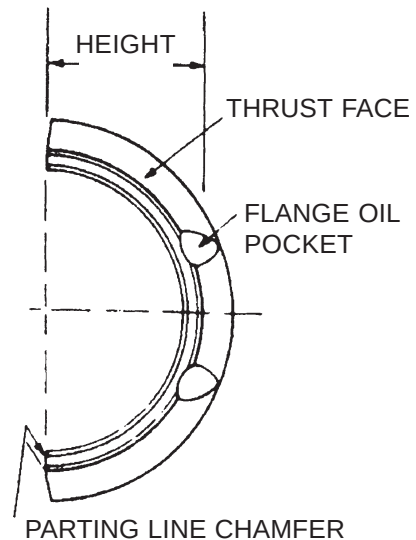
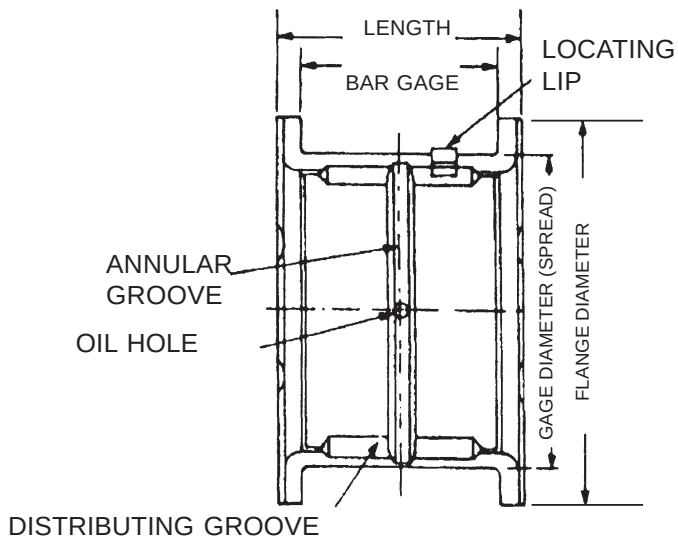
Shaft Crankpins Diameter	Recommended Crankshaft end Clearance
50 to 70 mm	0.10 to 0.15 mm
71 to 90 mm	0.15 to 0.20 mm
more than 90 mm	0.20 to 0.25 mm

BEARING NOMENCLATURE

Rod and Main Bearing



Thrust Bearing



Concentric Wall Bearing
(uniform wall thickness)



Eccentric Wall Bearing
(wall heavier at than at parting faces)

INSTRUÇÕES PARA USO DESTE MANUAL

A Federal-Mogul tem agrupado neste manual toda a informação técnico-comercial que consideramos de uso fácil e prático para retificadores, vendedores e mecânicos.

O lado esquerdo de página possui informações de catálogo, enquanto o lado direito contém informações técnicas.

INFORMAÇÕES RELATIVAS AOS GRUPOS

Cada grupo se divide em 9 colunas. Cada coluna, da esquerda à direita, tem a seguinte informação:

A.- CILINDROS, DIÂMETRO, CURSO E CILINDRADA

O primeiro número indica a quantidade de cilindros que tem o motor.

Imediatamente abaixo se encontram as dimensões do diâmetro interno e do curso, em milímetros, seguidos da cilindrada, em centímetros cúbicos e/ou em polegadas cúbicas, quando também pode-se conhecer o motor por esta última referência.

B.- NÚMERO DE PEÇA E POSIÇÃO

Esta coluna indica a aplicação, de cada peça ou jogo na posição correspondente, exemplo:

Bielas Jogo de bronzinas de biela
Centrais Jogo de bronzinas centrais
Arruelas de encosto Jogo de arruelas de encosto
Comando de válvulas Jogo de buchas para eixo comando de válvulas

As posições dos munhões do virabrequim e dos eixos de comando de válvulas, estão indicados na ordem em que aparecem, a contar da frente do motor:

Nº 1 - Bronzina dianteira
Nº 2-3 - Bronzinas intermediárias
Nº 4 - Bronzina traseira

C.- NÚMERO DO JOGO E DA POSIÇÃO

Os números nesta coluna são, nossos números de jogos ou peças.

Todas as bronzinas de bielas, centrais, comandos de válvulas e arruelas de encosto são fornecidos exclusivamente em jogo para um motor.

Nas bronzinas de bielas, o dígito anterior ao número indica a quantidade de pares que compõe cada jogo. Os números de peça correspondentes às posições estão detalhados com o propósito de somente identificar, já que estas não são fornecidas soltas, somente em jogo completo.

D.- DIÂMETRO STANDARD DO EIXO

Nesta coluna se indicam as dimensões dos limites mínimo e máximo correspondente aos munhões e moentes do virabrequin (bielas e centrais) e do eixo

comando de válvulas, nas medidas standard.

E.- DIÂMETRO DO ALOJAMENTO

Esta coluna mostra as dimensões com os limites mínimo e máximo do alojamento da bronzina.

F.- FOLGA DE ÓLEO

Esta coluna demonstra os limites mínimo e máximo especificados para a folga de óleo das bronzinas.

G.- ESPESSURA MÁXIMA DE BRONZINA STANDARD

Esta coluna contém a espessura máxima da bronzina, medido no centro.

H.- LARGURA TOTAL

É a máxima largura da bronzina, incluindo as flanges, quando são parte integrante da mesma.

I.- DIMENSÕES

Todas as dimensões são, expressas em milímetros. Quando originalmente os valores estiverem em polegadas, serão incluídas abaixo a equivalência correspondente.

Os dados contidos neste manual tem sido cuidadosamente recopiados e revisados com o objetivo de assegurar sua exatidão, no entanto, não, assumimos qualquer responsabilidade por reclamações originadas em função de possíveis erros ou omissões.

Os tipos e/ou materiais das bronzinas e outras peças aqui catalogadas se identificam por letras de sufixos que seguem imediatamente depois dos números de jogos ou peças.

A interpretação destas letras é o seguinte:

M Jogo de bronzinas centrais ou eixo comando de válvulas

BRONZINAS DE BIELAS E CENTRAIS

AP, APA Liga de alumínio-cádmio sobre respaldo de aço com película à base de chumbo-estanho.

AT, ATA Liga de alumínio com 6% de estanho.

B, AB, BA, BI Liga de babbitt sobre respaldo de bronze.

CA, CAA, CAB Liga de cobre-chumbo sobre respaldo de aço.

CP, CPA, CPB Liga de cobre-chumbo sobre respaldo de aço com película à base de chumbo-estanho.

RA, RAA, RAB Liga de alumínio com 20% de estanho sobre respaldo de aço.

SA, SB, SBI, SI Liga de babbitt sobre respaldo de aço.

ARRUELAS DE ENCOSTO

AIF, AFA, AFF Sem indicação específica do material.
BF, BFA, BFF

BUCHA PARA EIXO COMANDO DE VÁLVULAS

DR, DRI, DRO Sem indicação específica do material.

BUCHAS EM GERAL

Y, W Bucha bimetálica

X Bucha de bronze

J.- MEDIDAS

As bronzinas são fornecidas, além da medida standard, em sobremedidas com diferentes espessuras do metal base e/ou da liga de revestimento.

INFORMAÇÕES PRÁTICAS PARA O MECÂNICO

Estas páginas, juntamente com as da seção alfabética, são, dedicadas à informações pertinentes às bronzinas de motor e peças relacionadas.

Em toda troca de bronzinas, espera-se recuperar novamente a folga de óleo, especificada pelo fabricante. A folga de óleo é a diferença entre a medida do eixo e o diâmetro interno da bronzina, depois desta instalada. Para obter a folga de óleo apropriada, devem ser considerados sempre três elementos: o eixo, a bronzina e o alojamento, como fatores principais.

O VIRABREQUIM

Neste catálogo se indica a medida standard original dos munhões e moentes do virabrequim. Deve-se conhecer esta medida com exatidão, quando se efetua a retificação do eixo, para podermos substituir corretamente por uma bronzina de sobremedida.

Por exemplo, nominalmente, o virabrequim de uma determinada marca de veículo importado é 2 1/4" para os moentes das bielas. O retificador deve saber, necessariamente, que o tamanho é 2.250" em décimos, mas na realidade, este virabrequim, como um produto original, possui 2.2488/2.2498".

O diâmetro interno da bronzina standard para este moente mede entre 2.2505/2.2510". As sobremedidas que usualmente são fornecidas pelas fábricas de bronzinas medem 0.010", 0.020", 0.030". O retificador, a fim de adaptar corretamente estas bronzinas, devem retificar o virabrequim para 2.2388"/ 2.2398" que é a medida 0.010", e 2.2188"/2.2198" que é a medida 0.030". Se fosse retificado com a medida nominal de 2 1/4", teríamos eliminado a necessária folga de óleo.

DIÂMETRO DO ALOJAMENTO DAS BRONZINAS DE BIELAS E MANCAIS CENTRAIS

O diâmetro dos alojamentos das bronzinas de bielas e mancais centrais é, geralmente, igual ao diâmetro externo das correspondentes bronzinas.

Os alojamentos das bronzinas de uma maneira geral, se deformam de suas dimensões originais. As bielas cujos alojamentos foram ovalizados mais de 0.050 mm (0.002"), devem ser substituídas ou recondiçionadas

para a medida correta.

Os alojamentos ovalizados dos mancais centrais podem ser reparados brunindo-os, e instalandose bronzinas de sobremedida.

FOLGAS DE ÓLEO

As corretas folgas de óleo são especificadas pelo fabricante do motor.

Nas substituições de bronzinas, deve manter-se esta folga de óleo. Deve-se compensar o desgaste do virabrequim mediante o uso de bronzinas de sobremedida. Em certos casos pode haver disponível um material de bronzinas diferente do indicado pelo número da peça.

Se a substituição for feita com a bronzina correta, então, está garantida a folga de óleo que é requerida. Assim como as bronzinas de bielas e mancais centrais são fabricadas dentro dos limites de tolerância com a máxima precisão, também as buchas para eixo comando de válvulas são, produzidas de tal maneira que permitam manter a folga de óleo correta, assim que instaladas em seus respectivos alojamentos

INSTALAÇÃO DE BRONZINAS

Os furos de óleo dos bronzinas devem coincidir corretamente com os furos do bloco ou das bielas.

O virabrequim deve estar perfeitamente alinhado, e os munhões e moentes, depois de retificados, devem ser bem polidos. Após o brunimento, as bronzinas centrais ficam alinhadas, mas deve-se verificar, também, o alinhamento do virabrequim.

Tudo deve estar limpo, o interior do motor, o cárter de óleo, os condutos de óleo, os elementos dos conjuntos das bielas, o virabrequim, os alojamentos das bronzinas, etc. Uma pequena partícula de metal, de carbono, de areia ou de poeira, pode eliminar a folga de óleo em uma determinada área, o que ocasionará um começo de falhas nas bronzinas ou ranhuras no virabrequim, provocando seu desgaste prematuro.

Ao instalar novas bronzinas de biela, é recomendável usar novos parafusos e porcas nas bielas, como também usar um torquímetro para montá-las.

Em determinadas reparações é conveniente que as buchas de eixo comando de válvulas sejam brunidas depois de instaladas.

Isto é aconselhável quando os munhões foram retificados. As buchas desgastadas do eixo comando de válvulas, são com frequência a causa da perda de pressão e do alto consumo de óleo.

Podem ocorrer perdas excessivas de óleo nas buchas desgastadas da eixo comando de válvulas, ou até mesmo nas bronzinas de bielas e mancais centrais. É muito fácil e prático localizar bronzinas e buchas desgastadas, ou verificar se sua montagem está correta, utilizando um pré-lubrificador Federal-Mogul para motores.

Além de verificar se a folga de óleo está correta, devemos, também, observar se a superfície de contato da bronzina está sendo lubrificada. O óleo deve chegar em quantidade suficiente e a uma pressão adequada. O sistema completo de lubrificação, começando pelo

filtro de óleo, deve ser verificado sempre, juntamente com a bomba de óleo.

As bronzinas centrais, também, devem ser sempre verificadas para que não haja fugas excessivas de óleo, pois isto faria com que as bronzinas não tivessem uma lubrificação suficiente, assim como, excessivo, desgaste dos anéis e dos cilindros.

Uma prática aconselhável antes da instalação, é a de comparar as bronzinas novas com as que foram substituídas, verificando sua semelhança, no que diz respeito à ranhuras, furos de óleo, chavetas, etc., devendo todas essas características coincidirem exatamente.

Em caso de haver diferenças entre ambas as bronzinas, é conveniente averiguar o motivo, já que algumas vezes são introduzidas modificações para melhorar o seu rendimento. Se persistir dúvidas, consulte o departamento de assistência técnica da Federal-Mogul.

DETERMINAÇÃO DA FOLGA DE ÓLEO

	Milímetros	Polegada
Diâmetro mínimo do alojamento	67.183	2.6450
Espessura máxima da bronzina (x2)	-3.683	0.1450

Diâmetro mínimo interno montado	63.500	2.5000
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Diâmetro máximo do virabrequim	63.475	2.4990
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Folga mínima de óleo	0.025	0.0010
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	Milímetros	Polegada
Diâmetro mínimo do alojamento	67.196	2.6455
Espessura máxima da bronzina (x2)	-3.658	0.1440

Diâmetro mínimo interno montado	63.538	2.5015
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Diâmetro máximo do virabrequim	63.462	2.4985
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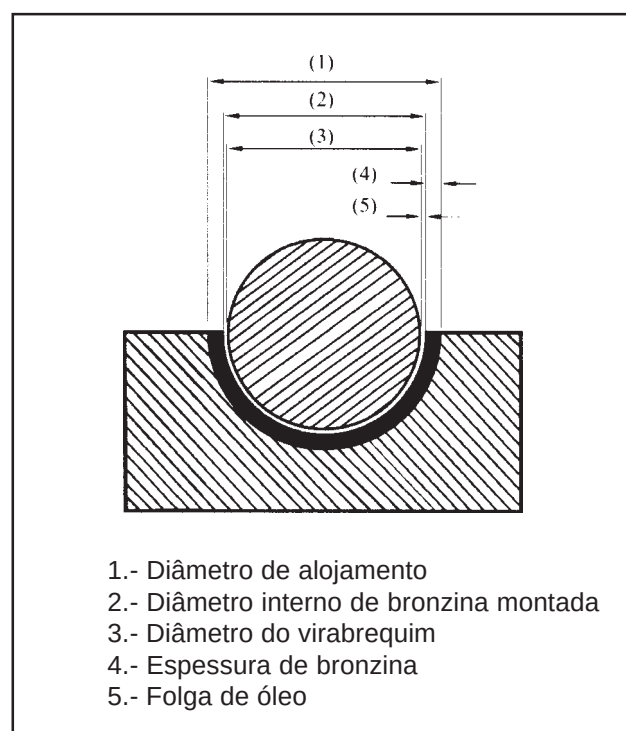
Folga mínima de óleo	0.076	0.0030
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BRONZINAS CENTRAIS COM FLANGE

As bronzinas centrais com flange de sobremedida, tem excesso de material nas faces das flanges para um melhor ajuste axial de eixo. Por conseguinte, é essencial medir com exatidão (em centésimos de milímetros ou em milésimos de polegadas) as distâncias entre as superfícies das flanges no virabrequim durante sua retificação.

As bronzinas com flange são fornecidas nas seguintes medidas em sua largura total:

Medidas	Largura
STD-0.10	STD
0.10 à 0.50	+0,30



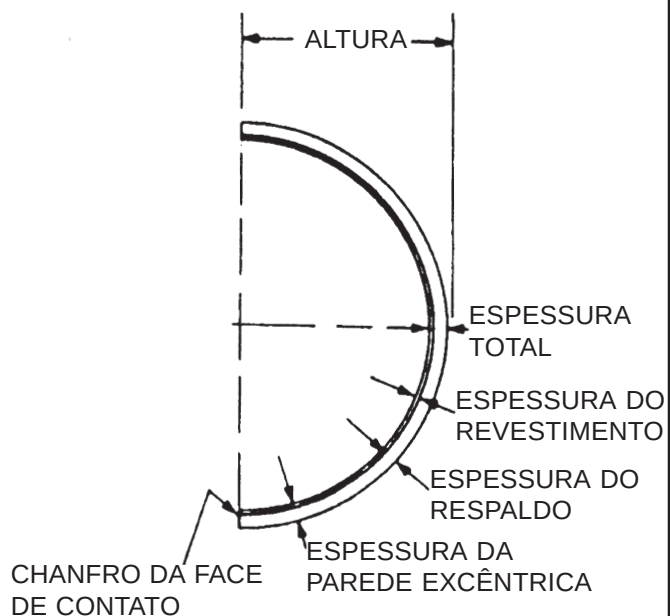
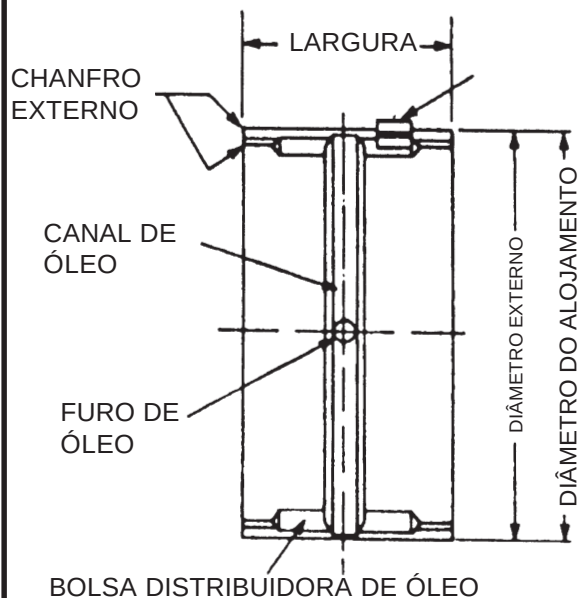
Bronzinas centrais com flange jogo axial recomendado

A folga axial recomendada pelo fabricante do motor deve ser verificada com lâminas calibradas. No caso destes valores não estarem disponíveis, podem ser utilizados os valores indicados no quadro abaixo:

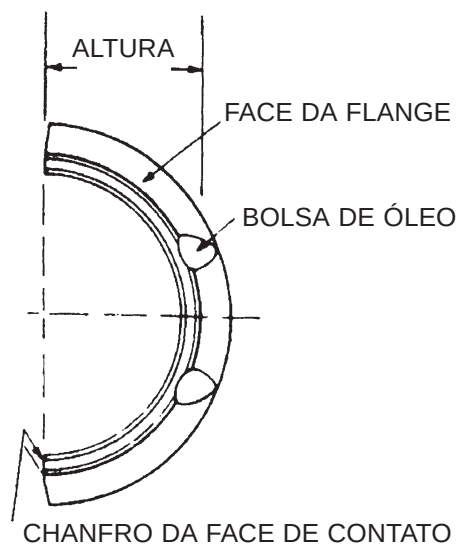
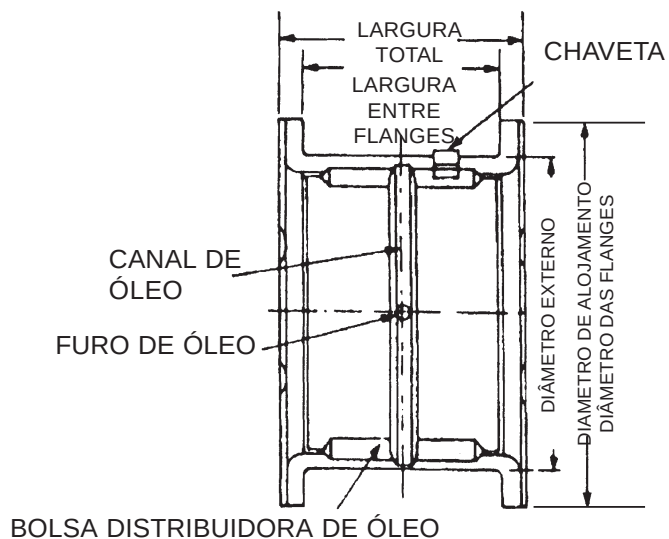
Diâmetro dos munhões do virabrequim	Folga axial
50 à 70 mm	0.10 à 0.15 mm
71 à 90 mm	0.15 à 0.20 mm
maior que 90 mm	0.20 à 0.25 mm

NOMENCLATURA DE BRONZINAS

Bronzinas de Bielas e Centrais.



BRONZINAS CENTRAIS COM FLANGE



Bronzina de parede concêntrica
(parede de espessura uniforme)



Bronzina de parede excêntrica
(parede más grossa no centro do que nas
faces de encosto)

CUADRO DE IDENTIFICACION		
CASTELLANO	INGLES	PORTUGUES
Axial	Thrust washer	Arruelas de encosto
Bancada	Main bearings	Centrais
Biela	Conn rod bearings	Biela
Buje de Biela	Camshaft bearings	Bucha de biela

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DS/DSC/DSI II	82
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Subaru

GL 1400/1600/1800, Elf, Leone 1200/1400/1600, EA61/62/63/64/71	82
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Suzuki

ST90 Van, F8A, F10A	82
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Toyota

Hi-Lux Diesel, Crown Diesel, Hi-ACE, L, 2L	83
Carina, Celica, Corolla, 1400, 1600, 1800 - 1978 / 1982	83
Celica 2000, 18R	83
Hi-Lux 18R	84

Volkswagen

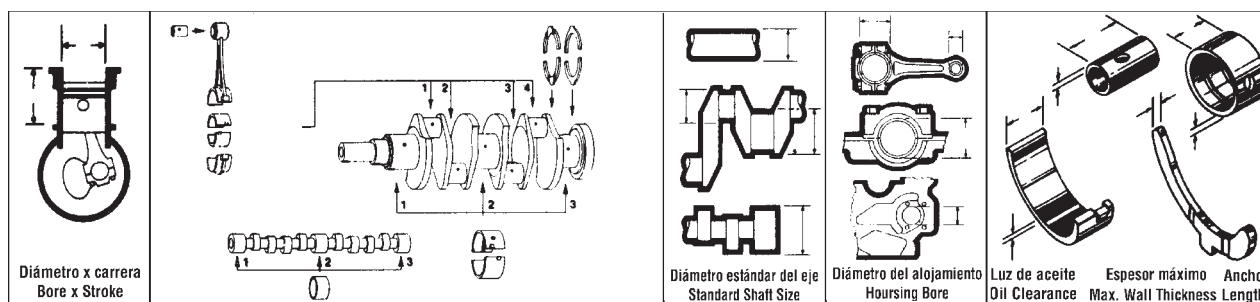
Gol 1.000	84
Fusca, Brasilia, Gol, Kombi, Variant, Karmann Guia, Furgoneta, 1.3, 1.5, 1.6	84
Passat, Voyage, Parati, Passat TS/LS, Gacel, 1.6, 1.5	85
Passat D, Pick-Up, Furgao, Gol, Parati, Voyage, Passat Saveiro, Apollo, Santana, Gol GTI, Logus, Quantum, Gacel, GS, Carat, Senda 1.6, 1.8, 2.0, 2.0i, 1.6 Diesel - 1985 /	85
Golf III/IV, Golf Cabrio, Golf Variant, Polo Classic, Pasat, Sharan, Vento, Bora, Beetle, Caravelle, Transporter 1.9 TD/TDI Eng. AAZ/ABL/ADF/AFN/AHU/ ARD/1Z Turbo Diesel - 1993 /	86
Golf III/IV, Pasat, Sharan, Vento, Caravelle, Bora VR6 (AAA/AES) - 1991 /	86



Cojinetes para motores

Bronzinas

Engine Bearings



Alfa Romeo



1

33 1.2/ 1.3/ 1.5/ 1.7

4	Biela	4-66325 RA	STD-10-20-30-40	49.992/50.000 1.9682/1.9685"	53.696/53.708 2.1140/2.1145"	0.023/0.069	1.836 .0723"	17.800 .0008"
80 x 67.2 1351 c.c.	Bancada	6446 M 66326 RA	STD-10-20-30-40	59.944/59.957 2.3600/2.3605"	63.663/63.673 2.5064/2.5068"	0.023/0.071	1.841 .0725"	23.400 .0213"
84 x 67.2 1490 c.c.	Axial	66326 AF		65.790/66.040 2.5902/5.6000"	82.420/82.680 3.2449/3.2551"		2.362 .0930"	

Alfa Romeo



2

145, 146, 1.6i/ 33 4Q 1.7i 16V

4	Biela	4-4075 AP	STD-0.25-0.50	49.984/50.000 1.9679/1.9685"	53.696/53.708 2.1140/2.1145"	0.023/0.076	1.836 .0723"	17.800 .0008"
87 x 72.2 1712 c.c.	Bancada	7178 4078 CP	STD-0.25-0.50	59.944/59.964 2.3600/2.3608"	63.663/63.673 2.5064/2.5068"	0.020/0.069	1.842 .0723"	23.400 .9213"
	Axial	66326 AF	STD-5	65.790/66.040 2.5902/2.6000"	82.420/82.680 3.2449/3.2551"		2.362 .0930"	

Alfa Romeo



3

145, 146, 155 Twin Spark 16V, 1.8 -2.0

4	Biela	4-4350 CP	STD-10-20	50.787/50.805 1.9995/2.0002"	53.901/53.913 2.122/2.1226"	0.028/0.082	1.534 .0604"	19.500 .0677"
82 x 82.7 1747 c.c.	Bancada	7255 M 4351 RA	STD-10-20		56.705/56.718 2.2352/2.2330"		1.843 .0726"	20.750 .8169"
83 x 91 1970 c.c.		4352 CP			56.705/56.718 2.2352/2.2330"		1.838 .0724"	20.750 .8169"
	Axial	4351 AF	STD	58.808/59.062 2.3153/2.3253"	73.740/73.486 2.903/2.8931"		2.358 .0928"	

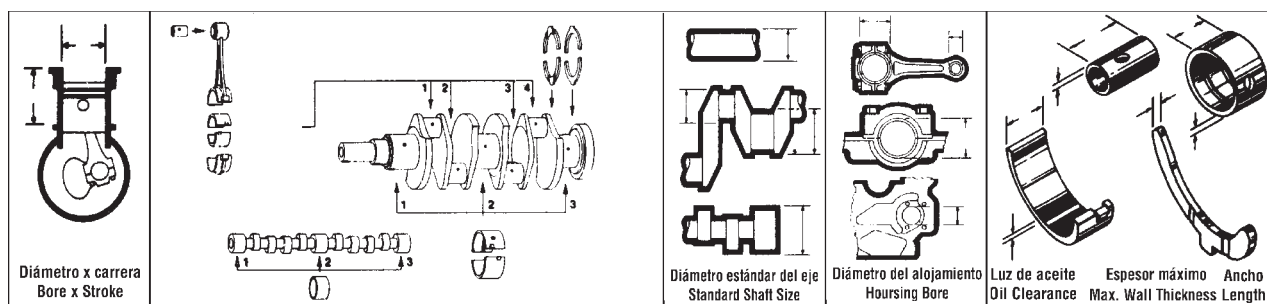
Alfa Romeo



4

155, 164 Twin Spark, 155, 4Q 16V 2.0

4	Biela	4-60010 CP	STD-10-20-30	49.988/49.998 1.9680/1.9684"	53.695/53.708 2.1140/2.1145"	0.021/0.068	1.838 .0724"	23.440 .9228"
84 x 90 1995 c.c.	Bancada	4687 M 60011 CP	STD-10-20-30-40	59.960/59.973 2.3606/2.3611"	63.657/63.676 2.5062/2.5069"	0.020/0.067	1.837 .0723"	25.530 1.0051"
		60012 CPA		59.960/59.973 2.3606/2.3611"	63.657/63.676 2.5062/2.5069"	0.020/0.067	1.837 .0723"	18.360 .0228"
	Axial	2777 AF		65.786/66.040 2.5900/2.6000"	82.423/82.677 3.2450/3.2550"		2.362 .0930"	



Alfa Romeo



5	155, 156, 166 2.5 V6/ 164, 166 24V							
6 88 x 68.3 2492 c.c. 93 x 72.6 2959 c.c.	Biela	6-4080 CP	STD-10-20	51.980/52.000 2.0465/2.0472"	55.511/55.524 2.1855/2.1860"	0.020/0.071 .0689"	1.749 .7520"	19.100
	Bancada	7179 M 4081 CP	STD-10-20	59.951/59.971 2.3603/2.3611"	63.657/63.676 2.5062/2.5069"	0.020/0.068 .0725"	1.841 .8799"	22.350
	Axial	4081 AF	STD-5-10	65.786/66.040 2.5900/2.6000"	82.423/82.677 3.2450/3.2550"	2.362 .0930"		

Bedford



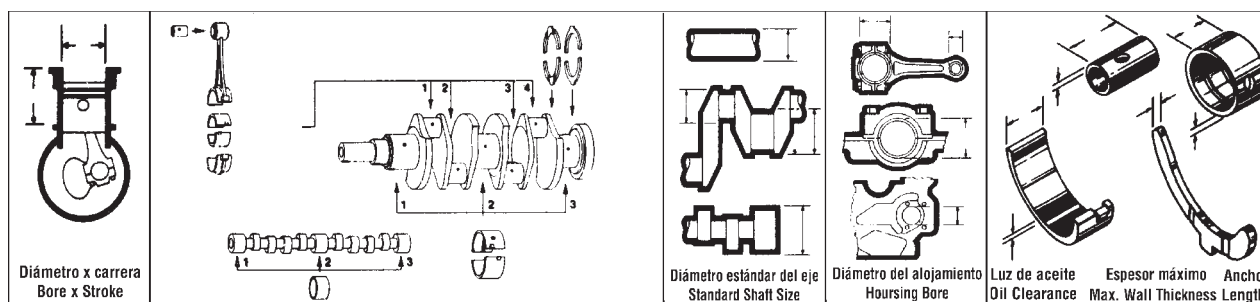
1	200D SERIE DKT							
4 98.43 x 107.95 3285 c.c. (200")	Bielas	4-2915 RA	STD-10-20-30-40-50	60.300/60.325 2.3740/2.3750"	63.995/64.008 2.5195/2.5200"	0.051/0.102 .0020/.0040"	1.810 .0714"	31.80 1.252"
	Bancadas	6470 M	STD-10-20-30-40-50					
	1	62086 RA		69.837/69.850 2.7495/2.7500"	74.079/74.092 2.9165/2.9170"	0.051/0.127 .0020/.0050"	2.083 .0820"	34.47 1.357"
	2-4	62087 RA		69.837/69.850 2.7495/2.7500"	74.079/74.092 2.9165/2.9170"	0.051/0.127 .0020/.0050"	2.083 .0820"	27.13 1.068"
	3	62088 RA		69.837/69.850 2.7495/2.7500"	74.079/74.092 2.9165/2.9170"	0.051/0.127 .0020/.0050"	2.083 .0820"	35.31 1.390"
	5	62089 RA		69.837/69.850 2.7495/2.7500"	74.079/74.092 2.9165/2.9170"	0.051/0.127 .0020/.0050"	2.083 .0820"	31.12 1.225"
	Semi Arand de Empuje	2919 BF	STD-5-10				2.362	
	Levas	1309 M	STD-2-5-10-15-20-30					
	1	2916 DR		54.254/54.267 2.1360/2.1365"	57.379/57.429 2.2590/2.2610"	0.510/0.102 .0020/.0040"	1.537 .0605"	36.76 1.447"
	2	2917 DR		51.968/51.981 2.0460/2.0465"	55.093/55.143 2.1690/2.1710"	.510/0.102 .0020/.0040"	1.537 .0605"	25.65 1.010"
	3	2918 DR		50.876/50.889 2.0030/2.0035"	54.000/54.051 2.1260/2.1280"	0.051/0.102 .0020/.0040"	1.537 .0605"	25.65 1.010"
	4	2919 DR		50.114/50.127 1.9730/1.9735"	55.093/55.143 2.1960/2.1710"	0.051/0.102 .0020/.0040"	2.464 .0970"	32.79 1.291"
Buje de Biela 4056 Y								

TORSION DE APRIETO: Bielas 7.6/ 8.3 mkg. Bancada 9 mkg.
Tapa de Cilindros: prisionero 12.4/ 13.8 mkg., tuerca 20/ 22 mkg.

Bedford



2	300, 300D, 350D, SERIES D, J, K, T, J4, J6 - 1957/1967							
6 98.43 x 107.95 4927 c.c. (300")	Bielas	6-2915 RA	STD-10-20-30-40-50	60.300/60.325 2.3740/2.3750"	63.995/64.008 2.5195/2.5200"	0.051/0.102 .0020/.0040"	1.810 .0714"	31.80 1.252"



Bedford



2 (continuación) 300, 300D, 350D, SERIES D, J, K, T, J4, J6 - 1957/1967

6 98.43 x 107.95 4927 c.c. (300")	Bancadas	6471 M	STD-10-20-30-40-50					
	1	62086 RA		69.837/69.850 2.7495/2.7500"	74.079/74.092 2.9165/2.9170"	0.051/0.127 .0020/.0050"	2.083 .0820"	34.47 1.357"
106 x 107.95 5755 c.c. (350")	2-3-5-6	62087 RA		69.837/69.850 2.7495/2.7500"	74.079/74.092 2.9165/2.9170"	0.051/0.127 .0020/.0050"	2.083 .0820"	27.13 1.068"
	4	62088 RA		69.837/69.850 2.7495/2.7500"	74.079/74.092 2.9165/2.9170"	0.051/0.127 .0020/.0050"	2.083 .0820"	35.31 1.390"
	7	62089 RA		69.837/69.850 2.7495/2.7500"	74.079/74.092 2.9165/2.9170"	0.051/0.127 .0020/.0050"	2.083 .0820"	31.12 1.225"
	Semi Arand. de Empuje	2919 BF	STD-5-10				2.362	
	Levas	1309 M	STD-2-5-10-20-30					
	1	2916 DR		54.254/54.267 2.1360/2.1365"	57.379/57.429 2.2590/2.2610"	0.051/0.102 .0020/.0040"	1.537 .0605"	36.76 1.447"
	2	2917 DR		51.968/51.981 2.0460/2.0465"	55.093/55.143 2.1690/2.1710"	0.051/0.102 .0020/.0040"	1.537 .0605"	25.65 1.010"
	3	2918 DR		50.876/50.889 2.0030/2.0035"	54.000/54.051 2.1260/2.1280"	0.051/0.102 .0020/.0040"	1.537 .0605"	25.65 1.010"
	4	2919 DR		50.114/50.127 1.9730/1.9735"	55.093/55.148 2.1690/2.170"	0.051/0.102 .0020/.0040"	2.464 .0970"	32.79 1.291"
Buje de Biela 4055 Y (nafta)								
Buje de Biela 4056 Y (diesel)								

TORSION DE APRIETO: Bielas 7.6/ 8.3 mkg. Bancadas 9 mkg.
Tapa de cilindros, prisionero 12.4/ 13.8 mkg., tuerca 20/ 22 mkg

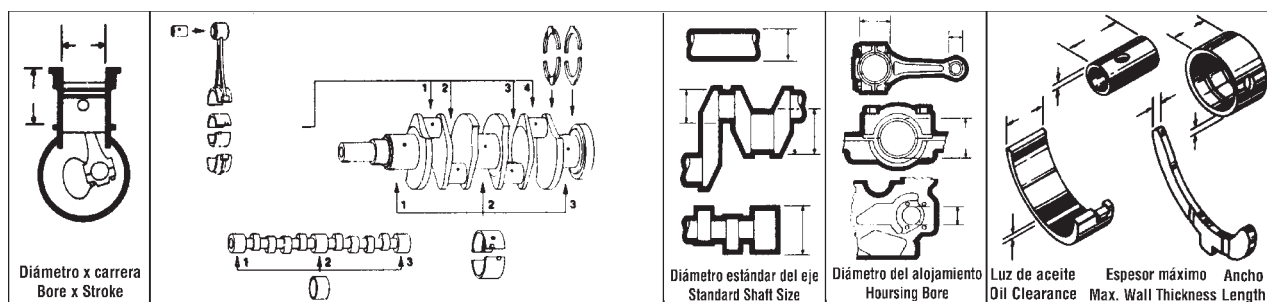
Borgward



1

V M 492 A

4 92 x 90 2392 c.c.	Biela	4-1255 CP	STD-0.10-0.20-0.30	53.904/53.924	57.563/57.582	1.798	23.25
	Bancada	7018 M	STD-0.10-0.20-0.30				
	1	1256 RA		62.980/63.000	67.025/67.050	2.005	32.25
	2-3-4	1257 RA		62.980/63.000	66.670/66.987	1.823	30.00
	5	1258 RA		69.980/70.000	75.005/75.030	2.750	32.25
Axial		1256 BF	STD-5-10				
Buje de Biela 4452 Y							



Chevrolet

2.0, 1.8, 1.6 - Monza SL/SLE/Classic/SR, Omega/ Omega-Suprema, Kadett Ipanema, Vectra GLS/CD, Vectra 16 V - 1987

1

4 86 x 86 1998 c.c.	Bielas	4-1490 CPA	STD-0.25-0.50-0.75 1.00	48.971/48.987 1.9280/1.9286"	52.000/52.012 2.0472/2.0477"	0.019/0.063 .0007/0.0025"	1.497 0.0600"	19.30 0.760"
Reemplaza a 4-1490 CP y 4-68140 CP								
	Bancada	7108 M	STD-0.25-0.50	57.983/57.996 2.2828/2.2883"	62.001/62.014 2.4410/2.4415"	0.018/0.058 .0007/0.0023"	1.994 .0785"	19.51 .768"
	1-2-4-5	68141 CP	0.75-1.00					
	3 ▲	* 68142 CP		57.983/57.996 2.2828/2.2883"	62.001/62.014 2.4410/2.4415"	0.018/0.058 .00070.0023"	1.994 .0785"	25.90 1.020"
		* 0.25						26.10 1.028"
		* 0.50						26.30 1.035"
		* 0.75						26.30 1.035"

Chevrolet

2

Opel K-180

4 98.43 x 82.50 2500 c.c. (151")	Bielas	4-2020 RA	STD-10-20-30-40	50.780/50.800 1.9992/2.000"	53.967/53.980 2.1247/2.1252"	0.015/0.079 .0006/0.0031"	1.576 .0620"	21.39 .842"
4 101.60 x 76.20 2512 c.c. (153")	Bancadas	4123 M	STD-10-20-30-40	58.377/58.402 2.2983/2.2993"	63.261/63.287 2.4906/2.4916"	0.013/0.074 .0005/0.0029"	2.426 .0955"	20.50 .807"
	1-2-3-4	2556 RA						
	5 ▲	2558 RA		58.377/58.402 2.2983/2.2993"	63.261/63.287 2.4906/2.4916"	0.013/0.074 .0005/0.0029"	2.426 .0955"	25.62 1.009"
	Levas	1556 M	SEMI					

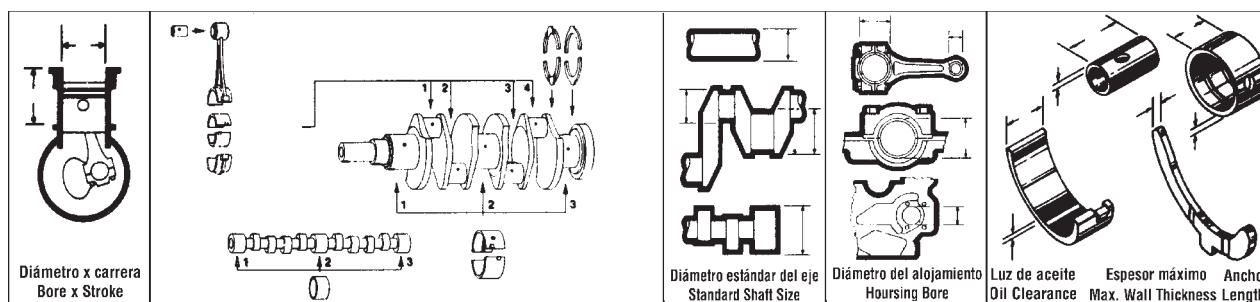
Chevrolet

194, 230,250,194 Cl. D., 400, Special, Rally Sports, Super, S. Sport, Chevy, Chevy S. Sport, Malibu, Pick Up C10, C30, A-10/A-20/C-10, Opala, Caravan, Veraneio, Bonanza - 1962/1978

3

6 90.49 x 82.55 3185 c.c. (194")	Bielas	6-2020 RA	STD-10-20-30-40	50.775/50.800 1.9990/2.0000"	53.967/53.980 2.1247/2.1252"	0.018/0.068 .00070.0027"	1.575 .0620"	21.39 .842"
9.843 x 82.55 3769 c.c. (230")	Bancadas	4124 M	STD-10-20-30-40-50	58.377/58.402 2.2983/2.2993"	63.261/63.287 2.4906/2.4916"	0.007/0.074 .0003/0.0029"	2.426 .0955"	20.50 .807"
	1-2-3-4-5-6	2556 RA						
98.43 x 89.66 4097 c.c. (250")	7 ▲	2558 RA		58.377/58.402 2.2983/2.2993"	63.261/63.287 2.4906/2.4916"	0.007/0.074 .0003/0.0029"	2.426 .0955"	25.63 1.009"
1er. TIPO HASTA APROX. JULIO 1977								

▲ Cojinete con pestaña - Flanged bearing - * Ancho diferente - Largura diferente



Chevrolet



194, 230, 250, 194 Cl. D., 400, Special, Rally Sports, Super, S. Sport, Chevy, Chevy S. Sport, Malibu, Pick Up C10, C30, A-10/A-20/C-10, Opala, Caravan, Veraneio, Bonanza - 1962/1978

3 (continuación)

6	Levas	1203 M	STD-5-10-15-20					
90.49 x 82.55	1-4	2287 DR		47.452/47.478	51.029/51.079	0.020/0.112	1.765	21.97
3185 c.c.				1.8682/1.8692"	2.0090/2.0110"	.008/.0044"	.0695"	.865"
(194")	2-3	2288 DR		47.452/47.478	50.775/50.825	0.020/0.112	1.638	21.97
9.843 x 82.55				1.8682/1.8692"	1.9990/2.0010"	.008/.0044"	.0645"	.865"
3769 c.c.	Levas	1557 M	STD-5-10-15-20					
(230")	1-2-3-4	2288 DR		47.452/47.478	50.775/50.825	0.020/0.112	1.638	21.97
98.43 x 89.66				1.8682/1.8692"	1.9990/2.0010"	.0008/.0044"	.0645"	.865"
4097 c.c.								
(250")								
				TORSION DE APRIETO: Bielas 4.1/ 4.8 mkg. Bancadas 8.3/ 9.7 mkg				
				Tapa de cilindros 12.4/ 13.8 mkg.				
	Bielas	4-3275 RA	STD-10	COMPETICION - DISEÑO OPCIONAL				

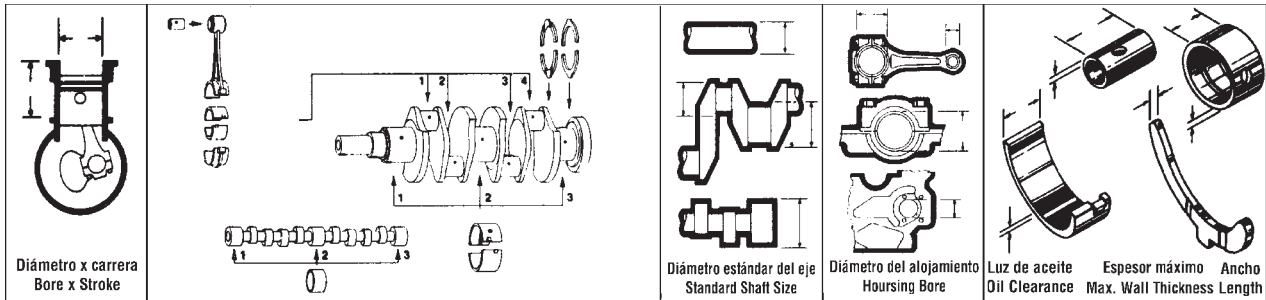
Chrysler



170, 225, De Soto D100, D400, Valiant, Polara, Coronado, GT, GTX, Dodge D100, D200, D400, D500, Fargo D100, D400 - 1964/1980

1

6	Bielas	6-2375 RA	STD-10-20-30-40	55.537/55.563	58.738/58.750	0.012/0.037	1.585	25.15
86.36 x 79.37				2.1865/2.1875"	2.3125/2.3130"	.0005/.0015"	.0624"	.990"
2788 c.c.								
(170")								
86.36 x 104.77	Bancadas	4092 M	STD-10-20-30-40					
3687 c.c.	1	2376 RA		69.837/69.863	74.740/74.752	0.012/0.037	2.436	26.39
(225")				2.7495/2.7505"	2.9425/2.9430"	.0005/.0015"	.959"	1.039"
	3 ▲	2377 RA		69.837/69.863	74.740/74.752	0.012/0.037	2.436	31.88
				2.7495/2.7505"	2.9425/2.9430"	.0005/.0015"	.959"	1.255"
	2-4	2378 RA		69.837/69.863	74.740/74.752	0.012/0.037	2.436	26.39
				2.7495/2.7505"	2.9425/2.9430"	.0005/.0015"	.959"	1.039"
	Levas	1187 M	STD-5-10-20-30					
	1	2376 DR		50.749/50.775	54.089/54.115	0.025/0.076	1.648	23.62
				1.9980/1.9990"	2.1295/2.1305"	.0010/.0030"	.0649"	.930"
	2	2322 DR		50.343/50.368	53.683/53.708	0.025/0.076	1.648	19.30
				1.9820/1.9830"	2.1135/2.1145"	.0010/.0030"	.0649"	.760"
	3	2378 DR		49.962/49.987	53.302/53.327	0.025/0.076	1.648	19.30
				1.9670/1.9680"	2.0985/2.0995"	.0010/.0030"	.0649"	.760"
	4	2379 DR		49.555/49.581	52.896/52.921	0.025/0.076	1.648	19.30
				1.9510/1.9520"	2.0825/2.0835"	.0010/.0030"	.0649"	.760"
				TORSION DE APRIETO: Bielas 6.2 mkg. Bancada 11.7 mkg				
				Tapa de cilindros 9 mkg.				



Chrysler



2 318-V8, Coupe GTX - 1970/1978

8 99.31 x 84.15 5210 c.c. (318")	Bielas	8-2130 CP	STD-10-20-30-40	53.949/53.975 2.1240/2.1250"	57.150/57.168 2.2500/2.2507"	0.018/0.074 .0005/.0029"	1.585 .0624"	21.53 .848"
	Bancadas	4923 M	STD-10-20-30-40					
	1-2-4	2131 RAA		63.487/63.513 2.4995/2.5005"	63.390/68.402 2.6925/2.6930"	0.015/0.069 .0006/.0027"	2.431 .0957"	22.28 .877"
	3 ▲	2132 RAA		63.487/63.513 2.4995/2.5005"	63.390/68.402 2.6925/2.6930"	0.015/0.069 .0006/.0027"	2.431 .0957"	29.26 1.152"
	5	2143 RAA		63.487/63.513 2.4995/2.5005"	63.390/68.402 2.6925/2.6930"	0.015/0.069 .0006/.0027"	2.431 .0957"	33.71 1.327"
Levas								
	1						2.159 .085"	22.35 .88"
	2			50.343/50.368 1.9820/1.9830"	53.683/53.708 2.1135/2.1145"	0.018/0.089 .0007/.0033"	1.648 .0649"	19.30 .760"
	3			49.962/49.987 1.9670/1.9680"	53.302/53.327 2.0985/2.0995"	0.018/0.089 .0007/.0033"	1.648 .0649"	19.30 .760"
	4			49.555/49.581 1.9510/1.9520"	52.696/52.921 2.0825/2.0835"	0.018/0.089 .0007/.0033"	1.648 .0649"	19.30 .760"
	5			39.637/39.662 1.5605/1.5615"	42.977/43.002 1.6920/1.6930"	0.018/0.089 .0007/.0033"	1.648 .0649"	23.88 .940"
TORSION DE APRIETO: Bielas 6.2 mkg. Bancada: 11.7 mkg Tapa de cilindros: 7 mkg final 11.7 mkg.								

Citröen



1 Xantia, Xsara, Berlingo D/TD, 1.9, XUD9

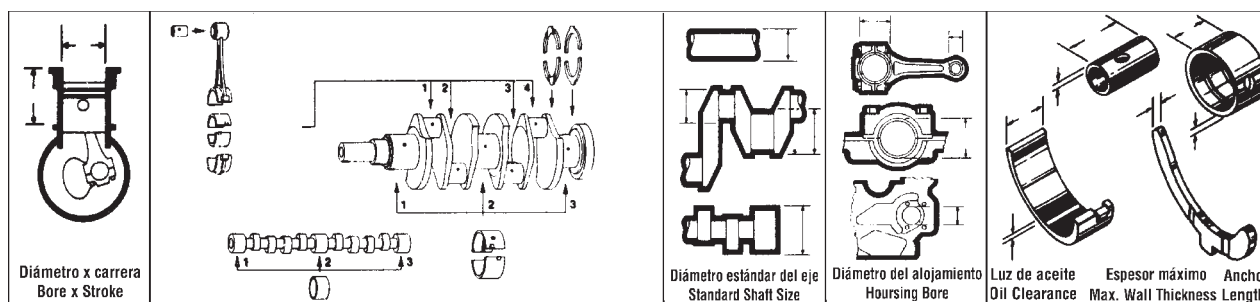
4 80 x 88 1769 c.c. 83 x 88 1905 c.c.	Biela	4-66725 RA	STD-0.30	49.984/50.000 1.9679/1.9865"	53.695/53.708 2.1140/2.1145"	0.035/0.088	1.830 0.0720"	20.050 0.7894"
	Bancada	6610 M	STD-0.30					
		66726 RA		59.981/60.000 2.3615/2.3622"	63.731/63.750 2.5091/2.5098"	0.041/0.103	1.845 .0726"	20.290 .7988"
	Axial	66721 BF	STD-10-15	67.050/67.300 2.6398/2.6496"	83.050/83.300 3.2697/3.2795"		2.330 .0917"	

Cummins



1 B/BT 6 cil. 5.9 Lts.

L6 4.016 x 4.724 359" (5.9L)	Biela	6-4205 CP	STD-.25-.50-.75	2.7161/2.7170"	2.8735/2.8745"	.0015/.0045"	.0774"	1.225"
	Bancada	7216 M	STD-.25-.50-.75					
	1-2-3-4-5-7	4206 RA		3.2672/3.2682"	3.4639/3.4652"	.0016/.0047"	.0970"	1.143"
	4	4207 RA		3.2672/3.2682"	3.4639/3.4652"	.0016/.0047"	.0970"	1.470"



Cummins



2

B/BT 4 cil. 3.9 Lts.

L4 4.016 x 4.724 239 ³ (3.9L)	Biela	4-4205 CP	STD-.25-.50-.75	2.7161/2.7170"	2.8735/2.8745"	.0015/.0045"	.0774"	1.225"
	Bancada	7251 M	STD-.25-.50-.75	2.7161/2.7170"	2.8735/2.8745"	.0015/.0045"	.0774"	1.225"
	1-2-3-4-5	4206 RA		3.2672/3.2682"	3.4639/3.4652"	.0016/.0047"	.0970"	1.143"
	4	4207 RA		3.2672/3.2682"	3.4639/3.4652"	.0016/.0047"	.0970"	1.470"

Daihatsu



1

AB, AD, Cuore, Max Cuore, Van L 40, 55 W - 1978/1982

2 71.6 x 68 547 c.c.	Bielas	2-66765 CP	STD-0.25-0.50-0.75	39.980/40.000	43.000/43.020	0.020/0.040	1.491	18.00
	Bancadas	6662 M	STD-0.25-0.50-0.75					
	1-2-3	66766 CP		41.979/42.000	46.000/46.020	0.022/0.050	1.989	19.50
	Axial	66676 BF	STD-5-10				1.989	

Daihatsu



2

CB, Charade, Hijet 550, 600 - 1978/1982

3 76 x 73 993 c.c.	Bielas	3-1180 RA	STD-0.25-0.50-0.75	39.980/40.000	43.000/43.020	0.020/0.040	1.491	18.00
	Bancadas	6697 M	STD-0.25-0.50-0.75					
	1-2-3-4	1186 RA		41.979/42.000	46.000/46.020	0.022/0.050	1.989	19.50
	Axial	66676 BF	STD-5-10				1.989	

Deutz

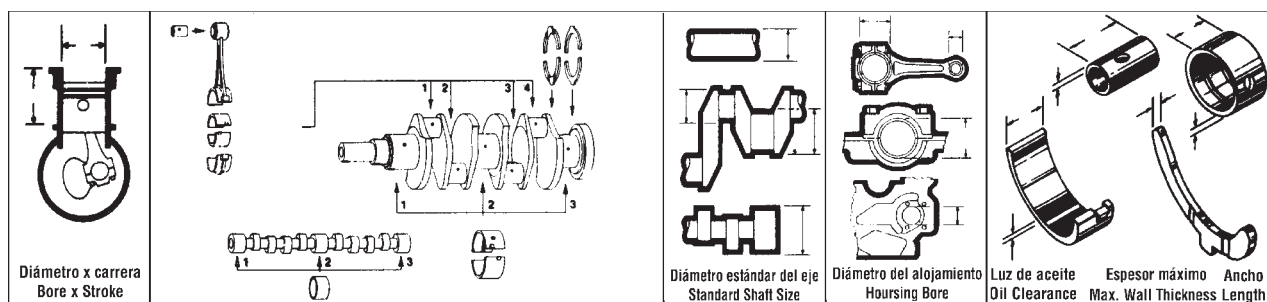


1

F3L 913 - 1978

3 102 x 125 3064 c.c.	Bielas	1-66585 CP (3 pares por motor)	STD-0.25-0.50	59.941/59.960	64.000/64.019	0.040/0.098	2.000	25.00
	Bancadas	1-66586 CP (4 pares por motor)	STD-0.25-0.50					
	1-2-3-4			69.971/69.990	74.500/74.519	0.050/0.112	2.289	27.00
	Axial	66586 BF	STD-5-10				2.985	
	Levas	1705 M						
	1-2-3-4-5	66586 DR		47.934/47.950	52.000/52.030	0.030/0.100	2.010	30.50

TORSION DE APRIETO: Bielas preaprieto 3 mkg., reaprieto 60° + 30° = 90°. Bancadas preaprieto 3 mkg., reaprieto 60° + 45° = 105°. Tapa de cilindros preaprieto 3 mkg., reaprieto 45° + 45° + 45° = 135°



Deutz



2 F4L 913 - 1978

4 102 x 125 4086 c.c.	Bielas	1-66585 CP (4 pares por motor)	STD-0.25-0.50	59.941/59.960	64.000/64.019	0.040/0.098	2.000	25.00
	Bancadas 1-2-3-4-5	1-66586 CP (5 pares por motor)	STD-0.25-0.50	69.971/69.990	74.500/74.519	0.050/0.112	2.289	27.00
	Axial	66586 BF	STD-5-10				2.985	
	Levas 1-2-3-4-5	1703 M 66586 DR		47.934/47.950	52.000/52.030	0.030/0.100	2.010	30.50

TORSION DE APRIETO: Bielas preaprieto 3 mkg, reaprieto 60° + 30° = 90°. Bancadas preaprieto 3 mkg., reaprieto 60° + 45° = 105°. Tapa de cilindros preaprieto 3 mkg., reaprieto 45° + 45° + 45° = 135°

Deutz



3 F5L 913 - 1978

5 102 x 125 5106 c.c.	Bielas	1-66585 CP (5 pares por motor)	STD-0.25-0.50	59.941/59.960	64.000/64.019	0.040/0.098	2.000	25.00
	Bancadas 1-2-3-4-5-6	1-66586 CP (6 pares por motor)	STD-0.25-0.50	69.971/69.990	74.500/74.519	0.050/0.112	2.289	27.00
	Axial	66586 BF	STD-5-10				2.985	
	Levas 1-2-3-4-5-6-7	1703 M 66586 DR		47.934/47.950	52.000/52.030	0.030/0.100	2.010	30.50

TORSION DE APRIETO: Bielas preaprieto 3 mkg, reaprieto 60° + 30° = 90°. Bancadas preaprieto 3 mkg., reaprieto 60° + 45° = 105°. Tapa de cilindros preaprieto 3 mkg., reaprieto 45° + 45° + 45° = 135°

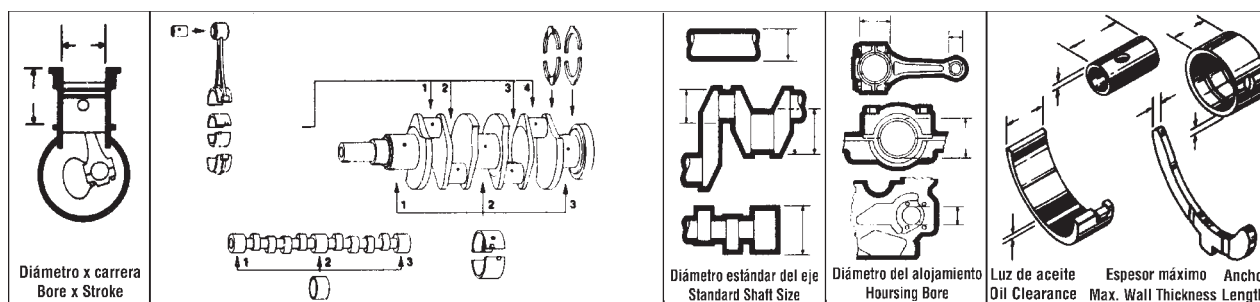
Deutz



4 F6L 913, DD1000, F 7000 - 1978

6 102 x 125 6128 c.c.	Bielas	1-66585 CP (6 pares por motor)	STD-0.25-0.50	59.941/59.960	64.000/64.019	0.040/0.098	2.000	25.00
	Bancadas 1-2-3-4-5-6-7	1-66586 CP (7 pares por motor)	STD-0.25-0.50	69.971/69.990	74.500/74.519	0.050/0.112	2.289	27.00
	Axial	66586 BF	STD-5-10				2.985	
	Levas 1-2-3-4-5-6-7	1705 M 66586 DR		47.934/47.950	52.000/52.030	0.030/0.100	2.010	30.50

TORSION DE APRIETO: Bielas preaprieto 3 mkg, reaprieto 60° + 30° = 90°. Bancadas preaprieto 3 mkg., reaprieto 60° + 45° = 105°. Tapa de cilindros preaprieto 3 mkg., reaprieto 45° + 45° + 45° = 135°



Deutz

5

BF6L 913 Turbo, F7000 - 1978

6 102 x 125 6128 c.c.	Bielas	6-66590 CP	STD-0.25-0.50	65-971/95.990	70.000/70.019	0.040/0.098	1.985	25.00
	Bancadas	6571 M	STD-0.25-0.50					
	1-2-3-4-5-6-7	66591 CP		74.971/74.990	79.000/79.019	0.050/0.108	1.985	25.00
	Axial	66591 BF	STD-5-10				2.985	
	Levas	1705 M						
	1-2-3-4-5-6-7	66586 DR		47.925/47.950	52.070/52.100	0.030/0.100	2.010	30.00
TORSION DE APRIETO: Bielas preaprieto 3 mkg, reaprieto 60° + 30° = 90° Bancada preaprieto 3 mkg, reaprieto 60° + 45° = 105° Tapa de cilindros preaprieto 3 mkg, reaprieto 45° + 45° + 45° = 135°								

Deutz

6

A/F 1114, A/F 2114, A46 - 1969/1978

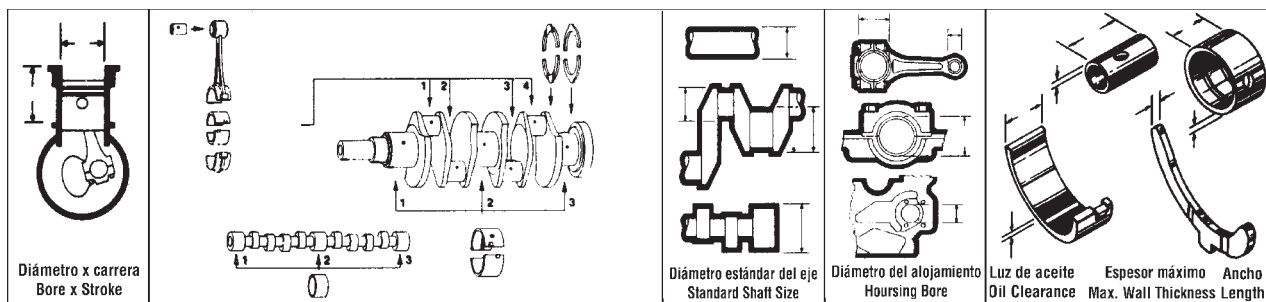
2 115 x 140 2908 c.c. 120 x 140 3167 c.c.	Bielas	61960 CPA	Ver notas 1 y 2	79.971/79.990	85.000/85.022	0.070/0.138	2.470	29.00
		61960 CP	Grupo 9					
		(2 pares por motor)						
	Bancadas	4593 M	STD-0.25-0.50-0.75					
	1-2	61961 CP		84.966/84.988	90.000/90.022	0.070/0.138	2.471	29.00
	3 ▲	61962 CP		84.966/84.988	90.000/90.022	0.070/0.138	2.471	39.82
	Levas							
	1-2-3			47.911/47.950	48.000/48.025	0.0750/0.139		
	Buje de Biela	4305 Y						
TORSION DE APRIETO: VER GRUPO 9								

Deutz

7

A/F3L 1114, A/F3L 2114, A65 - 1969/1978

3 115 x 140 4363 c.c. 120 x 140 4750 c.c.	Bielas	61960 CPA	Ver notas 1 y 2	79.971/79.990	85.000/85.022	0.070/0.138	2.470	29.00
		61960 CP	Grupo 9					
		(3 pares por motor)						
	Bancadas	4593 M	STD-0.25-0.50-0.75					
	1-2-3	61961 CP		84.966/84.988	90.000/90.022	0.070/0.138	2.471	29.00
	4 ▲	61962 CP		84.966/84.988	90.000/90.022	0.070/0.138	2.471	39.82
	Levas							
	1-2-3			47.911/47.950	48.000/48.025	0.0750/0.139		
	Buje de Biela	4305 Y						
TORSION DE APRIETO: VER GRUPO 9								



Deutz

8 A/F4L 1114, A/F4L 2114, A85 - 1969/1978

4 115 x 140 5817 c.c. 120 x 140 6333 c.c.	Bielas	61960 CPA 61960 CP (4 pares por motor)	Ver notas 1 y 2 Grupo 9	79.971/79.990	85.000/85.022	0.070/0.138	2.470	29.00
	Bancadas	4595 M	STD-0.25-0.50-0.75					
	1-2-3-4	61961 CP		84.966/84.988	90.000/90.022	0.070/0.138	2.471	29.00
	5 ▲	61962 CP		84.966/84.988	90.000/90.022	0.070/0.138	2.471	39.82
	Levas							
	1-2-3-4-5			47.911/47.950	48.000/48.025	0.0750/0.139		
	Buje de Biela	4305 Y	TORSION DE APRIETO: Bielas: Ajuste previo 2.5/ 3.5 mkg. Tensión final: 45° + 45° + 30° = 120° Bancada: Ajuste previo 2.5/ 3.5 mkg. Tensión final: 30° + 30° + 30° = 90° Tapa de cilindros: Ajuste previo: 2.5/ 3.5 mkg. Tensión final: 45° + 45° + 45° + 30° = 165°					

Deutz

9 A/F6L 1114, B/F6L 1114, A/F6L 2114, BF6L 2114, A130, A144, 150L12 - 1969/1978

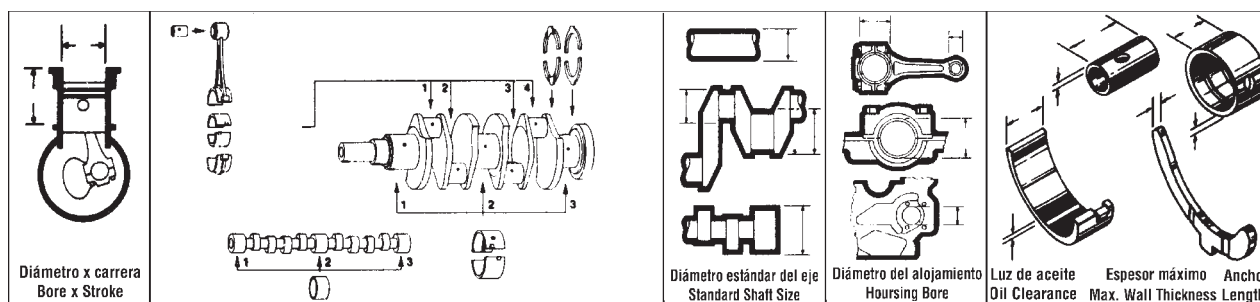
6 115 x 140 8725 c.c. 120 x 140 9500 c.c.	Bielas (1) (2)	61960 CPA 61960 CP (6 pares por motor)	STD-0.25-0.50-0.75	79.971/79.990	85.000/85.022	0.070/0.138	2.470	29.00
	Bancadas	4596 M	STD-0.25-0.50-0.75					
	1-2-3-4-5-6	61961 CP		84.966/84.988	90.000/90.022	0.070/0.138	2.471	29.00
	7 ▲	61962		84.966/84.988	90.000/90.022	0.070/0.138	2.471	39.82
	Levas							
	1-2-3-4-5-6-7			47.911/47.950	48.000/48.025	0.0750/0.139		
	Buje de Biela	4305 Y	TORSION DE APRIETO: Bielas: Ajuste previo 2.5/ 3.5 mkg. Tensión final: 45° + 45° + 30° = 120° Bancada: Ajuste previo 2.5/ 3.5 mkg. Tensión final: 30° + 30° + 30° = 90° Tapa de cilindros: Ajuste previo: 2.5/ 3.5 mkg. Tensión final: 45° + 45° + 45° + 30° = 165° (1) Primer tipo hasta 1970 Uña de fijación / (2) Segundo tipo hasta 1971 Perno de fijación					

Deutz

10 F8L 413, 200 D16, 200 D19, 200 D24, 200 RS12, 210 EV120-5020 - 1969/1977

8 120 x 125 11309 c.c.	Bielas	61960 CPA 71-2834/8	STD-0.25-0.50-0.75	74.971/74.990	80.000/80.019	0.060/0.118	2.475	28.00
	Bancadas	6468 M H-925/5	STD-0.25-0.50-0.75.					
	1-2-4-5	62076 CP		89.966/89.988	96.000/96.022	0.072/0.141		29.00
				89.966/89.988	96.000/96.022	0.072/0.141	2.965	32.00
	3 ▲	62077 CP		89.966/89.988	96.000/96.022	0.072/0.141	2.965	40.38
	Levas	1679 M	STD-5-10-20					
	1	62076 DR		59.910/59.940	64.000/64.019	0.060/0.137	2.000	39.00
	2-3-4-5	62077 DR		59.910/59.940	64.000/64.019	0.060/0.137	2.000	24.00

▲ Cojinete con pestaña - Flanged bearing



Deutz



10 (continuación)

F8L 413, 200 D16, 200 D19, 200 D24, 200 RS12, 210 EV120-5020 - 1969/1977

Buje de Biela 4305 Y

TORSION DE APRIETO:

Bielas: Ajuste previo 4 + 1 mkg. Tensión final: $60^\circ + 60^\circ = 120^\circ$

Bancada: Ajuste previo 5 + 1 mkg. Tensión final: $60^\circ + 60^\circ + 30^\circ = 150^\circ$

Vertical (M16) $30^\circ + 30^\circ = 60^\circ$. Horizontal (M14)

Tapa de cilindros: Ajuste previo: 3 mkg.. Orden 3-1-2 Tensión final: 60°

Orden 1-2-3. Tensión final 60°

Di Tella



1489 c.c., 1500, Magnette, Riley, Rural Traveller, Pick Up Argenta - 1960/1967

1

4
73.02 x 88.90
1489 c.c.

Bielas	4-2055 RA	STD-10-20-30-40	47.648/47.661	51.333/51.346	0.025/0.063	1.824	25.53
Bancadas 1-2-3	6024 M 2056 RA	STD-10-20-30-40	50.813/80.825	54.508/54.521	0.013/0.051	1.829	35.05
Axial	2056 BF	STD-5-10				2.362	
Levas 1	1307 M 2056 DR		43.434/45.447	48.641/48.666	0.025/0.100	1.842	37.56
2	2057 DR		43.910/43.923	47.117/47.142	0.025/0.100	1.842	28.83
3	2058 DR		41.218/41.230	44.425/44.450	0.025/0.100	1.842	20.88

TORSION DE APRIETO: Bielas 3.2/ 4.8 mkg. Bancada 9.7/ 10.4 mkg
Tapa de cilindros 4.1/ 4.8 mkg

Di Tella



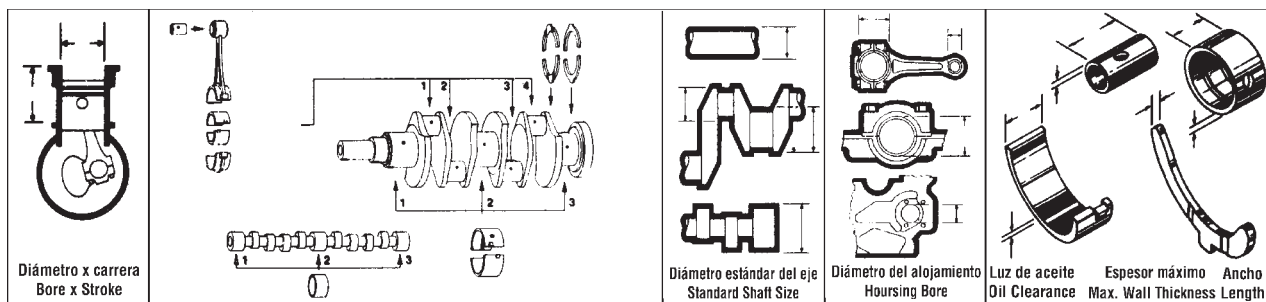
1622 c.c., Magnette, Morris, MG 1650 - 1965/1967

2

4
76.20 x 88.90
1622 c.c.

Bielas	4-2055 RA	STD-10-20-30-40	47.648/47.661	51.333/51.346	0.025/0.063	1.824	25.53
Bancadas 1-2-3	4587 M 2056 RAA	STD-10-20-30	50.813/80.825	54.508/54.521	0.013/0.051	1.829	35.05
Axial	2056 BF	STD-5-10				2.362	
Levas 1	1307 M 2056 DR		43.434/45.447	48.641/48.666	0.025/0.100	1.842	37.56
2	2057 DR		43.910/43.923	47.117/47.142	0.025/0.100	1.842	28.83
3	2058 DR		41.218/41.230	44.425/44.450	0.025/0.100	1.842	20.88

TORSION DE APRIETO: Bielas 3.2/ 4.8 mkg. Bancada 9.7/ 10.4 mkg
Tapa de cilindros 5.5/ 6.9 mkg



Dodge

1 **1498 c.c., 1500 - 1972/1981**

4 86.1 x 64.3 1498 c.c.	Bielas	4-3480 RA	STD-10-20-30-40	50.787/50.800 1.9995/2.000"	54.508/54.521 2.1460/2.1465"	0.023/0.061 .0009/.0024"	1.845 .0726"	20.75 .817"
	Bancadas	4963 M	STD-10-20-30-40					
	1-2-4-5	3481 RA		53.962/53.980 2.1245/2.1252"	57.683/57.704 2.2710/2.2780"	0.013/0.064 .0005/.0025"	1.848 .728"	25.53 1.005"
	3	3482 RA		53.962/53.980 2.1245/2.1252"	57.683/57.704 2.2710/2.2780"	0.013/0.064 .0005/.0025"	1.848 .728"	28.70 1.130"
	Axial	3482 BF	STD-5-10				2.362	
	Levas	1425 M	STD-5-10-15-20-30					
	1	3481 DR		49.136/49.154 1.9345/1.9352"	52.451/52.476 2.0650/2.0660"	0.030/0.076 .0013/.0030"	1.636 .0644"	19.30 .760"
	2	3482 DR		44.373/44.391 1.7470/1.7477"	47.701/47.727 1.8780/1.8790"	0.030/0.076 .0013/.0030"	1.642 .0646"	19.30 .760"
	3	3483 DR		39.611/39.629 1.5595/1.5602"	42.926/42.951 1.6900/1.6910"	0.030/0.076 .0013/.0030"	1.636 .0644"	19.30 .760"

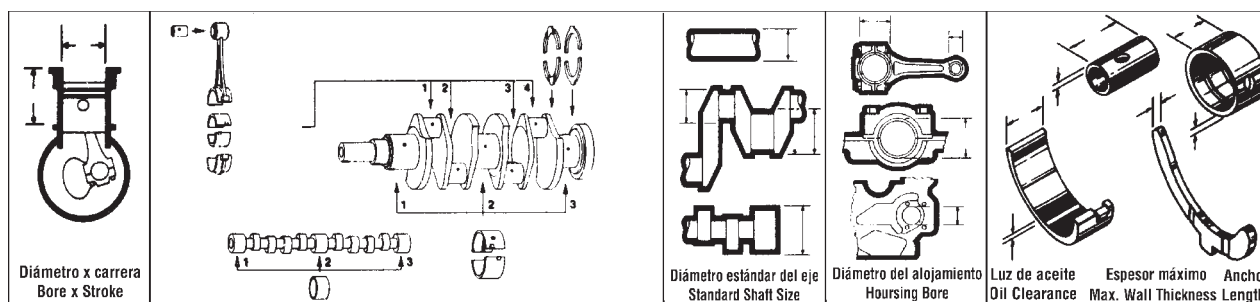
TORSION DE APRIETO: Bielas 4 mkg. Bancada 7.6 mkg
Tapa de cilindros 7.74 mkg

Dodge

2 **1798 c.c., 1500 1.8 - 1972/1981**

4 86.1 x 77.2 1798 c.c.	Bielas	4-62080 RA	STD-10-20-30	53.950/53.962 2.1240/2.1245"	57.683/57.696 2.2710/2.2715"	0.013/0.061 .0005/.0024"	1.843 .0726"	20.75 .817"
	Bancadas	6459 M	STD-10-20-30					
	1-2-4-5	62081 RA		60.300/60.317 2.3740/2.3747"	64.000/64.021 2.5197/2.5205"	0.013/0.064 .0005/.0025"	1.835 .722"	25.53 1.005"
	3	62082 RA		60.300/60.317 2.3740/2.3747"	64.000/64.021 2.5197/2.5205"	0.013/0.064 .0005/.0025"	1.835 .722"	28.70 1.130"
	Axial	62082 BF	STD-5-10				2.362	
	Levas	1425 M	STD-5-10-20-30					
	1	3481 DR		49.136/49.154 1.9345/1.9352"	52.451/52.476 2.0650/2.0660"	0.030/0.076 .0013/.0030"	1.636 .0644"	19.30 .760"
	2	3482 DR		44.373/44.391 1.7470/1.7477"	47.701/47.727 1.8780/1.8790"	0.030/0.076 .0013/.0030"	1.642 .0646"	19.30 .760"
	3	3483 DR		39.611/39.629 1.5595/1.5602"	42.926/42.951 1.6900/1.6910"	0.030/0.076 .0013/.0030"	1.636 .0644"	19.30 .760"

TORSION DE APRIETO: Bielas 4 mkg. Bancada 7.6 mkg
Tapa de cilindros 7.74 mkg



Fiat

1

767 c.c., 797 c.c., 600 D, E, R, S - 1962 / 1977

4	Bielas	4-3025 RA	STD-10-20-30-40-50	39.985/40.005	43.657/43.670	0.020/0.067	1.816	16.55
62 x 63.5 767 c.c.	Bancadas	4245 M	STD-10-20-30-40					
62 x 66 797 c.c.	1	3026 RA		50.780/50.795	54.507/54.520	0.030/0.070	1.841	20.62
	2	2847 RA		50.780/50.795	54.507/54.520	0.030/0.070	1.841	22.78
	3	3028 RA		50.780/50.795	54.507/54.520	0.030/0.070	1.841	20.62
	Axial	2847 BF	STD-5-10				2.360	
	Levas	1308 M	STD-5-10-15-20-30					
	1	3026 DR		37.975/38.000	A 47.990/48.000 B 48.000/48.010 C 48.010/48.020	0.025/0.062	4.975 4.980 4.985	27.50
	2	3027 DR		37.975/38.000	41.920/41.950	0.025/0.062	1.954	15.24
	3	3028 DR		30.975/31.000	35.921/35.951	0.026/0.063	2.453	15.24
	Buje de Biela 4100 Y							

TORSION DE APRIETO: Bielas 3.5 mkg. Bancada 6.2 mkg
Tapa de cilindros 2.8/ 3 mkg

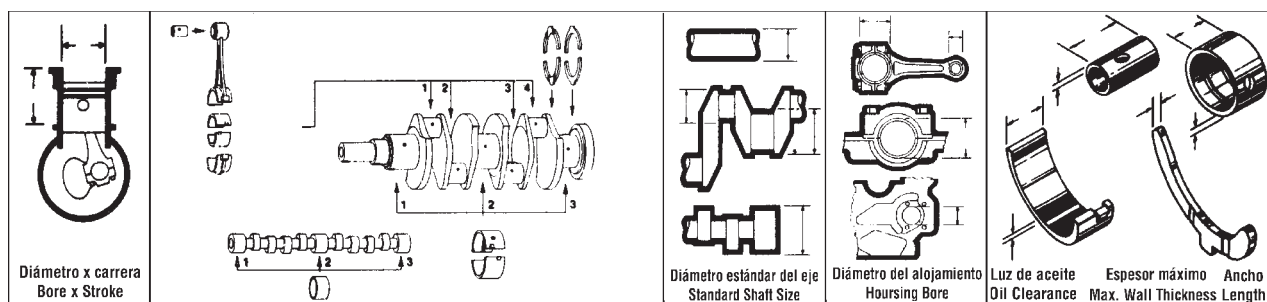
Fiat

2

843 c.c., 903 c.c., 600 S, 133 - 1977/1982

4	Bielas	4-3025 RA	STD-2-10-20-30-40-50	39.985/40.005	43.657/43.670	0.020/0.067	1.816	16.55
62 x 63.5 843 c.c.	Bancadas	6562 M	STD-10-20-30					
65 x 68 903 c.c.	1-3	62006 RA		50.785/50.805	54.507/54.520	0.028/0.073	1.841	20.62
	2	62007 RA		50.785/50.805	54.507/54.520	0.030/0.070	1.841	22.78
	Axial	2847 BF	STD-5-10				2.360	
	Levas	1702 M	STD-5-10-20-30					
	1	3027 DRI		37.975/38.000	B 50.500/50.510 C 50.510/50.520 D 50.700/50.710 E 50.710/50.720	0.025/0.075		27.50
	2	2849 DR		43.348/43.373	46.420/46.450	0.046/0.091		15.25
	3	3028 DR		30.975/31.000	35.921/35.951	0.026/0.071		15.25

TORSION DE APRIETO: Bielas 4 mkg. Bancada 7 mkg
Tapa de cilindros 5 a 6 mkg



3 1089 c.c., 1100 - 1953/1963

4 68 x 75 1089 c.c.	Bielas	4-2815 RA	STD-10-20-30-40-50	39.988/40.008	43.657/43.670	0.017/0.070	1.816	24.38
	Bancadas	5055 M	STD-10-20-30-40					
	1-2	2836 RA		41.988/42.008	45.698/45.710	0.020/0.072	1.834	27.99
	3	2837 RA		41.988/42.008	45.698/45.710	0.020/0.072	1.834	34.54
4 76 x 71.50 1300 c.c.	Levas	1249 M	STD-10-20					
	1	2836 DR		35.975/36.000	46.000/46.025	0.040/0.085	4.981	32.00
	2	2837 DR		38.975/39.000	A 44.920/44.930 B 44.930/44.940 C 44.940/44.950	0.040/0.085	2.952	15.00
	3	2838 DR		37.975/38.000	A 43.920/43.930 B 43.930/43.940 C 43.940/43.950	0.040/0.085	2.948	25.00
	TORSION DE APRIETO: Bielas 3.3 mkg. Bancada 6.2 mkg							
	Tapa de cilindros 8 mkg							



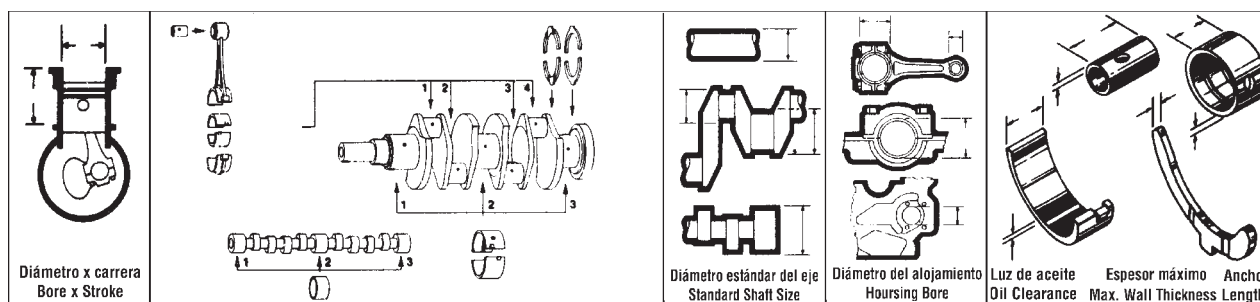
4 Spazio Diesel TRD, Duna SD, 147, Oggi, Uno, Mille, Europa, Panorama, Fiorino, Elba Prêmio, Pick Up, Rally E, Uno S/CS 4T DIESEL, 1.0, 1.05, 1.3, 1.5 - 1984

4 76 x 54.80 994 c.c.	Bielas	4-68090 RA	STD10-20-30-40	43.998/44.008 1.7322/1.7326"	47.130/47.142 1.8555/1.8560"	0.025/0.049 .0009/.0019"	1.543	18.50
	Bancadas	6818 M	STD-10-20-30-40				.0607"	.728"
4 76 x 57.80 1050 c.c.	1-2-4-5	68091 RA		48.199/48.209 1.8976/1.8980"	51.921/51.934 2.0441/2.0446"	0.029/0.053 .0011/.0021"	1.831	17.65
	3	68092 RA		48.199/48.209 1.8976/1.8980"	51.921/51.934 2.0441/2.0446"	0.029/0.053 .0011/.0021"	1.831	25.25
4 76 x 71.50 1300 c.c.	Axial	68092 BF	STD-5-10				2.360	.0929"
	Levas	1674 M	SEMI					
4 76 x 82.50 1500 c.c.		68093 DR		29.990/30.015 1.1807/1.1817	33.026/33.056 1.30202/1.3014"			22.00 .866"
	Buje de Biela 4-62093 Y							
	Buje							
	Eje Auxiliar	68091 DR1 68092 DR1						



5 1481 c.c., 1500 Berlina, Familiar, Multicarga Coupe - 1963/1969

4 77 x 79.5 1481 c.c.	Bielas	4-60000 RA	STD-10-20-30-40-50	52.983/53.004	56.718/56.730	0.020/0.071	1.847	21.20
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Fiat

5 (continuación)
1481 c.c., 1500 Berlina, Familiar, Multicarga Coupe - 1963/1969

4 77 x 79.5 1481 c.c.	Bancadas	4562 M	STD-10-20-30-40					
	1	60001 RA		62.962/62.982	66.675/66.687	0.020/0.071	1.834	26.53
	2-3	60003 RA		62.962/62.982	66.675/66.687	0.020/0.071	1.834	29.53
	Axial	2516 BF	STD-5-10				2.360	
	Levas	1311 M	STD-2-5-10-20-30					
	1	60001 DR		35.975/36.000	52.502/52.527	0.025/0.089	8.228	27.47
	2	60003 DR		47.650/47.675	50.927/50.952	0.025/0.065	1.619	15.00
	3	60004 DR		46.253/46.278	49.304/49.365	0.025/0.065	1.524	18.50
	Buje de Biela 4115 X							

TORSION DE APRIETO: Bielas 6.6 mkg. Bancada 10.5 mkg
Tapa de cilindros 9 mkg

Fiat

6
1625 c.c., 1500 Familiar Multicarga, 1600 Berlina Sport - 1969/1972

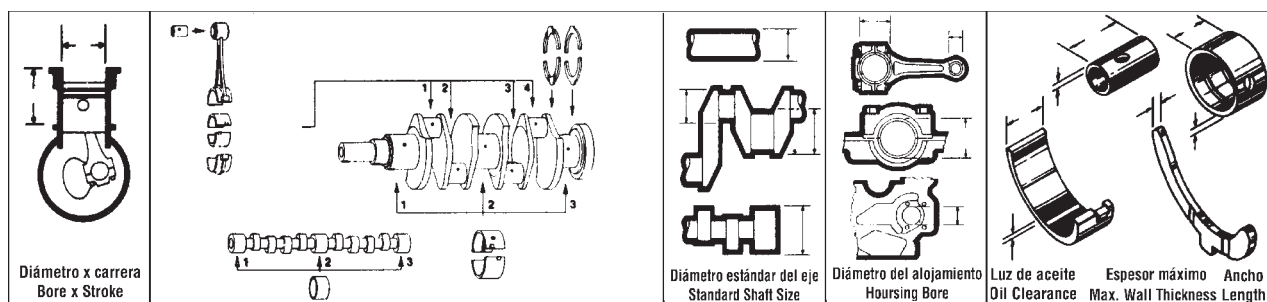
4 78 x 85 1625 c.c.	Bielas	4-66295 RA	STD-10-20-30-40	52.992/53.013	56.718/56.730	0.011/0.072	1.842	18.30
	Bancadas	6441 M	STD-10-20-30-40					
	1	66296 RA		62.497/62.967	66.675/66.687	0.040/0.090	1.834	26.16
	2	66297 RA		62.497/62.967	66.675/66.687	0.040/0.090	1.834	28.42
	3	60003 RA		62.497/62.967	66.675/66.687	0.040/0.090	1.834	29.53
	Axial	2516 BF	STD-5-10				2.360	
	Levas	1311 M	STD-5-10-15-20-30					
	1	60001 DR		37.975/38.000	52.502/52.527	0.025/0.089	8.228	27.47
	2	60003 DR		47.650/47.675	50.927/50.952	0.045/0.090	1.619	15.00
	3	60004 DR		46.253/46.278	49.340/49.365	0.046/0.091	1.524	18.50
	Buje de Biela 4115 X							

TORSION DE APRIETO: Bielas 11 mkg. Bancada 10.5 mkg
Tapa de cilindros 9 mkg

Fiat

7
1116 c.c., 1290 c.c., 1301c.c., 128L Berlina, Familiar, IAVA, 1300TV, Europa, Super Europa, 147L, Spazio TR/TR5, Brio, Duna S/SC - 1971/1990

4 80 x 55.5 1116 c.c.	Bielas	4-60985 RA	STD-10-20-30-40	45.498/45.518	48.630/48.646	0.036/0.086	1.538	19.20
	Bancadas	4659 M	STD-10-20-30-40					
	1-2-4-5	60988 RA		50.775/50.795	54.507/54.520	0.050/0.095	1.831	20.25
	3	60987 RA		50.775/50.795	54.507/54.520	0.050/0.095	1.831	20.75
86 x 55.5 1290 c.c.								
86.4 x 55.5 1301 c.c.								



Fiat

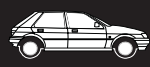


1116 c.c., 1290 c.c., 1301c.c., 128L Berlina, Familiar, IAVA, 1300TV, Europa, Super Europa, 147L, Spazio TR/TR5, Brio, Duna S/SC - 1971/1990

7 (continuación)

	Axial	60987 BF	STD-5-10				2.360
4	Levas						
80 x 55.5 1116 c.c.	1			29.944/29.960	30.009/30.034	0.049/0.090	
86 x 55.5 1290 c.c.	2			47.935/47.950	48.000/48.025	0.050/0.090	
	3			48.135/48.150	48.200/48.225	0.050/0.090	
86.4 x 55.5 1301 c.c.	4			48.335/48.350	48.400/48.425	0.050/0.090	
	5			48.535/48.550	48.600/48.625	0.050/0.090	
Buje de Biela 4325 Y ó 4325 X (sólo para 1290 c.c.)							
TORSION DE APRIETO: Bielas 5 mkg. Bancada 8.5 mkg Tapa de cilindros 9.5 mkg							

Fiat



1608 c.c., 125 Berlina, Familiar, Potenciado, Sport, Multicarga, SL, Mirafiori - 1972/1981

8

4 80 x 80 1608 c.c.	Bielas	4-61645 RA	STD-10-20-30-40	48.224/48.234 48.234/48.244	51.330/51.346	0.026/0.060	1.534 1.530	19.20
	Bancadas	4659 M	STD-10-20-30-40					
	1-2-4-5	60698 RA		50.775/50.795	54.507/54.520	0.050/0.095	1.831	20.25
	3	60987 RA		50.775/50.795	54.507/54.520	0.050/0.095	1.831	20.75
	Axial	60987 BF	STD-5-10				2.360	
	Levas (*)							
	1			29.944/29.960	30.009/30.034	0.049/0.090		
	2			45.735/45.751	45.800/45.825	0.049/0.090		
	3			46.135/46.151	46.200/46.225	0.049/0.090		
	Árbol Auxiliar							
	1			48.031/48.038	51.120/51.150	0.046/0.091		
	2			38.929/38.954	42.030/42.060	0.046/0.091		
TORSION DE APRIETO: Bielas 5.2 mkg. Bancada 8.2 mkg Tapa de cilindros 7.7 mkg (*) Motor con doble árbol de levas. No usa cojinetes.								

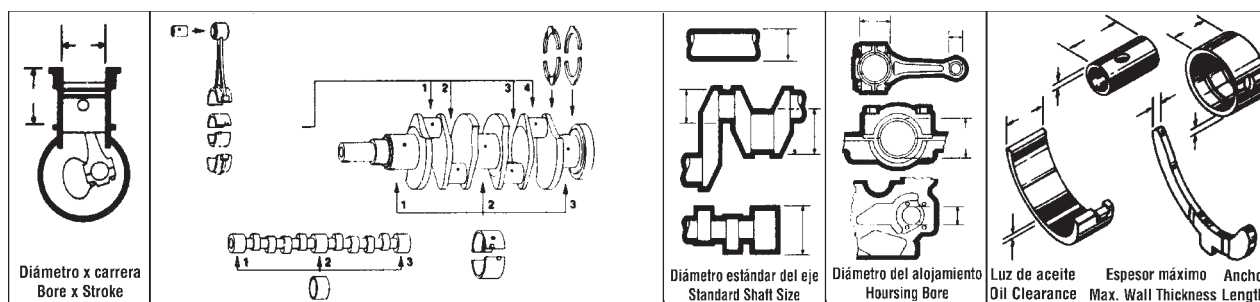
Fiat



Tipo 1.4, 1.6, Uno, Duna, 147 Vivace, Fiorino, Tipo 1.6 i - 1990 /

9

4 80.5 x 67.4 1372 c.c.	Bielas	4-1645 RA	STD-10-20-30-40	45.508/45.518	48.629/48.646		1.541	19.20
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9 (continuación)

Tipo 1.4, 1.6, Uno, Duna,
147 Vivace, Fiorino, Tipo 1.6 i - 1990 /

4 80.5 x 67.4 1372 c.c.	Bancadas	7084 M	STD-10-20-30-40				
	1-2-4-5	1645 RA		50.780/50.800	54.507/54.520	1.844	20.25
	3	1647 RA		50.780/50.800	54.507/54.520	1.844	20.25
	Axial	60987 BF	STD-5-10			2.360	
86.4 x 67.4 1580 c.c.	Buje de Biela	4-62039 X	SEMI	21.539/21.641	23.939/23.972		25.40
	1-2-3-4						



10

149 A 2 038, Regatta 100 S - 1985/1987

4 84 x 71.5 1585 c.c.	Bielas	4-66315 RA	STD-10-20-30-40-50-60	48.224/48.244	51.330/51.346	1.532	19.50
	Bancadas	6445 M	STD-10-20-30-40-50-60				
	1-2-4-5	66316 RA		52.985/53.005	56.717/56.730		20.75
	3	66317 RA		52.985/53.005	56.717/56.730		24.75
	Axial	62037 AF	STD-5			2.360	
	Levas	1677 M	STD-0.25-0.50				
	1	62036 DR		48.084/48.104	51.120/51.150		14.15
	2	62038 DR		43.904/43.924	46.920/46.950		16.00
	3	60968 DR		36.926/36.946	39.962/39.992		15.00

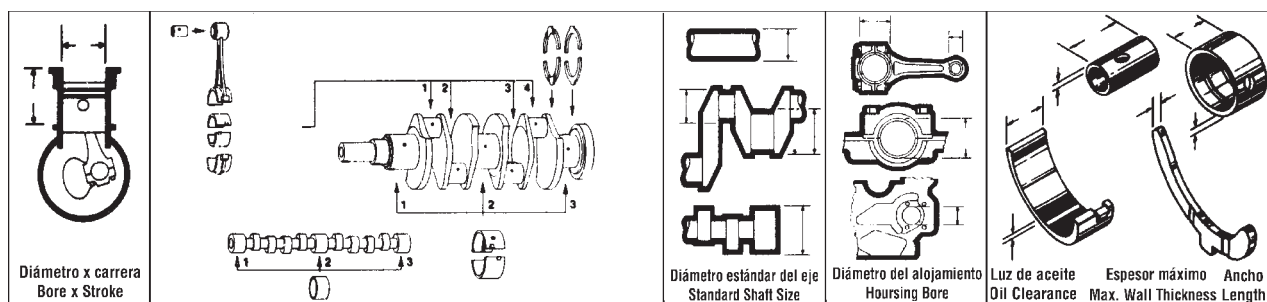


11

Tempra, 2.0 16V, Duna, Uno, Palio, Siena Diesel 1.7 Aspirado

4 84 x 90 1995 c.c.	Bielas	4-66240 RA	STD-10-20-30-40	50.782/50.802	53.987/53.913	1.529	19.50
	Bancadas	6445 M	STD-10-20-30-40-50-60				
	1-2-4-5	66316 RA		52.985/53.005	56.717/56.730	0.029/0.087	1.841 20.75
	3	66317 RA		52.985/53.005	56.717/56.730	0.029/0.087	1.841 24.25
4 82.6 x 79.2 1697 c.c.	Axial	62037 AF				2.360	
	Levas	1677 M					
	1	62038 DR		48.084/48.104	51.120/51.150		14.15
	2	62038 DR		43.904/43.924	46.920/46.950		16.00
	3	60988 DR		36.926/36.946	39.962/39.992		15.00
	Buje de Biela	4100 Y					

TORSION DE APRIETO: Bielas 3.5 mkg. Bancada 6.2 mkg
Tapa de cilindros 2.8/ 3 mkg



Fiat

12 **Palio, Siena, EL/HL, Turbo Diesel 1.7**

4 82.6 x 79.2 1697 c.c.	4-1860 CP	STD-.010-.020	50.787/50.807 1.9995/2.0003"	53.897/53.913 2.1219/2.1226"	0.020/0.078		
1-2-3-5	7191 M	STD-.010-.020-.030	52.990/53.010 2.0862/2.0870"	56.717/56.730 2.2330/2.2335"	0.023/0.080		

Fiat

1500 c.c, Super Europa, 1.5, Regatta 85 S, SC, Weekend, Duna SCV, Prêmio SL/CSL, Elba S/CSL, Uno 1 GR-1.5R Pick – Up LX, Prêmio, Elba, Uno CSL - 1985 – 1991

4 86.4 x 63.9 1498 c.c.	Bielas	4-66580 RA	STD-10-20-30-40	45.508/45.528	48.629/48.649	0.030/0.078	1.541	19.97
	Bancadas	6565 M	STD-10-20-30-40					
	1-2-4-5	66581 RA		50.775/50.795	54.507/54.520	0.032/0.077	1.840	20.25
	3	66582 RA		50.775/50.795	54.507/54.520	0.032/0.077	1.840	20.75
	Axial	60987 BF	STD-5-10				2.360	

Fiat

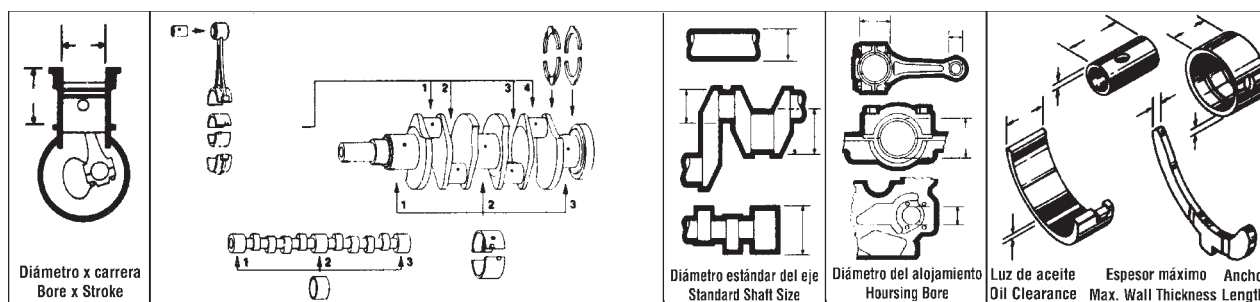
14 **1929 c.c., Ducato Turbo Diesel/Aspirado 1.9 D/TD**

4 82.6 x 90 1929 c.c.	4-1860 CP	1860 CP	STD-.010-.020	50.787/50.807 1.9995/2.0003"	53.897/53.919 2.1219/2.1228"	0.020/0.085	1.536	19.50
	7191 M	4128 CP	STD-.010-.020-.030	52.990/53.010 2.0862/2.0870"	56.717/56.730 2.2329/2.2335"	0.023/0.081	1.842	20.75
		4129 CP	STD-.010-.020-.030	52.990/53.010 2.0862/2.0870"	56.717/56.730 2.2329/2.2335"	0.023/0.081	1.842	25.25
	62037 BF		STD-.005	59.000/59.250 2.3228/2.3327"	73.600/73.850 2.8976/2.9075"		2.362	.0930"

Fiat

15 **2.5, 2.8 Ducato, Diesel**

4 93 x 90 2445 c.c.	4-1870 CP	STD-.010-.020	56.520/56.535 2.2252/2.2258"	60.333/60.345 2.3753/2.3758"	0.020/0.072		
93 x 92 2499 c.c.	7252 M	STD-.010	76.187/76.200 1.9995/2.0003"	80.586/80.607 3.1727/3.1735"	0.038/0.097		
94.40 x 100.00 2.800 c.c.	AEM 92980	STD-0.10-0.20	86.182/86.208 3.3930/3.3940"	90.588/90.614 3.5665/3.5675"		2.164	30.950



16 1135 c.c., U-25, 211 R - 1957/1966

2 85 x 100 1135 c.c.	Bielas	4-60080 RA	STD-10-20-30-40	58.723/58.743	62.408/64.442	0.020/0.065	1.822	24.71
	Bancadas	4519 M	STD-10-20-30-40					
	1-2 (Sup)	60084 CP		62.989/63.009	66.675/66.685	0.025/0.065	1.822	31.88
	1-2 (Inf)	60082 CPA		62.989/63.009	66.675/66.685	0.025/0.065	1.822	31.88
	Levas							
	1			49.975/50.000	56.000/56.030	0.025/0.090		
	2			34.975/35.000	54.186/54.126	0.025/0.090		
TORSION DE APRIETO: Bielas 6 mkg. Bancada 16 mkg (U-25); 14 mkg (211-R). Tapa de cilindros 2.8/ 3 mkg								



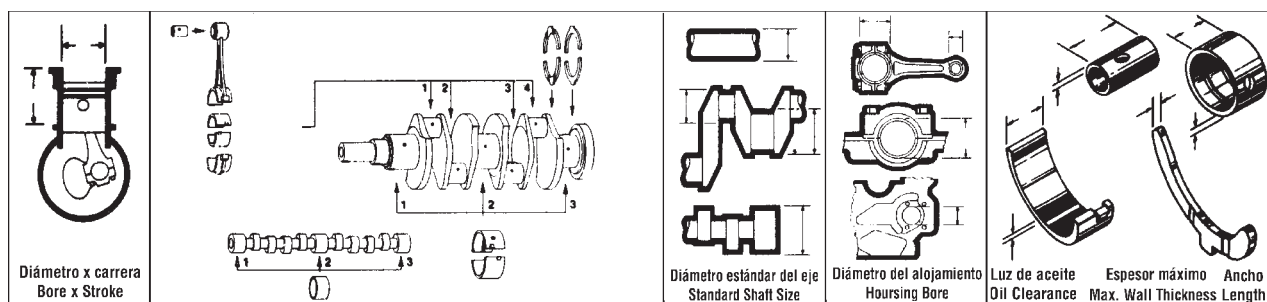
17 2270c.c., 411-R, 431-R, 481-R - 1959/1965

4 85 x 100 2270 c.c.	Bielas	4-60080 RA	STD-10-20-30	58.723/58.743	62.408/62.420	0.021/0.065	1.822	24.71
	Bancadas	4520 M	STD-10-20-30					
	1-3-5	6008 CPA		62.989/63.0095	66.675/66.687	0.022/0.0660	1.822	31.88
	2-4	60083 CP		62.989/63.0095	66.675/66.687	0.022/0.0660	1.822	24.62
	Levas							
	1			49.975/50.000	56.000/56.030	0.025/0.090		
	2			(*) 48.005/48.030		0.040/0.120		
	3			(*) 33.005/33.030		0.040/0.120		
TORSION DE APRIETO: Bielas 6mkg. Bancada 16 mkg Tapa de cilindros 9/ 10 mkg (*) A partir del motor N° 006558, hasta el motor N° 0065557 estos valores disminuyen en 0,030 mm.								



18 4397 c.c., 4156 c.c., 3456 c.c., Someca M45, M50, Supersom 55, 450, 650, 700S U - 1962/1971

4 100 x 100 3456 c.c.	Bielas	4-60115 CP	STD-10-20-30-40	68.832/69.855	73.713/73.729	0.074/0.114	1.892	36.51
	Bancadas	4521 M	STD-10-20-30-40					
	1-3	60116 RA		76.153/76.176	80.626/80.646	0.095/0.145	2.178	42.50
	2-4	60117 RA		76.153/76.176	80.626/80.646	0.095/0.145	2.178	32.50
	5	60118 RA		76.153/76.176	80.626/80.646	0.095/0.145	2.178	53.97
105 x 120 4156 c.c.	Axial	60116 BF	STD-5-10-20				2.360	
108 x 120 4397 c.c.	Levas							
	1-2-3-4-5			45.950/45.975	46.050/46.089	0.075/0.140		
Buje de Biela 4116 Y								
TORSION DE APRIETO: Bielas 10 mkg. Bancada 14 mkg Tapa de cilindros 22 mkg								



Fiat

19 CN3/CN4, 400 E, V, U 500, Super, 600 E - 1969/1980

3 110 x 120 3421 c.c.	Bielas	4-60115 CP	STD-10-20-30-40	69.860/69.878	73.713/73.729	0.074/0.114	1.892	36.51
	Bancadas	6411 M	STD-10-20-30-40					
	1-4	60116 RA		76.202/76.200	80.626/80.646	0.050/0.100	2.178	42.50
110 x 130 3705 c.c.	2-3	60117 RA		76.202/76.200	80.626/80.646	0.050/0.100	2.178	32.50
	Axial	60116 BF	STD-5-10-20				2.360	

MOTORES CON COJ. DE LEVAS

Levas	1596 M	STD-10-20-30						
1	62061 DR		45.950/45.975	49.000/49.025	0.080/0.140	1.473	36.00	
2-3	62082 DR		45.950/45.975	49.000/49.025	0.080/0.140	1.473	24.50	
4	62063 DR		45.950/45.975	49.000/49.025	0.080/0.140	1.473	28.00	

Buje de Biela 4285 Y

TORSION DE APRIETO: Bielas 12.5/ 13.5 mkg. Bancada 14 mkg
Tapa de cilindros 23/ 24 mkg

Fiat

20 7412 c.c., Tractor 900E, 1100E (motor CP3)

6 110 x 130 7412 c.c.	Bielas	4-60115 CP	STD-10-20-30-40	69.860/69.878	73.713/73.729	0.074/0.114	1.892	36.51
	Bancadas	6409 M	STD-10-20-30-40					
	1-4-7	60116 RA		76.202/76.220	80.626/80.646	0.050/0.100	2.178	42.50
	2-3-5-6	60117 RA		76.202/76.220	80.626/80.646	0.050/0.100	2.178	32.50
	Axial	60116 BF	STD-5-10				2.360	
	Levas	1594 M	STD-10-20-30					
	1	62061 DR		45.950/45.975	49.000/49.025	0.080/0.140	1.473	36.00
	2-3-4-5-6	62062 DR		45.950/45.975	49.000/49.025	0.080/0.140	1.473	24.50
	7	62063 DR		45.950/45.975	49.000/49.025	0.080/0.140	1.473	28.00

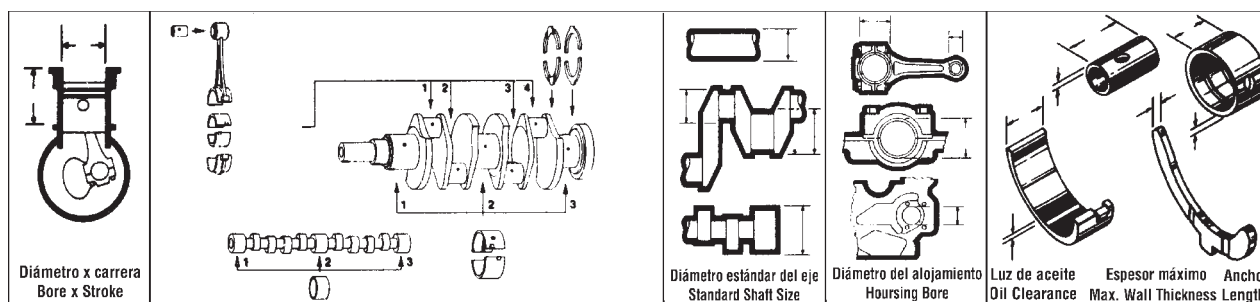
Buje de Biela 4285 Y

TORSION DE APRIETO: Bielas 12.5/ 13.5 mkg. Bancada 13.5/ 14.5 mkg
Tapa de cilindros 23/ 24 mkg

Fiat

21 5401 c.c. (C03), 700E, 800, E - 1971/1980

4 115 x 130 5401 c.c.	Bielas	4-62065 CP	STD-10-20-30	72.482/72.500	76.689/76.713	0.058/0.103	2.070	35.43
	Bancadas	6410 M	STD-10-20-30-40					
	1-3-5	62086 CP		79.782/79.800	84.206/84.226	0.050/0.100	2.178	42.49
	2-4	62087 CP		79.782/79.800	84.206/84.226	0.050/0.100	2.178	32.49
	Axial	62067 BF	STD-5-10				2.362	
	Levas							
	1-2-3-4-5			45.950/48.975	49.055/49.090	0.080/0.140		



21 (continuación)

5401 c.c. (C03), 700E, 800, E - 1971/1980

4 115 x 130 5401 c.c.	MOTORES CON COJ. DE LEVAS							
	Levas	1595 M	STD-10-20-30					
	1	62066 DR		48.950/48.975	52.000/52.025	0.080/0.140		36.00
	2-3	62067 DR		45.950/45.975	49.000/49.025	0.080/0.140		24.50
	4	62068 – DR		48.950/48.975	52.000/52.025	0.080/0.140		28.00
	Equilibrador armónico (*)	7010 M	STD-10-20-30					
	3 Coj.	5276 CP		34.920/34.935	38.626/38.646	0.047/0.094	1.822	30.00
	1 Coj.	5277 CP		34.920/34.935	38.626/38.646	0.047/0.094	1.822	40.00
	Buje de Biela	4286 Y						
TORSION DE APRIETO: Bielas 15/ 16 mkg. Bancada 20/ 21 mkg Tapa de cilindros (espárragos M16x 1.5) 25 mkg (espárragos M18x 1.5) 33 mkg (*) Usa 2 juegos N° 7010 M por motor.								



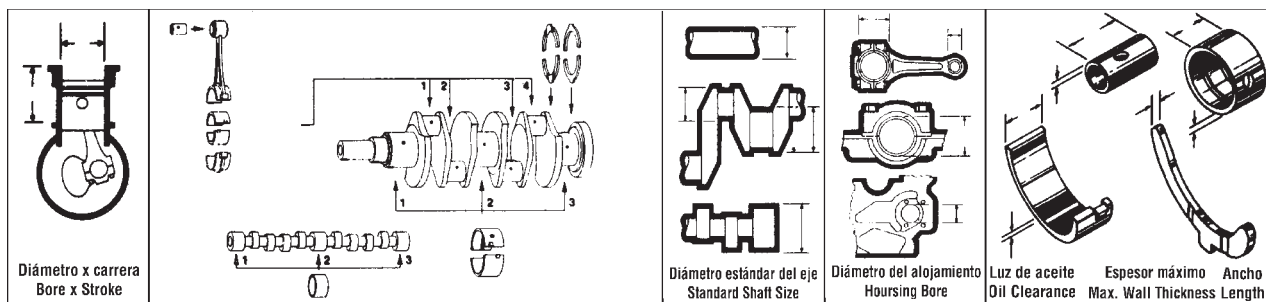
22

55R, 60R, 780, 780R - 1955/1971

4 122 x 140 6546 c.c.	Bielas	4-60085 RA	STD-10-20-30	83.134/83.159	87.299/87.323	0.064/0.126	2.038	44.25
	Bancadas	4522 M	STD-10-20-30-40					
125 x 140 6872 c.c.	1-3	60086 RA		92.087/92.112	96.570/96595	0.078/0.140	2.190	42.40
	2-4	60087 RA		92.087/92.112	96.570/96595	0.078/0.140	2.190	32.40
	5	60088 RA		92.087/92.112	96.570/96595	0.078/0.140	2.190	64.40
	Axial	60087 BF	STD-5-10				2.360	
	Levas							
	1			54.970/55.000	62.000/62.030	0.070/0.140		
	2			51.970/52.000	58.000/58.030	0.070/0.135		
	3			39.975/40.000	46.000/46.025	0.070/0.135		
Buje de Biela 4124 Y				TORSION DE APRIETO: Bielas 19 mkg (55/60R) Tornillo Auto Bloc.21.5 mkg.. Tornillo NO Auto Bloc. 19/ 221 mkg (780R-780) Bancada 25 mkg, (Ø18) 12.5 mkg (Ø 14) (NO Auto Bloc.) (55/60R). 30 mkg (Ø 18); 14 mkg, (Ø 14) (780R-780). Tapa de cilindros 30 mkg. (55/60R). 780R-780. Ver Nota.				

Nota:

Torsiones de aprieto tapa cil. 780-780R: Tuerca de 30 mm de altura, con arandela, para espárrago R100 y junta tejida, 34/36 mkg.
Tuerca de 30 mm de altura, sin arandela, para espárrago R80 y junta de amianto y cobre, 30/ 32.5 mkg.
Tuerca de 26 mm de altura, sin arandela, para espárrago R80 y junta de amianto y cobre, 28 mkg.



Fiat

23

221 A, 619N1, T1; 697N, T, B; Omnib. 319N - 1972/

6 137 x 156 13798 c.c.	Bielas	6-62020 CP	STD-10-20-30	84.713/84.735	88.482/88.504	0.077/0.133	1.835	41.25
	Bancadas	6375 M	STD-10-20-30					
	1-7	62021 CP		102.879/102.901	107.976/108.001	0.0850/0.144	2.495	37.41
	4	62022 CP		102.879/102.901	107.976/108.001	0.0850/0.144	2.495	54.38
	2-3-5-6	62023 CP		102.879/102.901	107.976/108.001	0.0850/0.144	2.495	34.82
	Levas	1563 M	STD-40					
	1	62021 DR		62.500/62.530	68.515/68.550	0.069/0.160	2.956	42.24
	2	62022 DR		62.000/62.030	68.015/68.050	0.069/0.160	2.956	32.26
	3	62023 DR		61.500/61.530	67.515/67.550	0.070/0.160	2.956	32.35
	4	62024 DR		61.000/61.030	67.015/67.050	0.070/0.160	2.956	32.26
Axial		62022 BFA	STD-5-10				2.362	
Buje de Biela 4283 Y								

TORSION DE APRIETO: Bielas 33 mkg. Bancada 37 mkg
(Lubricar con aceite)
Tapa de cilindros 30 mkg. (Lubricar con aceite)

Fiat - Iveco

24

150 Turbo 6 cil.

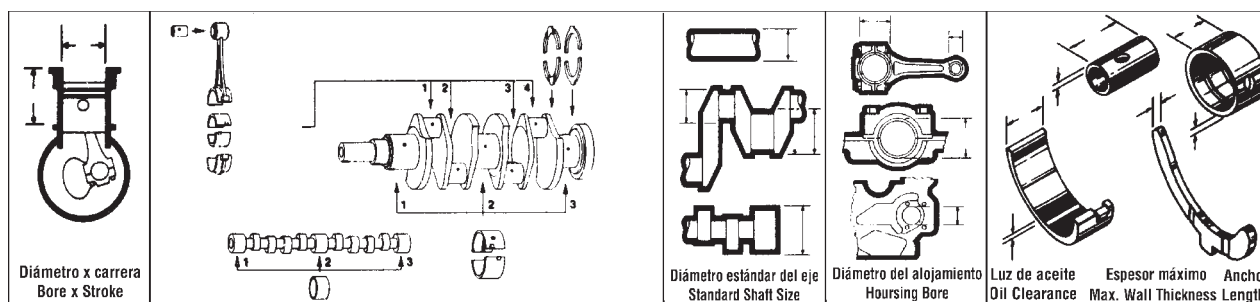
4 140 x 115 3908 c.c.	AEB91975	STD-.010-.020-.030-.040	63.725/63.744 2.5089/2.5096"	67.407/67.422 2.6538/2.6544"	0.033/0.085 .0013/.0033"	1.806 .07115"	28.70 1.1299"
	AEM91976	STD-.010-.020-.030-.040	79.791/79.810 3.1414/3.1421"	84.200/84.230 3.3150/3.3161"	0.032/0.105 .0013/.0041"	2.167 .0853"	25.00 .9843"
6 104 x 115 5861 c.c.	AEW95075	STD-.005-.010-.020	87.779/88.033 3.4559/3.4659"	111.510 nom 4.3902" nom		3.429 max .1350" max	
SE VENDEN POR PARES							

Ford

1

Escort - 1988/
Escort ,Corcel II, Belina II, Del Rey, Scala, Pampa 4 x 4 , Escort L/GL, Verona LX

4 77 x 83.5 1555 c.c.	Bielas	4-2950 RAA	STD-0.25-0.50-0.75-1.00	43.980/43.964 1.7315/1.7309"	47.614/47.625 1.8746/1.8750"	0.020/0.077 .0008/.0030"	1.801 .0713"	17.75 .699"
	Bancadas	6828 M	STD-0.25-0.50-0.75-1.00					
	1-3	68111 RA		54.785/54.805 2.1569/2.1577"	59.000/59.016 2.3228/2.3235"	0.010/0.077 .004/.0030"	2.092 .0824"	24.50 .965"
	2-4-5	68112 RA		54.785/54.805 2.1569/2.1577"	59.000/59.016 2.3228/2.3235"	0.010/0.077 .0004/.0030"	2.092 .0824"	19.50 .768"
	Axial	68111 BF	STD-5-10					



Ford

2 Ka, Fiesta (Endura 1.3)

4 74 x 75.5 1299 c.c.	Bielas	4-4165 RA	STD-25-50-75	40.990/41.010	43.990/44.010	0.080/0.145	1.450	17.950
	Bancadas	7200 M	STD-25-50-75	56.990/57.000	60.623/60.636	0.020/0.057	1.807	18.690
		66526 AF	STD-0.20	63.600	78.100		2.850	

Ford

3 Escort, Mondeo, Focus, Zetec 1.6, 1.8, 2.0, 16V

4 76 x 88 1597 c.c.	Bielas	4-4135 RA	STD-0.25	46.890/46.910 1.8461/1.8469"	49.890/49.910 1.9642/1.9650"	0.030/0.094	1.475	20.100 .0581" .7913"
	Bancadas	7193 M	STD-0.25					
	80.6 x 88 1796 c.c.	1-2-4-5 4136 RA		57.980/58.000 1.2827/2.2835"	62.287/62.300 2.4522/2.4528"	0.024/0.083	2.131	19.210 .0839" .7563"
84.8 x 88 1988 c.c.	3	4137 RA		57.980/58.000 1.2827/2.2835"	62.287/62.300 2.4522/2.4528"	0.024/0.083	2.131	24.680 .0839" .9717"

Ford

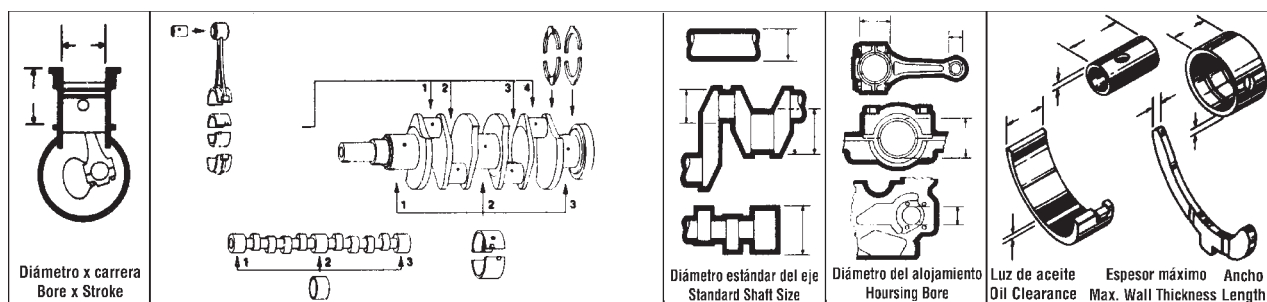
4 Sierra 1.6, GL - 1984 /

4 81.3 x 76.95 1597 c.c.	Bielas	4-3460 RA	STD-0.25-0.50-0.75-1.00	51.980/52.000	55.000/55.020	0.006/0.064	1.497	20.00
	Bancadas	6667 M	STD-0.25-0.50-0.75	56.970/56.990	60.620/60.640	0.010/0.064	1.835	23.60
	1-2-3-4-5	3461 RAB						
TODAS LAS POSICIONES RANURADAS								
MOTOR 2260								
	Bancadas	7013M	STD-0.25-0.50-0.75	56.970/56.990	60.620/60.640	0.010/0.064	1.835	23.60
	1-2-3-4-5	3461 RAC						
UÑA CON DIFERENTE UBICACION Y DIMENSIONES COJINETES INFERIORES LISOS								
	Axial	3461 BF	STD-5-10				2.35	

Ford

5 Fiesta, Escort, Courier, Focus Diesel 1.8

4 82.5 x 82 1753 c.c.	Bielas	4-4160 RA	STD-0.25-0.50	48.970/48.990 1.9280/1.9287"	52.000/52.020 2.0472/2.0480"	0.020/0.085	1.495	21.590 .0589" .8500"
	Bancadas	7367 M	STD-0.25-0.50	(Reemplaza a 7198 M) C/ TRABA				
	1	4161 RA		53.970/53.990 2.1248/2.1256"	57.683/57.696 2.2710/2.2715"	0.032/0.090	1.830	22.250 .0720" .8760"
	2-5	4162 RA		53.970/53.990 2.1248/2.1256"	57.683/57.696 2.2710/2.2715"	0.032/0.090	1.830	24.00 .0720" .9449"
	Axial	2786 BF	STD	59.346/59.690 2.3365/2.3500"	74.498/74.752 2.9330/2.9430"		2.362	.0930"



Ford



Versallies, Royale, Del Rey, Verona GLX, Escort Ghia, XR3, Belina, Pampa 4 x 2, 1.8, 2.0, 2.0i

6	4 81.00 x 86.40 1781 82.50 x 92.80 1984 c.c.	Bielas	4-68150 CP	STD-0.25-0.50-0.75-1.00 STD/0.75 mm	47.758/47.778	50.600/50.619	0.010/0.08	1.406	19.00
		Bancada	6838 M	SIN PESTAÑA STD-0.25-0.50-0.75-1.00 STD/0.75 mm					
		1-2-3-5	68096 RA		53.968/53.978	59.000/59.019	0.010/0.08	2.503	18.50
		4	68096 RAA		53.968/53.978	59.000/59.019	0.010/0.08	2.503	18.50
		Bancada	6819 M	PESTAÑA + 0.50 STD-0.25-0.50-0.75-1.00					
		1-2-5	68096 RA		53.940/53.960	59.000/59.019	0.025/0.060	2.503	18.50
		3 ▲	68097 CP		53.940/53.960	59.000/59.019	0.025/0.060	2.503	25.40
		4	68096 RAA						
		Bancada	6834 M	PESTAÑA STD STD-0.25-0.50-0.75-1.00					
		1-2-5	68096 RA		53.958/53.978	59.000/59.019	0.025	2.503	18.50
7	89.32 x 79.40 1990 c.c. 96.00 x 79.40 2299 c.c.	3 ▲	68097 CP		53.958/53.978	59.000/59.019	0.025	2.503	24.93
		4	68096 RAA						
		Axial	68097 BF	STD-5-10				1.975	
		Levas	N118/5 13-3317	STD-0.25-0.50	25.939/25.960 1.0212/1.0220"	28.250/28.271 1.1122/1.1130"	0.014/0.076 .0005/.0029"	1.138 .0448"	17.00 .6692"
		Buje de Biela	4-68099 YA	SEMI	24.000 .945"	27.000/27.021 1.062/1.0634"	0.016/0.029 .0006/.0011"	1.780 .0700"	25.00 .984"
		Buje de Biela	1726 M 68096 DR	STD	42.959/42.975 1.6913/1.6919"	46.000/46.025 1.8110/1.8120"	0.025/0.066 .0009/.0026"	1.770 .0697"	15.00 .590"
			68097 DR		41.959/41.975 1.6519/1.6526"	45.000/45.025 1.7717/1.7726"	0.025/0.066 .0009/.0026"	1.770 .0697"	15.00 .590"

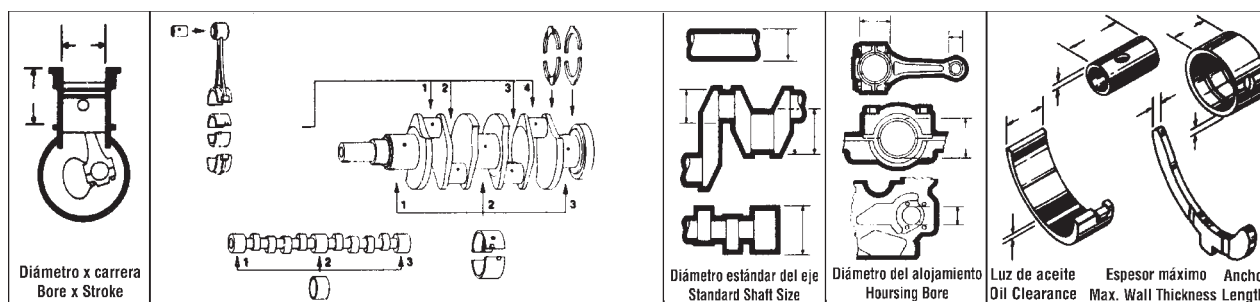
Ford



Taunus L, GXL, Ghia; Sierra 2.3, XR4, 2000, 2300

7	89.32 x 79.40 1990 c.c. 96.00 x 79.40 2299 c.c.	Bielas (*)	4-3545 RAA	STD-10-20-30-40	51.970/52.000	55.170/55.190	0.020/0.066	1.585	20.32
		Bancadas	4979 M	STD-10-20-30-40					
		1-2-4-5	3546 RA		60.904/60.935	65.791/65.822	0.020/0.066	2.428	24.12
		3 ▲	3547 RA		60.904/60.935	65.791/65.822	0.020/0.066	2.428	30.35
		Levas	1443 M	STD					
		1-2-3-4	3547 DR		44.990/45.010	48.275/48.301	0.025/0.084	1.621	17.85
		Arbol Auxiliar	1460 M	ST/STD					
		1	3548 DR		41.960/41.986	45.644/45.669	0.030/0.096	1.821	12.80
		2	3549 DR		41.960/41.986	45.644/45.669	0.030/0.096	1.571	12.80

TORSION DE APRIETO: Bielas 4.1/ 4.9 mkg.
Bancada: 11 / 12.5 mkg. Tapa de cilindros 11/ 12.5 mkg
(*) Biela con orificio.
Sustituye a N° 4-3545 CP



Ford

8 Falcon 144, 170, 187 - 1960 / 1969

6 88.90 x 63.50 2364 c.c. (144") 88.90 x 74.676 2780 c.c. (170")	Bielas	6-2380 RA	STD-2-10-20-30-40	53.929/53.949 2.1232/2.1240"	56.871/56.891 2.2390/2.2398"	0.025/0.066 .0010/.0026"	1.458 .0574"	20.57 .810
	Bancadas	4222 M	STD-10-20-30-40					
	1-2-4	2382 RA		57.104/57.125 2.2482/2.2490"	60.990/61.011 2.4012/2.4020"	0.020/0.066 .0008/.0026"	1.930 .0760"	25.91 1.020"
	3 ▲	2384 RA		57.104/57.125 2.2482/2.2490"	60.990/61.011 2.4012/2.4020"	0.020/0.066 .0008/.0026"	1.930 .0760"	32.28 1.271"
		Levas	1186 M	STD-5-10-20-30-				
		1	2381 DR	45.961/45.987 1.8095/1.8105'	50.241/50.267 1.9780/1.9790"	0.025/0.076 .0010/.0030"	2.113 .0832"	18.54 .730"
		2	2382 DR	45.961/45.987 1.8095/1.8105'	49.860/49.886 1.9630/1.9640"	0.025/0.076 .0010/.0030"	2.113 .0832"	18.54 .730"
		3	2383 DR	45.961/45.987 1.8095/1.8105'	49.479/49.505 1.9480/1.9490"	0.025/0.076 .0010/.0030"	2.113 .0832"	18.54 .730"
		4	2384 DR	45.961/45.987 1.8095/1.8105'	49.098/49.124 1.9330/1.9340"	0.025/0.076 .0010/.0030"	2.113 .0832"	18.54 .730"

TORSION DE APRIETO: Bielas 2.6/ 3.3 mkg,
Bancada: 8.3/ 9.68 mkg, Tapa de cilindros, apretar en tres etapas
1) 6.9/ 7.6 mkg; 2) 8.3/ 8.9 mkg; 3) 9.7/ 10.4 mkg.

Ford

9 Transit 2.5 Diesel

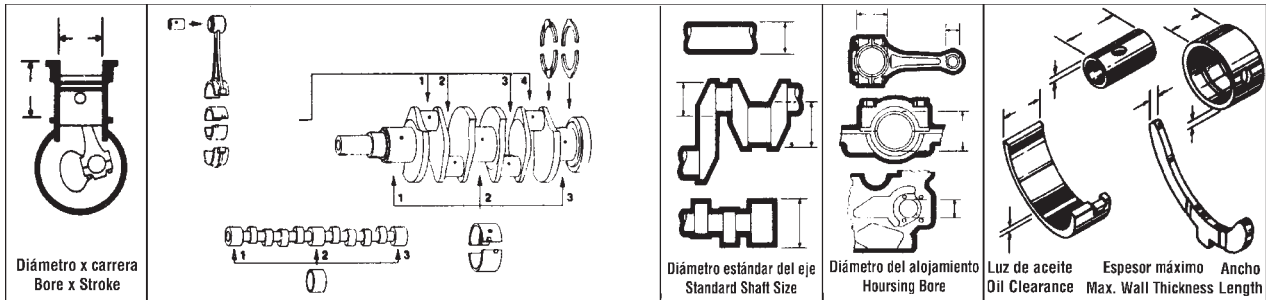
4 93.67 x 90.54 2496 c.c.	Bielas	4-66490 RA	STD-0.10-0.20-0.30-0.40	59.980/60.000	63.506/63.520	0.030/0.069	1.736	25.410
	Bancadas	7085 M	STD-0.10-0.20					
	1-2-4			69.990/70.000	74.000/74.010	0.056/0.101	1.972	23.139
		4	1686 AF	STD	78.750/79.000	94.000/94.250	2.850	

Ford

10 Falcon, F100 "Econo 6", Ranchero, F150, Fairlane - 188-221

6 93.47 x 74.68 3075 c.c. (188") 93.47 x 87.88 3618 c.c. (221") 90.42 x 79.248 3052 c.c. (187")	Bielas	6-2380 RA	STD-2-10-20-30-40	53.929/53.949 2.1232/2.1240"	56.871/56.891 2.2390/2.2398"	0.025/0.066 .0010/.0026"	1.458 .0574"	20.57 .810
	Bancadas	4953 M	STD-10-20-30-40					
	1-2-3-4-6-7	2382 RA		57.104/57.125 2.2482/2.2490"	60.990/61.011 2.4012/2.4020"	0.020/0.066 .0008/.0026"	1.930 .0760"	25.91 1.020"
	5 ▲	2384 RA		57.104/57.125 2.2482/2.2490"	60.990/61.011 2.4012/2.4020"	0.020/0.066 .0008/.0026"	1.930 .0760"	32.28 1.271"
		Levas	1352 M	STD-5-10-15-20-30				
		1-2-3-4	2384 RA	45.961/45.987 1.8095/1.8105'	50.241/50.267 1.9780/1.9790"	0.025/0.076 .0010/.0030"	2.113 .0832"	18.55 .730"

TORSION DE APRIETO: Bielas 2.6/ 3.3 mkg,
Bancadas 8.3/ 9.68 mkg, Tapa de cilindros apretar en tres etapas
1) 6.9/ 7.6 mkg; 2) 8.3/ 8.9 mkg; 3) 9.7/ 10.4 mkg.


Ford

11
Explorer 4.0i V6 OHV, Explorer 4.0i V6 SOHC

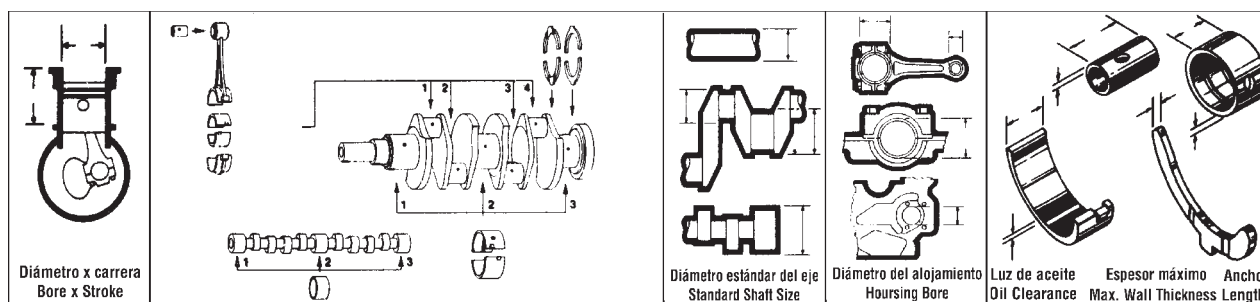
6 100 x 84 3958 c.c.	Bielas	71-2185/6	STD.010-.020-.030-.040	53.980/54.000 2.1252/2.1260"	56.820/56.840 2.2370/2.2378"	0.020/0.071	1.407 .0554"	16.48 .6488"
	Bancadas	H 816/4 02-2186	STD.010-.020-.030	56.980/57.000 2.2433/2.2441"	60.620/60.640 2.3866/2.3874"	0.020/0.071	1.806 .0711"	21.60 .8504"
6 100 x 84 4011 c.c.		02-2187 H		56.980/57.000 2.2433/2.2441"	60.620/60.640 2.3866/2.3874"	0.020/0.071	1.806 .0711"	26.26 1.0339"
	Axial	02-2187	Anchura entre el gualdera del cigüeñal al lugar de ubicación de los cojinetes de empuje STD 26.390/1.0390"- 26.440/1.0409" .010 26.390/1.0390"- 26.440/1.0409" .020 26.390/1.0390"- 26.440/1.0409" .030 26.390/1.0390"- 26.440/1.0409" .040 26.390/1.0390"- 26.440/1.0409"					

Ford

12
Fairlane, F-100, AF700 (Motor V8-292)

8 88.90 x 78.74 3908 c.c. (239")	Bielas	8-2000 RAB	STD-2-10-20-30-40-50	55.575/55.596 2.1880/2.1888"	58.725/58.745 2.3120/2.3128"	0.013/0.064 .0005/.0025"	1.562 .0615"	18.95 .746"
	Bancadas	884 M	STD-10-20-30-40-50					
92.08 x 78.74 4192 c.c. (256")	1-2-4-5	2001 RA		63.449/63.470 2.4980/2.4988"	68.356/68.377 2.6912/2.6920"	0.015/0.076 .0006/.0030"	2.441 .0961"	23.04 .907"
	3 ▲	2002 RA		63.449/63.470 2.4980/2.4988"	68.356/68.377 2.6912/2.6920"	0.015/0.076 .0006/.0030"	2.441 .0961"	28.45 1.120"
92.25 x 83.82 4785 c.c. (292")	Levas (1)	1696 M	STD-5-10-15-20-30					
	1	2003 DR		48.908/48.933 1.9255/1.9265"	52.261/52.286 2.0575/2.0585"	0.025/0.076 .0010/.0030"	1.648 .0649"	22.45 .884"
	3	2004 DRI		48.908/48.933 1.9255/1.9265"	52.261/52.286 2.0575/2.0585"	0.025/0.076 .0010/.0030"	1.648 .0649"	16.00 .630"
	2-4-5	2054 DR		48.908/48.933 1.9255/1.9265"	52.261/52.286 2.0575/2.0585"	0.025/0.076 .0010/.0030"	1.648 .0649"	16.00 .884"
	Levas (2)							
	1			52.070/52.095 2.0500/2.0510"	55.410/55.436 2.1815/2.1825"	0.018/0.084 .0007/.0033"	1.648 .649"	22.35 .880"
	2-3-4-5			52.070/52.095 2.0500/2.0510"	55.410/55.436 2.1815/2.1825"	0.018/0.084 .0007/.0033"	1.648 .649"	15.75 .620"

▲ Cojinete con pestaña - Flanged bearing



Ford



12 (continuación)

Fairlane, F-100, AF700 (Motor V8-292)

Levas (3)

1

48.908/48.933	52.261/52.286	0.025/0.076	1.648	22.35
1.9255/1.9265"	2.0575/2.0585"	.0010/.0030"	.0649"	.880"

2-3-4-5

48.908/48.933	52.261/52.286	0.025/0.076	1.648	15.75
1.9255/1.9265"	2.0575/2.0585"	.0010/.0030"	.0649"	.620"

IDENTIFICACION DE JUEGOS DE LEVAS			
AUTOMOVILES		CAMIONES	
Motor	Año	Motor	Año
(1) 272"	56-57	(1) 272"	56-57
(1) 292"	56-74	(1) 292"	58-74
(2) 239"	1954	(2) 239" primeros	1954
(3) 256", 272", 292"	1955	(3) 239", 256"	54-55

TORSION DE APRIETO: Bielas 5.5/ 6.2 mkg.
Bancada 11/ 12.4 mkg. Tapa de cilindros 8.9/ 10.4 mkg

General Motors



1

Corsa 1.0 OHV

4
72 x 61
993 c.c.

Bielas

71-2195/4

STD-0.25-0.50-0.75-1.00

44.971/44.987	48.000/48.012	0.020/0.068	1.499	19.20
1.7705/1.7711"	1.8898/1.8902"		.0590"	.7559"

Bancada

H-634/3

STD-0.25-0.50-0.75

1

72-2196

53.997/54.010	58.000/58.013	0.026/0.076	1.982	25.00
2.1259/2.1264"	2.2835/2.2840"		.0780"	.9843"

2

72-2197 H

54.007/54.020	58.000/58.013	0.020/0.067	1.982	28.90
2.1263/2.1268"	2.2835/2.2840"		.0780"	1.1378"

3

72-2196

54.007/54.020	58.000/58.013	0.020/0.067	1.982	22.50
2.1263/2.1268"	2.2835/2.2840"		.0780"	.8858"

General Motors



2

Corsa 1.0 Wind, 1.4, 1.6

4
77.6 x 73.4
1389 c.c.

Bielas

4-4210 CP

STD-0.25-0.50

42.971/42.987	46.000/46.012	0.020/0.072	1.497	16.60
1.6918/1.6924"	1.8110/1.8115"		.0589"	.6535"

Bancada

7218 M

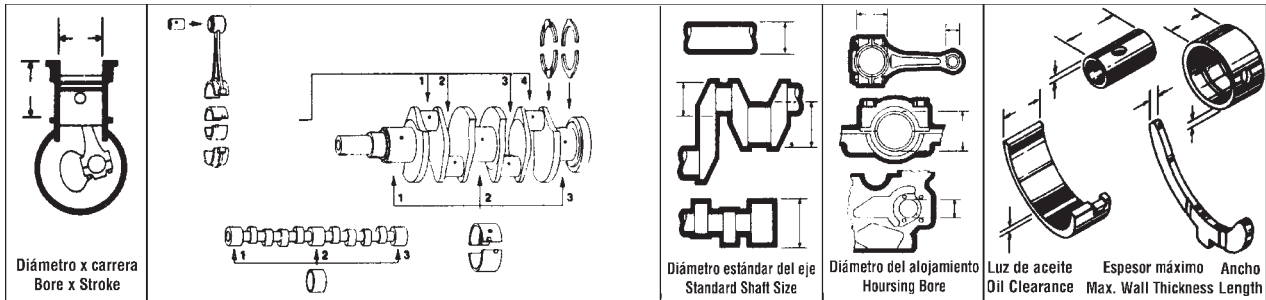
STD-0.25-0.50

72-3505

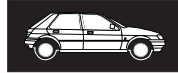
54.972/54.985	59.000/59.013	0.020/0.063	2.001	19.50
2.1643/2.1648"	2.3222/2.3233"		.0788"	.7677"

72-3506 H

54.972/54.985	59.000/59.013	0.020/0.063	2.001	19.50
2.1643/2.1648"	2.3222/2.3233"		.0788"	.7677"



General Motors



3 Astra, Combo, Corsa, Vectra 1.7 D/TD/TDS,
Eng. Isuzu 4 EE 1 TC (Opel = X 17DT) Diesel/ Turbo Diesel

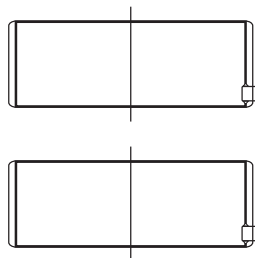
4 79 x 86 1686 c.c.	Bielas	4-4285 RA	STD	45.936/45.956 1.80851/1.8093"	48.976/49.000 1.92821/1.9291"	0.020/0.089	1.500 .0591"	16.70 .6575"
	Bancadas	7235 M	STD 0.25					
	1-5	4281 RAU		51.918/51.938 2.0440/2.0448"	55.976/56.000 2.2038/2.2047"	0.024/0.089	2.007 .0790"	15.45 .6083"
	1-5	4282 RAL		51.918/51.938 2.0440/2.0448"	55.976/56.000 2.2038/2.2047"	0.024/0.089	2.007 .0790"	15.45 .6083"

General Motors



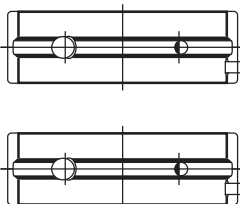
4 Astra, Combo, Vectra 1.7 D/TD
Eng. X17 DR/DTL Diesel/ Turbo Diesel

4 82.5 x 79.5 1700 c.c.	Bielas	71-3821/4	STD-0.25	48.971/48.987 1.9280/1.9286"	52.000/52.012 2.0472/2.0477"	0.020/0.066	1.510 .0594"	19.30 0.7598"
		Medio cojinete marcado "SPUTTER" = medio cojinete superior Medio cojinete inferior = 1.480						

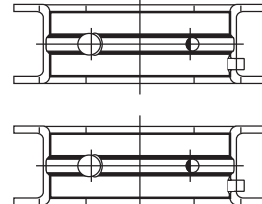


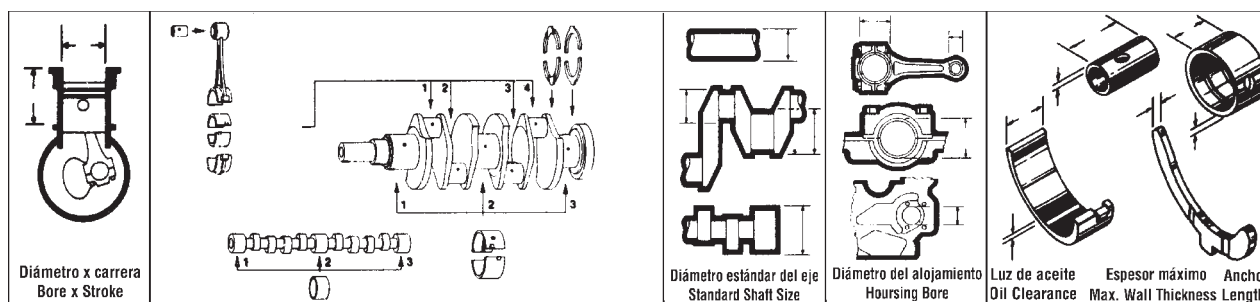
	Bancadas	H 059/5	STD-0.25					
	1-2-4-5	02-3822		57.974/57.995 2.2824/2.2833"	62.000/62.013 2.4409/2.4415"	0.020/0.062	2.001 .0788"	19.50 .7677"
	3	02-3823 H		57.974/57.995 2.2824/2.2833"	62.000/62.013 2.4409/2.4415"	0.020/0.062	2.001 .0788"	25.90 1.0197"

02-3822



02-3823





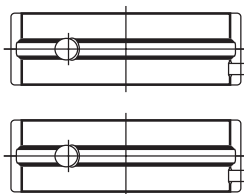
General Motors



Calibra, Astra, Omega, Vectra 2.0i 16V
Eng. X20 XEV (Ecotec) (Family II)

5

4 86 x 86 1998 c.c.	Bielas	71-3626/4	STD-0.25-0.50	48.917/48.987 1.9280/1.9286"	52.000/52.012 2.0472/2.0477"	0.020/0.066	1.500 .0591"	19.30 .7598"
	Bancadas	H 982/5	STD-0.25-0.50	57.982/57.995 2.2828/2.2833"	62.000/62.013 2.4409/2.4415"	0.020/0.066	1.995 .0785"	19.50 .7677"
		02-3532						
		02-3533 H		57.982/57.995 2.2828/2.2833"	62.000/62.013 2.4409/2.4415"	0.020/0.066	1.995 .0785"	25.90 1.0197"



Ancho de Pestaña
STD en medida STD
0.2 mm en medida 0.25
0.4 mm en medida 0.50

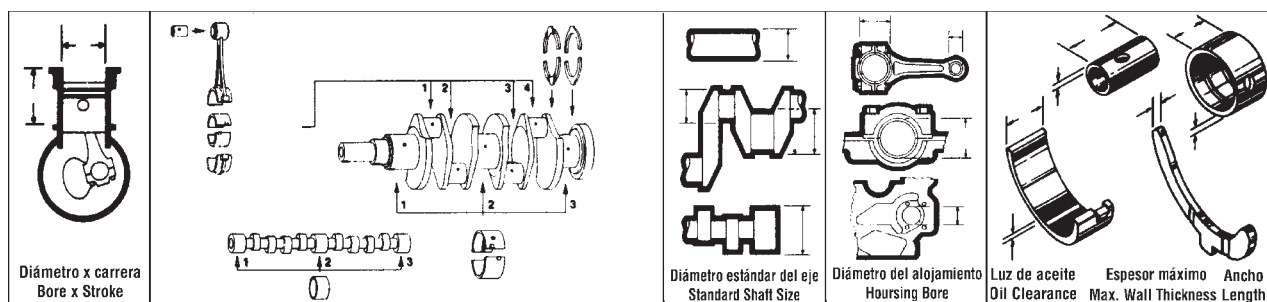
General Motors



Chevette, Chevette Hatch, Marajo, Chevy 500, 1.4, 1.6, OHV

6

4 82 x 66.20 1400 c.c.	Bielas	4-68065 RA	STD-0.25-0.50-0.75-1.00	45.958/45.984 1.8094/1.8104"	49.000/49.012 1.9291/1.9296"	0.018/0.080 .0007/.0031"	1.499 .0590"	19.05 .750"
	Bancada	6813 M	STD-0.25-0.50-0.75-1.00	Hasta 1987				
4 82 x 75.40 1600 c.c.	1-2-3-4	68066 RA		50.987/51.013	54.987/55.013	0.024/0.100	1.975	18.00
	5 ▲	*68078 RA		50.987/51.013	54.987/55.013	0.024/0.100	1.975	23.90
		7327 M	STD-0.25-0.50-0.75-1.00	Desde 1987				
	1-2-3	68066 RA		50.987/51.013	54.987/55.013	0.024/0.100	1.975	18.00
	4	68066 RAB		50.987/51.013	54.987/55.013	0.024/0.100	1.975	18.00
	5 ▲			50.987/51.013	54.987/55.013	0.024/0.100	1.975	23.90
		7327 MA		Pestaña +0.9 mm				



General Motors



Monza SL/SLE/Classic/SR, Ômega, Ômega-Suprema,
Kadett Ipanema, Vectra LGS/CD, Vectra 16V, 2.0 - 1987 /

7

4 86 x 86 1998 c.c.	Bielas	4-1490 CPA	STD-.025-0.50-0.75-1.00	48.971/48.987 1.9280/1.9286"	52.000/52.012 2.0472/2.0477"	0.019/0.063 .0007/.0025"	1.497 .0600"	19.30 .760"
	Bancada	7108 M						
	1-2-4-5	68141 CP		57.983/57.996 2.2828/2.2833"	62.001/62.014 2.4410/2.4415'	0.018/0.058 .0007/.0023"	1.994 .0785"	19.51 .768"
	3 ▲	68142 CP		57.983/57.996 2.2828/2.2833"	62.001/62.014 2.4410/2.4415'	0.018/0.058 .0007/.0023"	1.994 .0785"	25.90 1.020"

Honda



1

Civic 1.2, 1.3, EB, EJ, EN

4 72 x 76 1238 c.c. 72 x 82 1335 c.c.	Bielas	4-67340 CP	STD-0.25-0.50-0.75-1.00	39.977/39.999 1.5739/1.5748"	42.999/43.025 1.6929/1.6939"	0.020/0.070 .0591"	1.501 .6890"	17.50
	Bancada	6792 M	STD-0.25-0.50	49.977/49.999 1.9676/1.9685"	53.990/54.023 1.1256/1.1269"	0.026/0.073 .0787"	1.999 .7870"	19.899
	Axial	67341 AF	STD				2.489 .0980"	

Honda

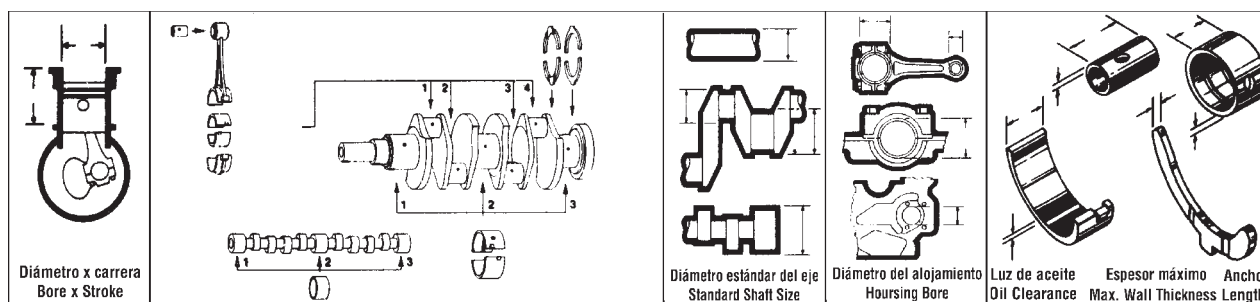


2

Accord, Prelude, 1.6, 1.8, EK, EL, EP, EF

4 74 x 86.5 1488 c.c. 74 x 93 1599 c.c. 77 x 94 1750 c.c.	Bielas	4-67345 CP	STD-0.25-0.50-0.75	41.976/41.999 1.6526/1.6535"	45.001/45.024 1.7717/1.7726"	0.013/0.071 .0005/.0028"	1.501 .0591"	17.50 .689"
	Bancada	6784 M	STD-0.25-0.75					
	1-2-3-4-5	67341		49.977/50.000 1.9676/1.9685"	54.000/54.023 2.1260/2.1269"	0.013/0.053 .0005/.0021"	2.009 .0791"	19.99 .787"
	Bancada	6793 M	STD-0.25-0.50-0.75					
	1-2-3-4-5	67341 CPB		50.005/50.041 1.9687/1.9705"	54.041/54.077 2.1276/2.1290"		2.006 .0790"	19.99 .787"
	Axial	67341 AF	STD				2.464 .097"	

TORSION DE APRIETO: Bielas 3.2 mkg., Bancada 6.5 mkg
Tapa de cilindros 6 mkg.



Honda



3

Accord, Prelude 2.0, B20A

4 75 x 90 1590 c.c.	Biela	4-1825 CP	STD-0.25	47.976/47.998 1.8888/1.8897"	51.001/51.021 2.0079/2.0087"	0.020/0.061	1.509 .0594"	19.48 .7669"
	Bancada	7102 M	STD-0.25	54.975/54.996 2.1644/2.1625"	59.004/59.024 2.3230/2.3238	0.026/0.060	2.006 .0790"	19.98 .7870"
	Axial	1826 AF	STD				2.489 .098"	

Honda



4

Accord, Prelude 2.0 - 2.2 16V, F22

4 85 x 95 2156 c.c.	Biela	4-1825 CP	STD-0.25	47.976/47.998 1.8888/1.8897"	51.001/51.021 2.0079/2.0087"	0.020/0.061	1.509 .0594"	19.48 .7669"
	Bancada	7103 M	STD-0.25	49.976/50.000 1.9676/1.9685"	54.000/54.024 2.1260/2.1269"	0.026/0.071	1.845 .0726"	20.29 .7988"
	Axial	1203 BF	STD				2.500 .0985"	

Ika

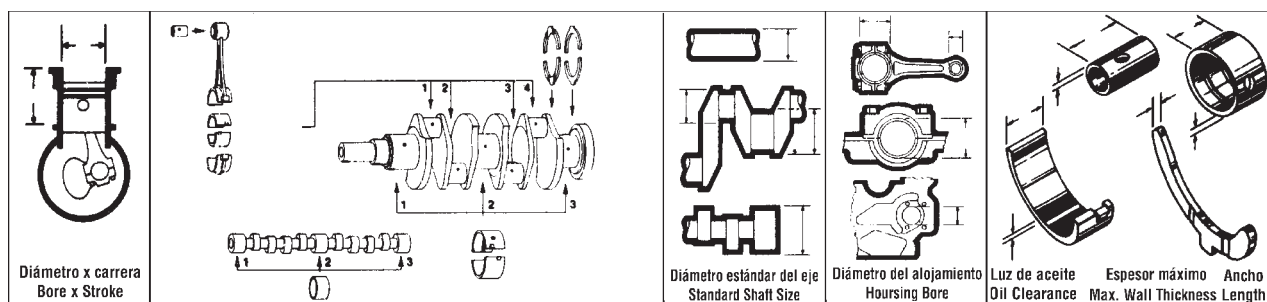


1

Bergantin, Jeep - 1956 / 1978, Motor Continental 4L-151

4 84.14 x 111.13 2470 c.c. (151")	Bielas	4-1750 RAA	STD-2-10-20-30-40-50	52.372/52.393 2.0619/2.0627"	55.537/55.550 2.1865/2.1870"	0.015/0.050 .0006/.0020"	1.572 .0619"	28.70 1.130"
	Bancadas	4588 M	STD-10-20-30-40-50					
	1	1751 RA		60.310/60.330 2.3744/2.3752"	65.062/65.080 2.5615/2.5622"	0.015/0.060 .0006/.0024"	2.366 .0931"	27.10 1.067"
	2	1752 RA		60.310/60.330 2.3744/2.3752"	65.062/65.080 2.5615/2.5622"	0.015/0.060 .0006/.0024"	2.366 .0931"	33.45 1.317"
	3 ▲	1753 RA		60.310/60.330 2.3744/2.3752"	65.062/65.080 2.5615/2.5622"	0.015/0.060 .0006/.0024"	2.366 .0931"	41.45 1.632"
	Levas	1093 M	STD-2-5-10-15-20-30-40					
	1	1207 DR		45.961/45.987 1.8095/1.8105"	49.199/49.212 1.9370/1.9375"	0.025/0.076 .0010/.0030"	1.600 .0630"	23.82 .938"
	2	1208 DR		44.379/44.412/ 1.7482/1.7485"	47.612/47.625 1.8745/1.8750"	0.025/0.076 .0010/.0030"	1.595 .0628"	41.27 1.625"
	3	1209 DR		31.686/31.712 1.2475/1.2485"	34.912/34.925 1.3745/1.3750"	0.025/0.076 .0010/.0030"	1.595 .0628"	29.46 1.160"
	Buje de Biela 4207-X							

TORSION DE APRIETO: Bielas 5.5/ 6.2 mkg. Bancada 11.7/ 13.1 mkg
Tapa de cilindros 4.8/ 6.2 mkg



Ika



Kaiser Carabela, Bergantin, Rambler, Estanciera, Baqueano, Jeep Gladiator - 1957 / 1964, Motor Continental 6L-226

2

4 84.14 x 111.13 3705 c.c. (226")	Bielas	6-1750 RAA	STD-2-20-30-40-50	52.372/52.393 2.0619/2.0627"	55.537/55.550 2.1865/2.1870"	0.015/0.050 .0006/.0020"	1.572 .0619"	28.70 1.130"
	Bancada	751 M	STD-10-20-30-40-50					
	1	1751 RA		60.310/60.330 2.3744/2.3752"	65.062/65.080 2.5615/2.5622"	0.015/0.060 .0006"/.0024"	2.366 .0931"	27.10 1.067"
	2-3	1752 RA		60.310/60.330 2.3744/2.3752"	65.062/65.080 2.5615/2.5622"	0.015/0.060 .0006"/.0024"	2.366 .0931"	33.45 1.317"
	4 ▲	1753 RA		60.310/60.330 2.3744/2.3752"	65.062/65.080 2.5615/2.5622"	0.015/0.060 .0006"/.0024"	2.366 .0931"	41.45 1.632"
	Levas	1088 M	STD-2-5-10-15-20-30-40					
	1	1206 DR		47.562/47.587 1.8725/1.8738"	50.787/50.800 1.9995/2.0000"	0.025/0.076 .0010/.0030"	1.595 .0628"	22.32 .879"
	2	1207 DR		45.961/45.987 1.8095/1.8105"	49.199/49.212 1.9370/1.9375"	0.025/0.076 .0010/.0030"	1.600 .0630"	23.82 .938"
	3	1208 DR		44.379/44.412 1.7482/1.7485"	47.612/47.625 1.8745/1.8750"	0.025/0.076 .0010/.0030"	1.595 .0628"	41.27 1.625"
	4	1209 DR		31.686/31.712 1.2475/1.2485"	34.912/34.925 1.3745/1.3750"	0.025/0.076 .0010/.0030"	1.595 .0628"	29.46 1.160"

Buje de Biela 4207-X

TORSION DE APRIETO: Bielas 5.5/ 6.2 mkg. Bancada 11.7/ 13.1 mkg
Tapa de cilindros 4.8/ 6.2 mkg.; 4.8/ 5.5 mkg (Aluminio)

Ika



Rambler, Torino, Estanciera, Jeep BA-1N, Gladiator Jeep T80, Jeep 1000 - 1965 / 1975, OHC-181, OHC-230

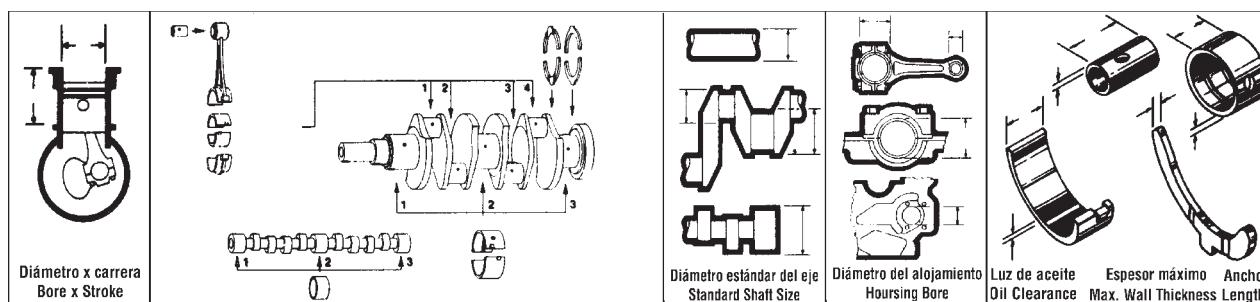
3

6 84.94 x 87.31 2966 c.c. (181")	Bielas	6-2700 RA	STD-2-10-20-30-40-50	53.372/52.393 2.0619/2.0627"	55.537/55.550 2.1865/2.1870"	0.015/0.063 .0006/.0025"	1.565 .0616"	24.77 .975"
	Bancadas	4532 M	STD-10-20-30-40-50					
	1	2701 RA		60.317/60.338 2.3747/2.3755"	65.062/65.080 2.5615/2.5622"	0.012/0.063 .0005/.0025"	2.357 .0928"	25.40 1.000"
84.94 x 111.13 3770 c.c. (230")	2-3	2702 RA		60.317/60.338 2.3747/2.3755"	65.062/65.080 2.5615/2.5622"	0.012/0.063 .0005/.0025"	2.357 .0928"	30.25 1.191"
	4 ▲	2703 RAA		60.317/60.338 2.3747/2.3755"	65.062/65.080 2.5615/2.5622"	0.012/0.063 .0005/.0025"	2.357 .0928"	38.76 1.526"
	Levas							
	1			50.711/50.736 1.9965/1.9975"	50.787/50.813 1.9995/2.005"	0.050/0.100 .0020/.0039"		
	2			47.536/47.561 1.8715/1.8725"	47.612/47.638 1.8745/1.8755"	0.050/0.100 .0020/.0039"		
	3			44.361/44.386 1.7465/1.7475"	44.437/44.463 1.7495/1.7505"	0.050/0.100 .0020/.0039"		
	4			34.836/34.861 1.3715/1.3725"	34.912/34.938 1.3745/1.3755"	0.050/0.100 .0020/.0039"		

Buje de Biela 4207-X

TORSION DE APRIETO: Bielas 6 mkg. Bancada 12.5 mkg
Tapa de cilindros 13 mkg.

▲ Cojinete con pestaña - Flanged bearing



Ika



4

Rambler, Torino, Jeep 1000 - 1974 / 1981, OCH-230-7B

6 84.94 x 111.13 3770 c.c. (230")	Bielas	6-62055 RA	STD-0.25-0.50-0.75-1.00	52.376/52.396	55.537/55.550	0.017/0.052	1.567	22.56
	Bancadas	6408 M	STD-0.25-0.50-0.75-1.00					
	1-2-3-4-5-6	62056 RA		64.378/64.394	69.124/69.142	0.022/0.060	2.358	21.42
	7 ▲	62057 RA		64.378/64.394	69.124/69.142	0.022/0.060	2.358	21.42
1	Levas	1593 M	STD/STD-1.00 mm					
	1	62057 DR		30.949/30.975	37.975/38.000	0.021/0.080	3.515	33.00
	2-3-4	62056 DR		30.949/30.975	37.975/38.000	0.021/0.080	3.515	25.00

TORSION DE APRIETO: Bielas 5.5/ 6.5 mkg. Bancada 11.7/ 13.1 mkg
Tapa de cilindros 11.7/ 13.1 mkg.

Indenor - Peugeot Diesel



1

XD 4.88

4 88 x 80 1948 c.c.	Bielas	4-2870 RA	STD-0.30-0.50-0.80-1.00	49.984/50.000	53.695/53.708	0.051/0.092	1.822	29.00
	Bancadas	4268 M	STD-0.30-0.50-0.80-1.00					
	1-3-5	2871 RAA		54.994/55.009	58.731/58.750	0.052/0.098	1.835	30.00
	2-4	2873 RAA		54.994/55.009	58.731/58.750	0.052/0.098	1.835	22.00
	Axial	2873 BF	STD-5-10				2.330	
	Levas (*)							
	1			41.925/41.950	42.000/42.039	0.050/0.114		
	2			41.925/41.950	42.000/42.039	0.050/0.114		
	3			41.925/41.950	42.000/42.039	0.050/0.114		
	Buje de Biela 4274-Y							

TORSION DE APRIETO: Bielas 6 mkg. Bancada 10 mkg.
Tapa de cilindros 7 mkg.
(*) No usa originalmente cojinetes de levas.

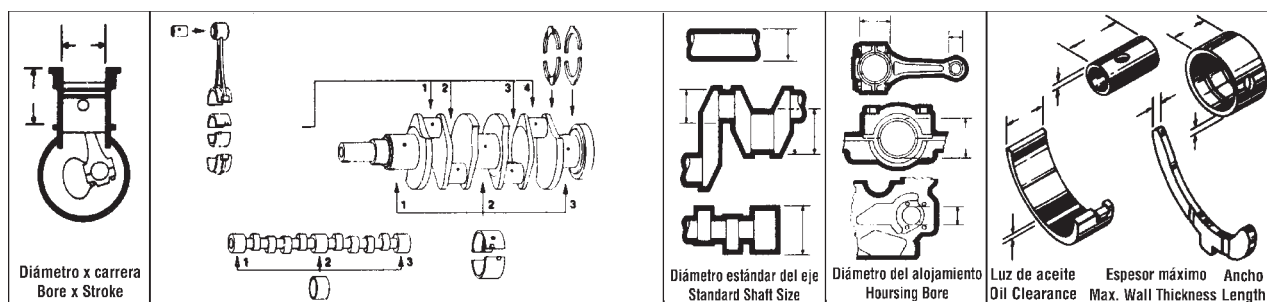
Indenor - Peugeot Diesel



2

XD3, XD2

4 94 x 83 2304 c.c.	Bielas	4-66690 RA	STD-0.30-0.50-0.80	54.994/55.021 2.1651/2.1662"	58.737/58.750 2.3125/2.3130"		1.838	29.00
	Bancadas	6594 M	STD-0.30-0.50-0.80-1.00					
	1-3-5	66691 RA		59.994/60.021 2.3620/2.3630"	63.731/63.750 2.5091/2.5098"		1.835	30.00
	2-4	66692 RA		59.994/60.021 2.3620/2.3630"	63.731/63.750 2.5091/2.5098"		1.835	24.00
2498 c.c.	Axial	66691 BF	STD-0.20 mm				2.330	
							.092"	



Indenor - Peugeot Diesel



3 XDP 6.9

6 94 x 83 3168 c.c.	Bielas	6-2870 RA	STD-0.30-0.50-0.80-1.00	49.984/50.000	53.695/53.708	0.051/0.092	1.822	29.00
	Bancadas	4269 M	STD-0.30-0.50-0.80-1.00					
	1-4-7	2871 RAA		54.994/55.009	58.731/58.750	0.052/0.098	1.835	30.00
	2-3-5-6	2872 RA		54.994/55.009	58.731/58.750	0.052/0.098	1.835	30.00
	2-3-5-6	2873 RAA						
	Axial	2873 BF	STD-5-10				2.330	
Levas (*)								
	1			41.925/41.950	42.000/42.039	0.050/0.114		26.00
	2-4			41.925/41.950	42.000/42.039	0.050/0.114		20.00
	3			41.925/41.950	42.000/42.039	0.050/0.114		21.00
	5			41.925/41.950	42.000/42.039	0.050/0.114		26.00

Buje de Biela 4274-Y

TORSION DE APRIETO: Bielas 6 mkg. Bancada 10 mkg

Tapa de cilindros 7 mkg.

(*) No usa originalmente cojinetes de levas.

Isuzu



1 Trooper II - 4ZD1

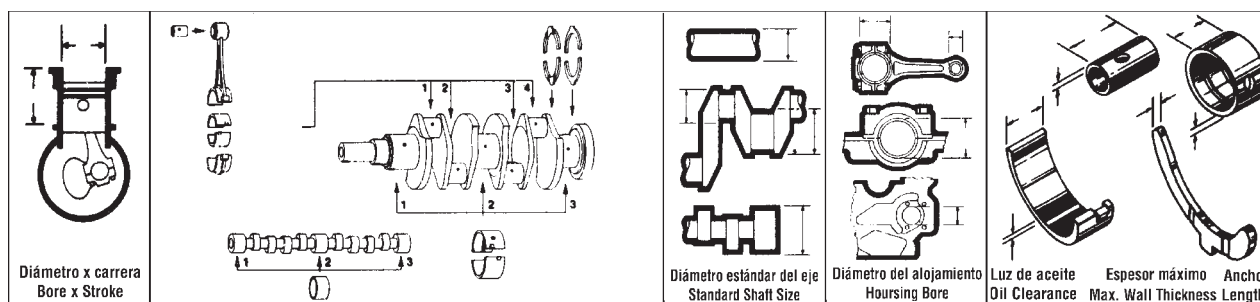
4 89 x 89 2254 c.c.	Bielas	4-3570 CPB	STD-0.25-0.50	48.925/48.940 1.9262/1.9268"	51.986/51.998 2.0467/2.0472"	0.020/0.070	1.514	22.131 .0596" .8713"
	Bancadas	6631 M	STD-0.25-0.50	55.920/55.930 2.2016/2.2020"	59.979/59.999 2.3614/2.3622"	0.031/0.086	2.009	21.869 .0791" .8610"
	Axial	3571 BF	STD	63.800 2.5118"	80.000 3.1496"		2.360 .0929"	
	Levas	1747 M	STD	29.949/29.984 1.1791/1.1797"	32.979/32.999 1.2984/1.2992"		1.501 .0591"	20.139 .7829"

Isuzu



2 Rodeo, Trooper, 2.BTA, 4JBI, 4JBITA

L4 93 x 102 2772 c.c.	Biela	4-1900 RA	STD-0.25-0.50	2.0834/2.0839"	2.2047/2.2047"	.0010/.0027"	.0596"	1.012"
	Bancada	7112 M	STD-0.25-0.50					
	1-2-3-4-5	3962 RA	Set Only	2.7526/2.7532"	2.9124/2.9134"	.0012/.0034"	.0790"	1.017"
	3	3954 BF	Std Only				.098"	



John Deere



1

1420, 2330 - 1970 / 1980

3 82 x 75 2696 c.c.	Bielas	1-3325 CP (3 pares)	STD-10-20-30-40	69.799/69.825 2.7480/2.7490"	73.660/73.685 2.900/2.9010"	0.025/0.102 .0010/.0040"	1.905 .0745"	30.12 1.186"
	Bancadas 1-2-3	6478 M 3326 CP	STD-10-20-30	79.324/79.350 3.1230/3.1240"	84.445/84.480 3.3246/3.3260"	0.041/0.117 .0016/.0046"	2.532 .1000"	29.46 1.160"
	4 ▲	3327 CP		79.324/79.350 3.1230/3.1240"	84.445/84.480 3.3246/3.3260"	0.041/0.117 .0016/.0046"	2.532 .1000"	38.85 1.530"
	Levas 1-2-3			55.872/55.898 2.1997/2.2007"	55.987/56.012 2.2042/2.2052"	0.089/0.140 .0035/.0055"		
Buje de Biela 4255-Y								

TORSION DE APRIETO: Bielas 8/ 9.6 mkg. Bancada 11.8 mkg.
Tapa de cilindros 15 mkg.

John Deere



2

2420, 2530, 2730 - 1970 / 1980

4 102 x 110 3596 c.c. 106.5 x 110 3920 c.c.	Bielas	1-3325 CP (4 pares)	STD-10-20-30-40	69.799/69.825 2.7480/2.7490"	73.660/73.685 2.900/2.9010"	0.025/0.102 .0010/.0040"	1.905 .0745"	30.12 1.186"
	Bancadas 1-2-3-4	6479 M 3326 CP	STD-10-20-30	79.324/79.350 3.1230/3.1240"	84.445/84.480 3.3246/3.3260"	0.041/0.117 .0016/.0046"	2.532 .1000"	29.46 1.160"
	5 ▲	3327 CP		79.324/79.350 3.1230/3.1240"	84.445/84.480 3.3246/3.3260"	0.041/0.117 .0016/.0046"	2.532 .1000"	38.85 1.530"
	Levas 1-2-3			55.872/55.898 2.1997/2.2007"	55.987/56.012 2.2042/2.2052"	0.089/0.140 .0035/.0055"		
Buje de Biela 4255-Y								

TORSION DE APRIETO: Bielas 8/ 9.6 mkg Bancada 11.8 mkg
Tapa de cilindros 15 mkg.

John Deere

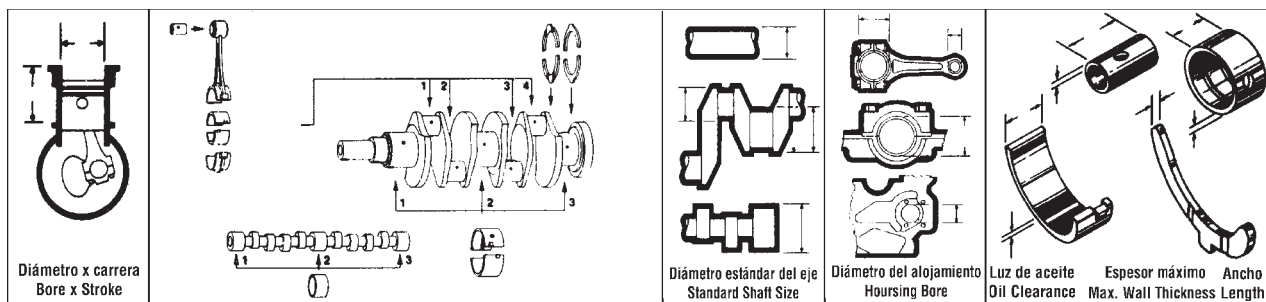


3

3420, 4420, 3530, 4530 - 1970 / 1980

6 98.04 x 109.98 4976 c.c. 102 x 110 5393 c.c. 106.5 x 110 5879 c.c.	Bielas	1-3325 CP (6 pares)	STD-10-20-30-40	69.799/69.825 2.7480/2.7490"	73.660/73.685 2.900/2.9010"	0.025/0.102 .0010/.0040"	1.905 .0745"	30.12 1.186"
	Bancadas 1-2-3-4-5-6	6480 M 3326 CP	STD-10-20-30	79.324/79.350 3.1230/3.1240"	84.445/84.480 3.3246/3.3260"	0.041/0.117 .0016/.0046"	2.532 .1000"	29.46 1.160"
	7 ▲	3327 CP		79.324/79.350 3.1230/3.1240"	84.445/84.480 3.3246/3.3260"	0.041/0.117 .0016/.0046"	2.532 .1000"	38.85 1.530"
	Levas 1-2-3			55.872/55.898 2.1997/2.2007"	55.987/56.012 2.2042/2.2052"	0.089/0.140 .0035/.0055"		
Buje de Biela 4255-Y								

TORSION DE APRIETO: Bielas 8/ 9.6 mkg Bancada 11.8 mkg
Tapa de cilindros 15 mkg.



Lada



1 2101, 2102, 2103, 2104, 2106, Niva 4x4

4 76 x 66 1198 c.c.	Biela	4-66415 RA	STD-10-20-30-40	47.814/47.834 1.8824/1.8832"	51.330/51.346 2.0209/2.0215"	0.035/0.096	1.730 .0681"	20.70 .8150"
79 x 66 1294 c.c.	Bancada	6481 M	STD-10-20-30-40					
76 x 80 1451 c.c.	1-2-4-5	66416 RA		50.785/50.805 1.9994/2.0002"	54.507/54.520 2.1459/2.1465"	0.039/0.098	1.831 .0721"	21.250 .8366"
79 x 80 1568 c.c.	3	60987 RA		50.785/50.805 1.9994/2.0002"	54.507/54.520 2.1459/2.1465"	0.039/0.098	1.831 .0721"	22.750 .8957"
82 x 71 1500c.c.								

Land Rover



1 Discovery, Defender 2.5 TA, Motor12J

4 90.5 x 88.9 2287 c.c.	Biela	4-65190 CP	STD-0.10-0.20	58.725/58.744	62.433/62.446	0.029/0.086	1.830	25.07
4	Bancada	8060M	STD-0.10-0.20					
	1-4			63.487/63.500	67.703/67.716	0.043/0.094	2.080	24.40
1	5			63.487/63.500	67.703/67.716	0.043/0.094	2.080	35.00
2		65192 BF	STD-0.10	70.739/70.993	85.598/85.852		2.340	

Maxion



1 Motor HST 2.5 L

4 90.48 x 97.00	Bielas	4-4795A	STD/0.25	58.725/58.744	62.433/62.446	0.023/0.067	1.827/1.833	25.00
	Bancadas	7372 M	STD/0.25	63.475/63.487	67.704/67.721	0.031/0.078	2.083/2.093	24.50

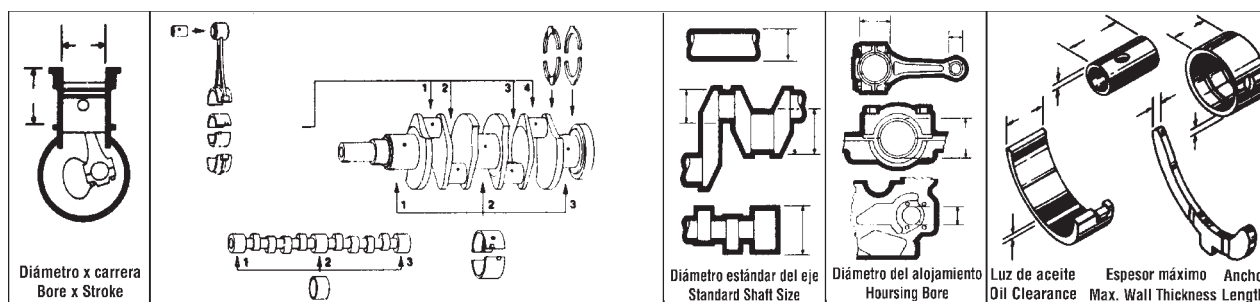
Mazda



1 323, Motor 1.3, 1.5, 1.4, TC, E1, Ford Laser

4 77 x 69.6 1269 c.c.	Biela	4-3910 RA	STD-0.25-0.50-0.75	39.950/39.954 1.5724/1.5730"	42.996/43.016 1.6928/1.6935"		1.509 .0594"	19.09 .669"
73 x 76 1272 c.c.	Bancada ▲	7207 M 3911 RA	STD-0.25-0.50-0.75	49.950/49.954 1.9661/1.9667"	53.999/54.013 2.1260/2.1265"	.0010/.0031"	2.014 .0793"	19.12 .753"
	Axial	3911 AF	STD				2.007 .0790"	

▲ Cojinete con pestaña - Flanged bearing



Mazda



2

Motor 1.0, 1.1, PB, TA, TB, TC

4 70 x 64 985 c.c. 73 x 76 1272 c.c.	Biela	4-67335 RA	STD-0.25-0.50-0.75					
	Bancada	6789 M	STD-0.25-0.50-0.75					
	1-2-4	67358 RA		62.938/62.953 2.4779/2.4785"	67.000/67.017 2.6378/2.6385"	0.028/0.085	2.009 .0791"	20.142 .7930"
	3-5	67357 RA		62.938/62.953 2.4779/2.4785"	67.000/67.017 2.6378/2.6385"	0.028/0.085	2.009 .0791"	23.139 .9110"
	Axial	67356 AF 67356 AFL 67356 AFU	STD	70.000 2.7559"	80.000 3.1496"		2.006 .0790"	

Mazda



3

Motor 1.6, 1.8, NA, VC, VB

4 78 x 83 1586 c.c. 80 x 88 1272 c.c.	Biela	4-1130 RA	STD-0.25-0.50	52.938/52.953 2.0842/2.0848"	55.999/56.019 2.2047/2.2055"	0.028/0.088	1.508 0.0594"	20.904 0.8230"
	Bancada	5027 M	STD-0.25-0.50					
	1-2-4	3566 RA		62.938/62.953 2.4779/2.4785"	67.000/67.017 2.6378/2.6385"	0.028/0.088	2.009 .0791"	18.389 .7240"
	3	3567 RA		62.938/62.953 2.4779/2.4785"	67.000/67.017 2.6378/2.6385"	0.028/0.088	2.009 .0791"	24.104 .9490"
	5	3568 RA		62.938/62.953 2.4779/2.4785"	67.000/67.017 2.6378/2.6385"	0.028/0.079	2.255 .0888"	24.104 .9490"
	Axial	3567AF 3567 AFL 3567 AFU	STD	70.000 2.7559"	86.000 3.3858"		2.057 .0810"	

Mazda

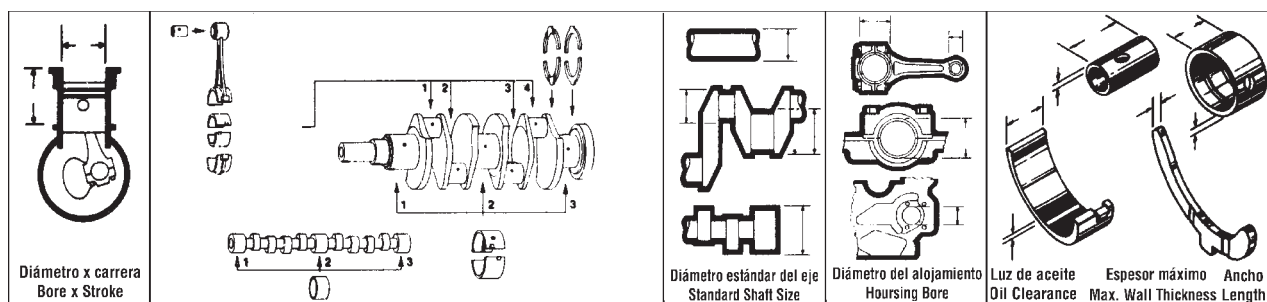


4

Motor 1.8, 2.0, VB, MA

4 78 x 94 1797 c.c. 80 x 88 1970 c.c.	Biela	4-4005 RA	STD-0.25-0.50-0.75-1.00	52.941/52.953 2.0843/2.0848"	55.999/56.019 2.2047/2.2055"	0.028/0.088	1.508 0.0594"	20.904 0.8230"
	Bancada	5027 M	STD-0.25-0.50					
	1-2-4	3566 RA		62.938/62.953 2.4779/2.4785"	67.000/67.017 2.6378/2.6385"	0.028/0.088	2.009 .0791"	18.389 .7240"
	3	3567 RA		62.938/62.953 2.4779/2.4785"	67.000/67.017 2.6378/2.6385"	0.028/0.088	2.009 .0791"	24.104 .9490"
	5	3568 RA		62.938/62.953 2.4779/2.4785"	67.000/67.017 2.6378/2.6385"	0.028/0.088	2.255 .0888"	24.104 .9490"
	Axial	3567AF 3567 AFL 3567 AFU	STD	70.000 2.7559"	86.000 3.3858"		2.057 .0810"	

▲ Cojinete con pestaña - Flanged bearing



Mercedes Benz



1 Compresores de Aire, OM-312, OM-314, OM-352, OM-321

1	Bielas	1-60545 RA	STD-10-20-30 STD/30	31.984/32.000 1.2592/1.2598"	35.001/35.016 1.3780/1.3786"	0.030/0.066 .0012/.0026"	1.487 .0585"	17.00 .669"
77 x 30 140 c.c.	TORSION DE APRIETO: Bielas 1.5 mkg + 0.20 mkg. Cilindro 3.5/ 4 mkg. Tapa de cilindro 2.5 mkg							

Mercedes Benz



2 Compresor aire OM -352, 352A, 364, 366, 366A

1	Biela	1-68125 RA	STD-0.25-0.50-0.75	37.984/32.000	35.000/35.016	0.020/0.068	1.490	21.00
94 x 30 208 c.c.								

Mercedes Benz



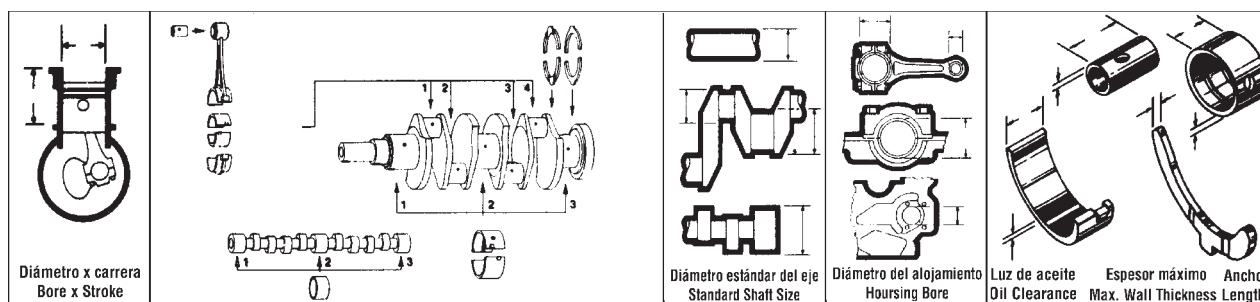
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3 170, 180 Unimog - 1949 / 1959, OM-636

4	Bielas	4-2720 CP 71-0271/4	STD-0.25-0.50-0.75-1.00	49.940/49.960	54.000/54.020	0.027/0.091	2.006	28.00
73.5 x 100 1696 c.c.	JUEGO DE COJINETES CON 1/2 PESTAÑA							
75 x 100 1767 c.c.	Bancadas	4584 M	STD-0.25-0.50-0.75					
	1	2721 CP		54.940/54.960	59.500/59.520	0.023/0.088	2.257	29.10
	2 (½ Sup) ▲	2722 CP		54.940/54.960	59.500/59.520	0.023/0.088	2.257	36.92
	2 (½ Inf)	2722 CP		54.940/54.960	59.500/59.520	0.023/0.088	2.257	31.10
	3	2723 CP		54.940/54.960	59.500/59.520	0.023/0.088	2.257	37.10
	JUEGO DE COJINETES CON PESTAÑA COMPLETA							
	Bancadas	H113/3	STD-0.25-0.50-0.75					
	1	72-0277		54.940/54.960	59.500/59.520	0.023/0.088	2.257	29.10
	2	72-0279		54.940/54.960	59.500/59.520	0.023/0.088	2.257	36.92
	3	72-0278		54.940/54.960	59.500/59.520	0.023/0.088	2.257	37.10
	Levas							
	1			27.967/27.980	50.500/50.530	0.041/0.061	11.256	40.00
	2			27.967/27.980	50.500/50.530	0.041/0.061	11.256	54.00
	3			27.967/27.980	50.500/50.530	0.041/0.061	11.256	51.20
	Buje de Biela 4105-Y TORSION DE APRIETO: Bielas 3.9 mkg. Bancada 8.1 mkg. Tapa de cilindros 8.1 mkg							

* Disponible en las marcas: Federal Mogul, Glyco y AE

▲ Cojinete con pestaña - Flanged bearing



Mercedes Benz



4

Van 180D, OM-616

4 91 x 92.4 2403 c.c.	Bielas	01-3040/4	STD-0.25-0.50-0.75-1.00	51.995/51.965 2.0455/2.0459"	55.600/55.619 2.1890/2.1897"	0.024/0.079	1.805 .0711"	26.00 1.0236"
4 87 x 92.4 2196 c.c.	Bancada	H 914/5	STD-0.25-0.50-0.75-1.00					
	1-2-4-5	02-1480		69.955/69.965 2.7541/2.7545"	74.500/74.519 2.9331/2.9338"	0.024/0.079	2.255 .0888"	27.00 1.0630"
	3 ▲	02-2774 H		69.955/69.965 2.7541/2.7545"	74.500/74.519 2.9331/2.9338"	0.024/0.079	2.255 .0888"	33.90 1.3346"
	Bancada	H951/5	STD-0.25-0.50-0.75-1.00					
	1-2-4-5			69.955/69.965	74.500/74.519	0.025/0.079	27.000	2.255
	Axial	A11314	STD-0.25-0.50-0.75-1.00					
	1			69.955/69.965	74.500/74.519	0.025/0.079	27.000	2.255
	2	A113/4	STD-0.10-0.20-0.40	78.025	91.500			2.200
	2	A113/4	STD-0.10-0.20-0.40	78.025	91.500			2.200

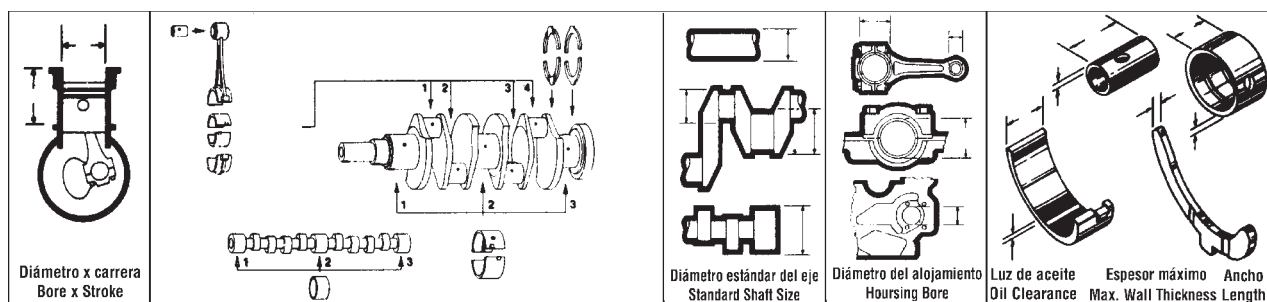
Mercedes Benz



5

OM-312, OM-321, L3500, L4500, L3250, - 1949 / 1964

6 90 x 120 4580 c.c.	Bielas	6-2725 CP	STD-0.25-0.50-0.75-1.00	59.970/60.000	66.000/66.019	0.0500/0.100	2.985	32.10
95 x 120 5100 c.c.	Biela	6-60205 CP	STD-0.25-0.50-0.75-1.00 STD/0.75-0.75/1.50 mm	59.990/60.015	66.000/66.019	0.050/0.1000	2.972	32.10
	Bancada (1)	4592 M	STD-0.25-0.50-0.75-1.00 STD/0.75-0.75/1.50 mm					
	1-2-3-5-6-7	60971 CP		69.970/70.000	78.000/78.019	0.050/0.100	3.985	24.10
	4 (½ Sup)	60972 CP		69.970/70.000	78.000/78.019	0.050/0.100	3.985	24.10
	4 (½ Inf.) ▲	60972 CP		69.970/70.000	78.000/78.019	0.050/0.100	3.985	31.81
	Bancada	4277 M	STD-0.25-0.50-0.75-1.00 STD-0.75-0.75/1.50 mm					
	1-2-3-5-6-7	2727 CP		72.970/73.000	78.000/78.019	0.0500/0.100	2.485	24.10
	4 (½ Sup)	2727 CP		72.970/73.000	78.000/78.019	0.0500/0.100	2.485	24.10
	4 (½ Inf) ▲			72.970/73.000	78.000/78.019	0.0500/0.100	2.485	31.81
	Levas	1315 M	STD-1.00 mm					
	1	61916 DR		55.941/55.960	60.000/60.030	0.030/0.079	2.010	26.10
	2	61917 DR		55.691/55.710	60.000/60.030	0.030/0.079	2.135	26.10
	3	61918 DR		55.441/55.460	60.000/60.030	0.030/0.079	2.260	26.10
	4	61919 DR		55.191/55.210	60.000/60.030	0.030/0.079	2.385	26.10
	Buje de Biela	4106-Y	STD-SEMI	29.994/30.000	33.000/00.025	0.016/0.035	1.920	34.57



Mercedes Benz *

6 OM-314/M-314-O - 1970 /1980

4 97 x 128 3783 c.c.	Bielas	4-61915 CP	STD-0.25-0.50-0.75-1.00	59.990/60.015	66.000/66.019	0.050/0.100	2.972	32.10
		4-61915 CPA	STD-0.25-0.50-0.75-1.00	59.990/60.015	66.000/66.019	0.050/0.100	2.972	29.10
		71-3572/4						
		AEB92167						
Bancada		6405 M	STD-0.25-0.50-0.75-1.00					
		H-712/5						
		AEM92338						
1-3-4-5		61916 CP		87.990/88.010	93.000/93.022	0.050/0.090	2.477	24.10
2 ▲		61917 CPA		87.990/88.010	93.000/93.022	0.050/0.090	2.477	31.81
ANCHO DE PESTAÑA + 0.80 MM / LARGURA FLANGE CON CANAL DE ACEITE / COM CANAL DE ÓLEO								
Levas		1653 M	STD-0.25-0.50-0.75					
1		2726 DRI	0.75 SEMI	55.691/55.710 2.1952/2.1933"	60.000/60.030 2.3622/2.3624"	0.016/0.105 .0006/.0041"	2.135 .0841"	26.20 1.031"
2		2728 DR		55.441/55.460 2.1827/2.1835"	60.000/60.030 2.3622/2.3634"	0.016/0.105 .0006/.0041"	2.260 .0890"	26.20 1.031"
3		2729 DRI		55.191/55.210 2.1729/2.1736"	60.000/60.030 2.3622/2.3634"	0.016/0.105 .0006/.0041"	2.385 .0939"	26.20 1.031"
Buje de Biela		4268-Y	STD-SEMI	35.993/36.000	39.000/39.025	0.030/0.047	2.100	34.50
		4268-YA	STD-SEMI	35.993/36.000	39.200/39.225	0.030/0.047	2.200	34.50

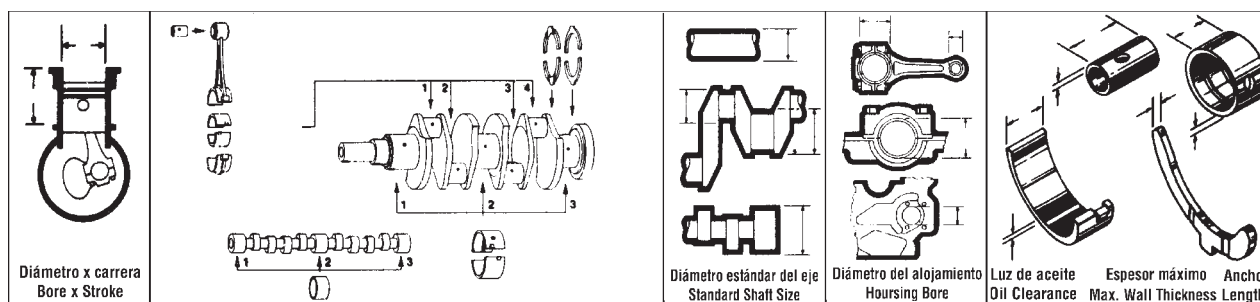
Mercedes Benz *

7 OM-352, OM-352A

6 97 x 128 5675 c.c.	Bielas	6-61915CP	STD-0.25-0.50-0.75-1.00	59.990/60.015	60.000/65.019	0.050/0.095	2.472	32.10
		6-61915CPA	STD-0.25-0.50-0.75-1.00	59.990/60.015	60.000/65.019	0.050/0.095	2.472	29.10
		71-3572/6						
		AEB92168						
Bancada		6296 M	STD-0.25-0.50-0.75-1.00-1.50					
		H-712/7	STD-0.25-0.50-0.75-1.00					
		AEB92124						
1-2-3-5-6-7		61916 CP		87.990/88.010	93.000/93.022	0.050/0.090	2.477	24.10
4 ▲		61917 CPA		87.990/88.010	93.000/93.022	0.050/0.090	2.477	31.81
Bancada		H-032/7	STD-0.25-0.50-0.75-1.00	(Pestaña +0.50 mm)				
		6296 MA	STD-0.25-0.50-0.75-1.00	(Pestaña +0.60 mm)				
1-2-3-5-6-7		61916 CP		87.990/88.010	93.000/93.022	0.050/0.090	2.477	24.10
4 ▲		61917 CPAA		87.990/88.010	93.000/93.022	0.050/0.090	2.477	31.81

* Disponible en las marcas: Federal Mogul, Glyco y AE

▲ Cojinete con pestaña - Flanged bearing



Mercedes Benz *

8 **OM-364 A**

4 97.50 x 133 3971 c.c.	Bielas	71-3572/4 AEB92167	STD-0.25-0.50-0.75-1.00	59.990/60.015	65.000/65.019	0.050/0.095	2.472	29.10
	Bancadas	H-048/5 AEM92173	STD-0.25-0.50-0.75-1.00					
	4	72-3768		87.990/88.010	93.000/93.022	0.050/0.090	2.477	24.10
	1 ▲	72.3602H		87.990/88.010	93.000/93.022	0.050/0.090	2.477	31.81

Mercedes Benz *

9 **OM-366/366A**

6 97.50 x 133 5957 c.c.	Bielas	6-61915 CPA 71-3572/6 AEB92168	STD-0.25-0.50-0.75-1.00	59.990/60.015	65.000/65.019	0.050/0.095	2.472	29.10
	Bancadas	7387 MA 4846 P	STD-0.25-0.50-0.75-1.00					
	6			87.990/88.010	93.000/93.022	0.050/0.090	2.477	24.10
	1 ▲	4847 P		87.990/88.010	93.000/93.022	0.050/0.090	2.477	31.81
	Bancadas	H-048/7 AEM92175	STD-0.25-0.50-0.75-1.00					
	6	72-3768		87.990/88.010	93.000/93.022	0.050/0.090	2.477	24.10
	1 ▲	72-3602 H		87.990/88.010	93.000/93.022	0.050/0.090	2.477	31.81

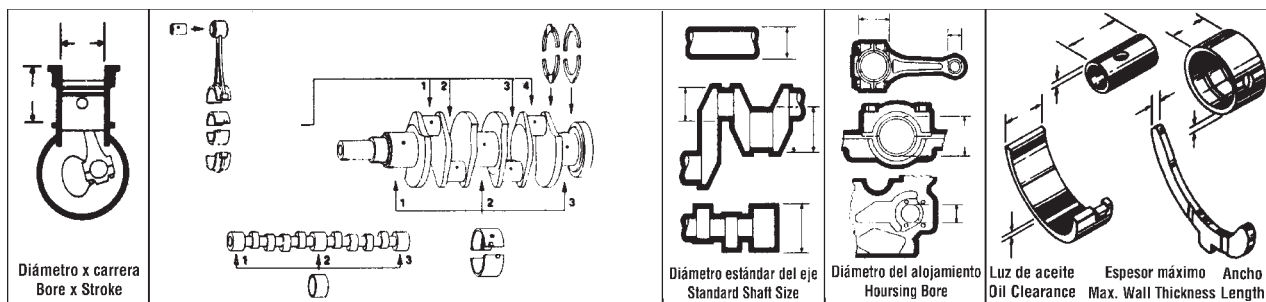
Mercedes Benz

10 **Eng. OM-904 LA 12V**

4 102 x 130 4247 c.c.	Bielas	71-3850/4	STD	69.995/70.015 2.7557/2.7565"	75.000/75.019 2.9528/2.9535"	0.029/0.093 .0976"	2.478	28.30 1.1142"
	Medio cojinete marcado "SPUTTER"= medio cojinete superior.							
	Bancadas	H 0675/5	STD0.25-0.50	85.990/86.010 3.3854/3.3862"	91.000/91.022 3.5827/3.5835"	0.040/0.107 .0974"	2.475	23.30 .09173"
	4	N 146/5	SEMI	67.941/67.960 2.6748/2.6756"	72.000/72.030 2.8346/2.8358"			20.00 .7874"
	1	N 146/5	SEMI	67.941/67.960 2.6748/2.6756"	72.000/72.030 2.8346/2.8358"			29.50 1.1614"

* Disponible en las marcas: Federal Mogul, Glyco y AE

▲ Cojinete con pestaña - Flanged bearing



Mercedes Benz



11

OM-355/5, OM-355/5A

5 128 x 150 9650 c.c.	Bielas	71-2549/5	STD-0.10-0.25-0.50-0.75- 1.00-1.25-1.50	77.980/78.000 3.0701/3.0709"	83.000/83.022 3.2677/3.2686"	0.064/0.136 .0025/.0053"	2.468 .0972"	41.75 1.6437"
	Bancada	H818/6	STD-0.10-0.25-0.50-0.75- 1.00-1.25-1.50					
	1-2-3-5	72-2550		94.980/95.000 3.7394/3.7401"	100.000/100.022 3.9370/3.9379"	0.060/0.132 .0024/.0052"	2.470 .972"	35.15 1.3839"
	4	72-2838		94.980/95.000 3.7394/3.7401"	102.000/102.022 4.0157/4.0166"	0.060/0.132 .0024/.0052"	3.470 .1366"	64.80 2.5512"
	6			94.980/95.000 3.7394/3.7401"	100.000/100.022 3.9370/3.9379"	0.060/0.132 .0024/.0052"	2.470 .972"	52.15 2.0531"
	Levas	N 82/6	SEMI					
	1	13-2553		53.851/53.870 2.1201/2.1290"	58.000/58.030 2.2835/2.2846"	0.030/0.079 .0012//.0031"	2.570 .1012"	33.0 1.2992"
	2	13-2554		53.751/53.770 2.1162/2.1169"	58.000/58.030 2.2835/2.2846"	0.030/0.079 .0012//.0031"	2.570 .1012"	30.0 1.1811"
	3	13-2554		53.551/53.570 2.1083/2.1090"	58.000/58.030 2.2835/2.2846"	0.030/0.079 .0012//.0031"	2.570 .1012"	30.0 1.1811"
	4	13-2555		53.651/53.670 2.1122/2.1130"	58.000/58.030 2.2835/2.2846"	0.030/0.079 .0012//.0031"	2.570 .1012"	55.0 2.1654"
	5	13-2556		53.451/53.470 2.1043/2.1051"	58.000/58.030 2.2835/2.2846"	0.030/0.079 .0012//.0031"	2.570 .1012"	30.0 1.1811"
	6	13-2557		53.251/53.270 2.0965/2.0972"	58.000/58.030 2.2835/2.2846"	0.030/0.079 .0012//.0031"	2.570 .1012"	30.0 1.1811"
	Buje de Biela	55-2538	SEMI	47.993/48.000 1.8895/1.8898"	52.000/52.030 2.0472/2.0484"	0.060/0.077 .0024/.0030"	2.250 .0886"	45.0 1.7717"
	Buje de Balancín	56-2537	SEMI	18.0 .7087"	20.000/20.052 .7874/.7894"	0.060/0.077 .0024/.0030"	1.260 .0496"	24.0 .9449"

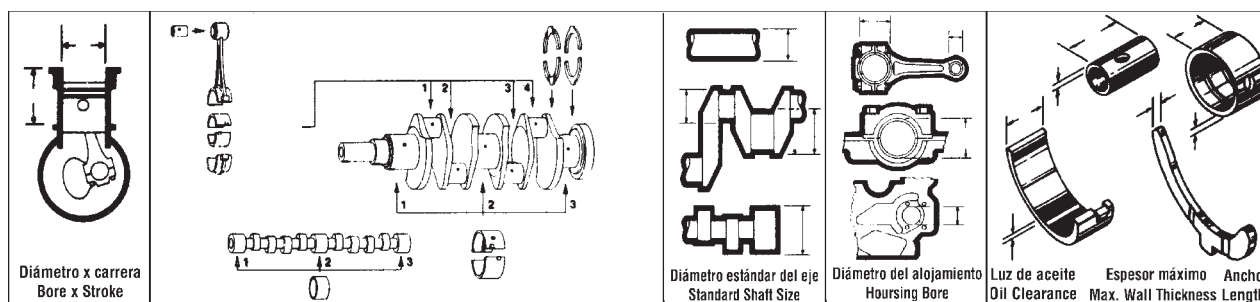
Mercedes Benz



12

OM-355/6A, OM-355/6LA

6 128 x 150 11850 c.c.	Bielas	71-2549/6	STD-0.10-0.25-0.50-0.75- 1.00-	77.980/78.000 3.0701/3.0709"	83.000/83.022 3.2677/3.2686"	0.064/0.136 .0025/.0053"	2.468 .0972"	41.75 1.6437"
	Bancada	H818/7	STD-0.10-0.25-0.50-0.75- 1.00-1.25-1.50					
	1-2-3-5-6	72-2550		94.980/95.000 3.7394/3.7401"	100.000/100.022 3.9370/3.9379"	0.060/0.132 .0024/.0052"	2.470 .972"	35.15 1.3839"
	4	72-2838		94.980/95.000 3.7394/3.7401"	102.000/102.022 4.0157/4.0166"	0.060/0.132 .0024/.0052"	3.470 .1366"	64.80 2.5512"
	7			94.980/95.000 3.7394/3.7401"	100.000/100.022 3.9370/3.9379"	0.060/0.132 .0024/.0052"	2.470 .972"	52.15 2.0531"



Mercedes Benz



12 (continuación)

OM-355/6A, OM-355/6LA

6 128 x 150 11850 c.c.	Levas	N 82/7 13-2553	SEMI	53.851/53.870 2.1201/2.1290"	58.000/58.030 2.2835/2.2846"	0.030/0.079 .0012/.0031"	2.570 .1012"	33.0 1.2992"
	2	13-2554		53.751/53.770 2.1162/2.1169"	58.000/58.030 2.2835/2.2846"	0.030/0.079 .0012/.0031"	2.570 .1012"	30.0 1.1811"
	3	13-2554		53.551/53.570 2.1083/2.1090"	58.000/58.030 2.2835/2.2846"	0.030/0.079 .0012/.0031"	2.570 .1012"	30.0 1.1811"
	4	13-2555		53.351/53.370 2.1004/2.1012"	58.000/58.030 2.2835/2.2846"	0.030/0.079 .0012/.0031"	2.570 .1012"	55.0 2.1654"
	5	13-2555		53.651/53.670 2.1122/2.1130"	58.000/58.030 2.2835/2.2846"	0.030/0.079 .0012/.0031"	2.570 .1012"	30.0 1.1811"
	6	13-2556		53.451/53.470 2.1043/2.1051"	58.000/58.030 2.2835/2.2846"	0.030/0.079 .0012/.0031"	2.570 .1012"	30.0 1.1811"
	7	13-2557		53.251/53.270 2.0965/2.0972"	58.000/58.030 2.2835/2.2846"	0.030/0.079 .0012/.0031"	2.570 .1012"	30.0 1.1811"
Buje de Biela 55-2538				47.993/48.000 1.8895/1.8898"	52.000/52.030 2.0472/2.0484"	0.060/0.077 .0024/.0030"	2.250 .0886"	45.0 1.7717"
Buje de Balancín 56-2537				18.0 .7087"	20.000/20.052 .7874/.7894"	0.060/0.077 .0024/.0030"	1.260 .0496"	24.0 .9449"

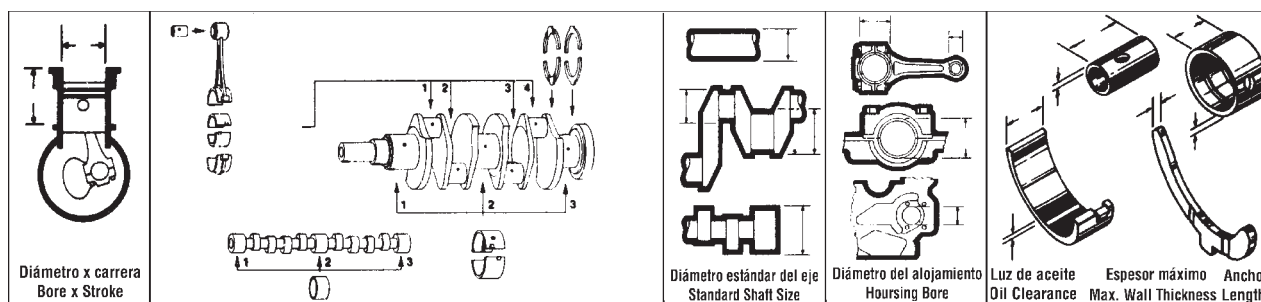
Mercedes Benz



13

OM-449/A/LA

5 128 x 155 9972 c.c.	Bielas	71-3099/5	STD-0.25-0.50	89.980/90.000 3.5425/3.5433"	95.000/95.022 3.7401/3.7410"	0.060/0.122 .0024/.0048"	2.478 .0976"	36.2 1.4252"
	Bancada 1-2-3-4-5	H881/7 72-3100	STD-0.25-0.50-0.75-1.00	103.980/104.000 4.0937/4.0945"	111.000/111.022 4.3702/4.3709"	0.060/0.126 .0024/.0050"	3.478 .1369"	36.0 1.4173"
	6	72-310H		103.980/104.000 4.0937/4.0945"	111.000/111.022 4.3702/4.3709"	0.060/0.126 .0024/.0050"	3.478 .1369"	45.81 1.8035"
	Levas 1	N109/6 13-3102	STD-SEMI	69.910/69.940 2.7523/2.7535"	75.000/75.030 2.9527/2.9539"	0.060/0.120 .0024/.0047"		36.0 1.4173"
	4	13-3103		69.910/69.940 2.7523/2.7535"	75.000/75.030 2.9527/2.9539"	0.060/0.120 .0024/.0047"		28.0 1.1024"
	1	53-3104		69.910/69.940 2.7523/2.7535"	75.000/75.030 2.9527/2.9539"	0.060/0.120 .0024/.0047"		39.0 1.5354"
	Buje de Biela	55-3328	SEMI	45.995/46.000 1.8108/1.8110"	50.600/50.630 1.9921/1.9933"	0.055/0.070 .0022/.0027"		38.7 1.5236"
	Buje de Balancín	56-3042		25.0 .9842"	28.000/28.052 1.1024/1.1044"	0.025/0.054 .0009/.0021"		17.8 .7008"



Mercedes Benz



14

OM-447LA

6 128 x 155 11966 c.c.	Bielas	71-3099/5	STD-0.25-0.50	89.980/90.000 3.5425/3.5433"	95.000/95.022 3.7401/3.7410"	0.060/0.122 .0024/.0048"	2.478 .0976"	36.2 1.4252"
	Bancada	H881/7	STD-0.25-0.50-0.75-1.00	103.980/104.000 4.0937/4.0945"	111.000/111.022 4.3702/4.3709"	0.060/0.126 .0024/.0050"	3.478 .1369"	36.0 1.4173"
	1-2-3-4-5	72-3100						
	6 ▲	72-310H		103.980/104.000 4.0937/4.0945"	111.000/111.022 4.3702/4.3709"	0.060/0.126 .0024/.0050"	3.478 .1369"	45.81 1.8035"
	Levas	N109/7	STD-SEMI	69.910/69.940 2.7523/2.7535"	75.000/75.030 2.9527/2.9539"	0.060/0.120 .0024/.0047"		36.0 1.4173"
	1	13-3102						
	4	13-3103		69.910/69.940 2.7523/2.7535"	75.000/75.030 2.9527/2.9539"	0.060/0.120 .0024/.0047"		28.0 1.1024"
	1	53-3104		69.910/69.940 2.7523/2.7535"	75.000/75.030 2.9527/2.9539"	0.060/0.120 .0024/.0047"		39.0 1.5354"
	Buje de Biela	55-3328	SEMI	45.995/46.000 1.8108/1.8110"	50.600/50.630 1.9921/1.9933"	0.055/0.070 .0022/.0027"		38.7 1.5236"
	Buje de Balancín	56-3042		25.0 .9842"	28.000/28.052 1.1024/1.1044"	0.025/0.054 .0009/.0021"		17.8 .7008"

Mercedes Benz



15

OM-346/A/H, OM-346

6 128 x 140 10803 c.c.	Bielas	71-2234/6	STD-0.25-0.50-0.75-1.00	73.980/74.000 2.9126/2.9134"	80.000/80.019 3.1496/3.1504"	0.059/0.124 0.1169"	2970 0.1169"	45.10 1.7756"
	Bancada	H-641/7	STD-0.25-0.50-0.75-1.00	93.980/94.000 3.7000/3.7008"	100.000/100.022 3.9370/3.9379"	0.059/0.127 .1169"	2.970 .1169"	37.10 1.4606"
	1-2-3-5-6	72-2235						
	4 ▲	72-2238H		93.980/94.000 3.7000/3.7008"	100.000/100.022 3.9370/3.9379"	0.070/0.137 .1167"	2.965 .1167"	68.81 2.7091"
	7	72-2237		93.980/94.000 3.7000/3.7008"	100.000/100.022 3.9370/3.9379"	0.059/0.127 .1167"	2.965 .1167"	55.10 2.1693"

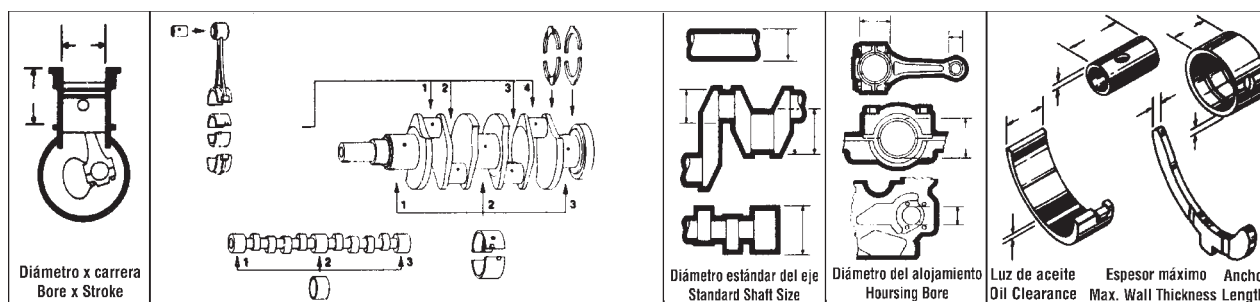
Mitsubishi



1

Galant, Lancer, GTO, Sapporo, 4G32, 4G33 - 1978 / 1982

4 73 x 98 1439 c.c.	Bielas	4-3485 CP	STD-0.25-0.50	44.981/45.001 1.7709/1.7717"	48.003/48.016 1.8899/1.8904"	0.015/0.076 .0006/.0030"	1.494 .0588"	23.60 .929"
	Bancadas	5103 M	STD-0.25-0.50	56.985/57.000 2.2435/2.2441"	61.000/61.021 2.4016/2.4024"	0.015/0.076 .0006/.0030"	1.994 .785"	23.09 .909"
	1-2-4-5	3486 CP						
	76.9 x 86 1597 c.c.	3 ▲		56.985/57.000 2.2435/2.2441"	61.001/61.021 2.4016/2.4024"	0.015/0.076 .0006/.0030"	1.994 .785"	28.96 1.140"



Mitsubishi



2

L100 Pick-Up Van, 2G23, G23

2 70 x 71	Bielas	2-66775 RA	STD-0.25-0.50-0.75	39.990	43.000/43.020	1.496	17.12
	Bancadas	6664 M	STD-0.25-0.50-0.75				
	1-2	66776 RA		43.000	47.000/47.020	1.989	20.13
	3	66777 RA		43.000	47.000/47.020	1.989	23.95

Mitsubishi



3

L-300, Montero, 2.4TA, 4A 55, 4A 55T

4 91.1 x 90 2477 c.c.	Bielas	4-66675 AP	STD-0.25-0.50-0.75	59.979/52.999 2.0858/2.0866"	55.999/56.016 2.2047/2.2055"	0.020/0.067	1.498	24.993 .0590"
	Bancadas	6593 M	STD-0.25-0.50-0.75					
	1-2-4-5	66776 RA		65.979/65.999 2.5976/2.5984"	69.999/70.020 2.7559/2.7567"	0.027/0.093	1.986	24.993 .0782"
	3	66777 RAH		65.979/65.999 2.5976/2.5984"	69.999/70.020 2.7559/2.7567"	0.027/0.093	1.986	30.962 .0782"

MWM



1

Motor Sprint

4 93 x 103 2800 c.c.	Biela	1-4890 P	STD-0.25-0.50-0.75-1.00	54.981/55.000	59.000/59.019	0.030/0.088	1.985	28.00
	Bancada	1-4891 P	STD-0.25-0.50-0.75-1.00	69.981/70.000	75.000/75.019	0.036/0.098	2.482	22.50
6 93 x 103 4200 c.c.		4891 P-I 4891 P-S	Semicanaleta Canaleta total y agujero diám. 5 mm					

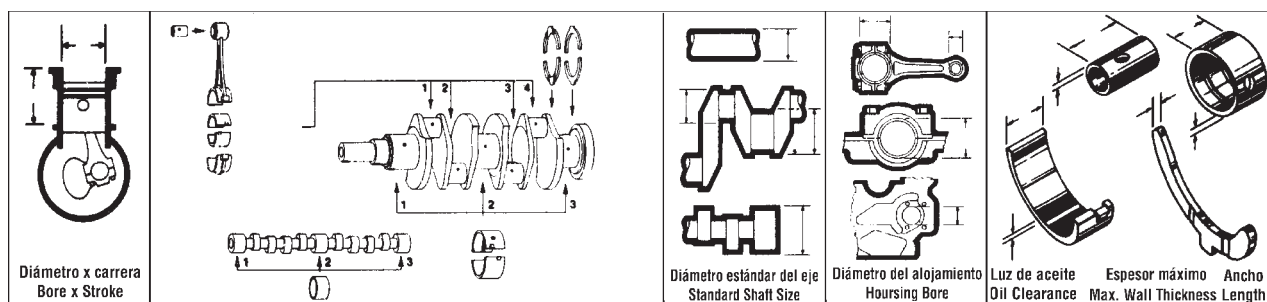
MWM



2

D225, D226, D229, D229

3-4-6 100 x 120 102 x 120 105 x 120	Bielas	1-65805 CP	STD-0.25-0.50-0.75 1.00	57.950/57.590 2.28152/2.2823"	62.000/62.019 2.4409/2.4417"	0.046/0.109 .0018/.0043"	1.992	27.1 1.0669"
	Bancada	1-65811 CP	STD-0.25-0.50-0.75-1.00	64.951/64.970 2.5571/2.5579"	71.000/71.019 2.7953/2.7960"	0.056/0.118 .0022/.0046"	2.987	28.0 1.1024"
	Bancada ▲	1-65812 CP		64.951/64.970 2.5571/2.5579"	71.000/71.019 2.7953/2.7960"	0.056/0.118 .0022/.0046"	2.987	35.88 1.4126"
	Levas	N105-3-4-6	SEMI					
	3-4-6	53-2952		42.940/42.960 1.6905/1.6913"	47.000/47.025 1.8504/1.8514"	0.030/0.016 .0012/.0041"	2.005	26.0 1.0236"
	Buje de Biela	55-2676	STD	31.993/32.000 1.2596/1.2598"	36.000/36.016 1.4173/1.4179"	0.030/0.073 .0012/.0029"	1.985	30.0 .0781"
				31.993/32.000 1.2596/1.2598"	36.000/36.016 1.4173/1.4179"	0.030/0.073 .0012/.0029"	2.300	30.0 .090"



MWM



3 Turbo camión VW 7.110S, TD-229/6, TD-229EC

4 102 x 120 3922 c.c.	Bielas	1-71-3447	STD-0.25-0.50-0.75-1.00	62.950/62.970 2.4783/2.4791"	67.000/67.019 2.6378/2.6385"	0.026/0.091 .0010/.0036"	2.000 .0787"	27.1 1.0669"
	Bancada	1-72-3448	STD-0.25-0.50-0.75-1.00	69.951/69.970 2.7540/2.7547"	75.000/75.019 2.9527/2.95435"	0.046/0.110 .0018/.0043"	2.490 .0980"	28.0 1.1024"
6 102 x 120 5883 c.c.	Axial	A116/4	STD					
	2	04-3449	STD	79.0 3.1102"	95.2 3.7480"		3.47 .1366"	
	2	04-3450	STD	79.0 3.1102"	95.2 3.7480"		3.47 .1366"	
	Levas	4476 Y	SEMI	42.940/42.960 1.6905/1.6913"	47.000/47.025 1.8504/1.8514"	0.030/0.105 .0012/.0041"	2.005 .0789"	26.0 1.0236"
	Buje de Biela	4478 Y	SEMI	34.995/35.000 1.3777/1.3779"	38.000/38.016 1.4961/1.4967"	0.040/0.089 .0016/.0035"	1.487 .585"	32.0 1.2598"

MWM



4 Motor 410

4 103 x 129 4297 c.c.	Biela	1-4885 P	STD-0.25-0.50-0.75-1.00	62.950/62.970	67.000/67.019	0.022/0.087	2.004	30.00
		4885 P-I 4885 P-S	Difiere posición de muesca Difiere posición de muesca					
	Bancada	1-4887 P	STD-0.25-0.50-0.75-1.00	85.942/85.964	92.000/92.022	0.036/0.106	3.000	28.10
		4886 P-I 4886 P-S	Canaleta Total y Agujero diámetro 6 mm - difiere de muesca de sup. Canaleta Total y Agujero diámetro 6 mm a 40°					

MWM



5 Motor 610

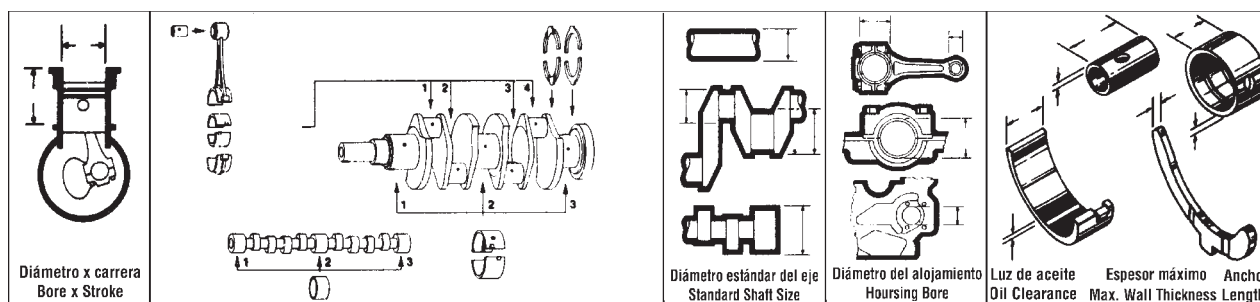
6 103 x 129 6445 c.c.	Biela	1-4885 P	STD-0.25-0.50-0.75-1.00	62.950/62.970	67.000/67.019	0.022/0.087	2.004	30.00
		4885 P-I 4885 P-S	Difiere posición de muesca de inferior Difiere posición de muesca de superior					
	Bancada	1-4886 P	STD-0.25-0.50-0.75-1.00	85.492/85.964	92.000/92.022	0.036/0.106	3.000	28.10
		4886 P-I 4886 P-S	Semicanaleta, muesca Canaleta total y agujero diám. 6 mm a 40° difiere posición de muesca inferior.					

Nissan



1 A12, 120A

4 73 x 70 1171 c.c.	Biela	4-67320 CP	SDT-0.25-0.50-0.75	44.955/44.968 1.7699/1.7704"	48.000/48.011 1.8898/1.8902"	0.024/0.073	1.503 .0592"	19.601 .7717"
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Nissan



1 (continuación)

A12, 120A

4	Bancada	7135 M	STD-0.25-0.50-0.75					
73 x 70 1171 c.c.	1-5	67321 CP		49.949/49.964 1.9665/1.9671"	53.664/53.670 2.1120/2.1130"	0.020/0.078	1.833 .0722"	21.998 .8661"
	2.4	67322`CP		49.949/49.964 1.9665/1.9671"	53.664/53.670 2.1120/2.1130"	0.020/0.078	1.833 .0722"	18.008 .7090"
	3	67323 CP		49.949/49.964 1.9665/1.9671"	53.664/53.670 2.1120/2.1130"	0.020/0.078	1.833 .0722"	26.949 1.0610"

Nissan



2

180 SX, Silvia

4	Biela	4-67320 CP	STD-0.25-0.50-0.75					
85 x 78 1770 c.c.				49.961/49.740 1.9670/1.9675"	52.999/53.012 2.0866/2.0871"	0.020/0.068 .0007/.0026"	1.503 .0592"	21.996 .8660"
	Bancada	7137 M	STD-0.25-0.50					
	1-5	67266 CP		54.942/54.952 2.1630/2.1635"	58.658/58.671 2.3094/2.3099"	0.033/0.081 .0012/.0031"	1.836 .0723"	26.009 1.0240"
	2-4	67267 CP		54.942/54.952 2.1630/2.1635"	58.658/58.671 2.3094/2.3099"	0.033/0.081 .0012/.0031"	1.836 .0723"	24.079 .9480"
	3 ▲	67268 CP		54.942/54.952 2.1630/2.1635"	58.658/58.671 2.3094/2.3099"	0.033/0.081 .0012/.0031"	1.836 .0723"	31.978 1.2590"

Nissan



3

L20, 200 A, Bluebird, Laurel, Fairlady

4	Biela	6-1245 CP	STD-0.25-0.50					
78 x 69.7 1998 c.c.				49.950/44.973 1.7697/1.7706"	47.993/48.013 1.8895/1.8903"	0.020/0.085 .0007/.0034"	1.501 .0591"	18.592 .7320"
	Bancada	7136 M	STD-0.25-0.50-0.75					
	1-7	67231 CP		54.942/54.995 2.1630/2.1636"	58.658/58.671 2.3094/2.3099"	0.028/0.081 .001/.0031"	1.836 .0723"	26.111 1.0280"
	2-3-5-6	67232 CP		54.942/54.995 2.1630/2.1636"	58.658/58.671 2.3094/2.3099"	0.028/0.081 .001/.0031"	1.836 .0723"	24.079 .9480"
	4 ▲	67233 CPA		54.942/54.950 2.1630/2.1634"	58.638/58.658 2.3086/2.3094"	0.020/0.081 .0008/.0031"	1.833 .0723"	31.978 1.2590"

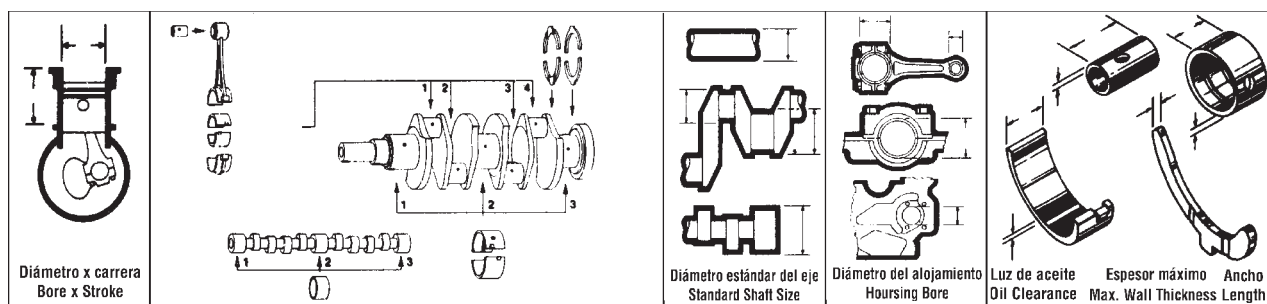
Nissan



4

TD25, TD27 Diesel, Pick Up Pathfinder, Terrano

4	Biela	4-4340 CP	STD-0.25-0.50					
96 x 92 2663 c.c.				56.910/56.940 2.2406/2.2417"	59.980/60.000 2.361/2.3622"	0.020/0.095	1.510 .0594"	26.00 1.0236"
	Bancada	7249 M	STD-0.25-0.50					
				70.900/70.925 2.7913/2.7923"	74.975/75.000 2.9518/2.9528"	0.049/0.124	2.000 .0787"	28.00 1.1024"



Nissan



4 (continuación)

TD25, TD27 Diesel, Pick Up Pathfinder, Terrano

4	Axial	67261 BF	STD			
96 x 92		67261 BFL		78.600	93.800	2.336
2663 c.c.				3.0945"	3.6929"	.0920"
		67261 BFU		78.600	93.800	2.336
				3.0945"	3.6969"	.0920"

Opel



1 1797, K180 - 1974 / 1978

4	Bielas	4-2020 RA	STD-2-10-20-30-40	50.775/50.800 1.9990/2.000"	53.967/53.980 2.1247/1.1252"	0.018/0.068 .0007/.0027"	1.575 .0620"	21.39 .842"
90.49 x 69.85 1797 c.c. (110")	Bancada 1-2-3-4	4123 M 2556 RA	STD-2-10-20-30-40	58.377/58.402 2.2983/2.2993"	63.261/63.287 2.4906/2.4916"	0.007/0.074 .0003/.0029"	2.426 .0955"	20.50 .807"
	5 ▲	2558 RA		58.377/58.402 2.2983/2.2993"	63.261/63.287 2.4906/2.4916"	0.007/0.074 .0003/.0029"	2.426 .0955"	25.63 1.009"
	Levas 1-3	1739 M 2287 DR	STD-5-10-15-20	47.452/47.478 1.8682/1.8692"	51.029/51.079 2.0090/2.0110"	0.020/0.112 .0008/.0044"	1.765 .0695"	21.97 .865"
	2	2288 DR		47.452/47.478 1.8682/1.8692"	50.775/50.825 1.9990/2.0010"	0.020/0.112 .0008/.0044"	1.765 .0695"	21.97 .865"
	Levas 1-2-3	1695 M 2287 DR	STD-5-10-15-20	47.452/47.478 1.8682/1.8692"	50.775/50.825 1.9990/2.0010"	0.020/0.112 .0008/.0044"	1.638 .0645"	21.97 .865"

1er. TIPO

2do TIPO Desde aprox. Julio de 1977

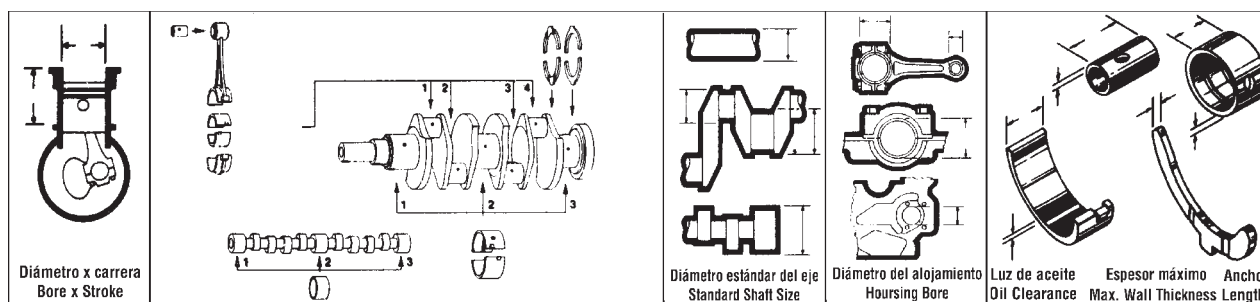
TORSION DE APRIETO: Bielas 4.1 / 4.8 mkg.
Bancada 8.3/ 9.7 mkg.
Tapa de cilindro 12.4/ 13.8 mkg.

Perkins



1 3-152, A3, 152, B3.152, D3.152

3	Bielas	3-2015 RAA	STD-10-20-30-40	57.112/57.125 2.2485/2.2490"	60.833/60.846 2.3950/2.3955"	0.046/0.084 .0018/.0033"	1.829 .0720"	30.23 1.190"
91.44 x 127 2500 c.c. (152")	Bancada 1	6528 M 66466 RA	STD-10-20-30-40	69.812/69.825 2.7485/2.7490"	74.079/74.092 2.9165/2.9170"	0.056/0.109 .0022/.0043"	2.096 .0825"	32.36 1.274"
	2-3	66467 RA		69.812/69.825 2.7485/2.7490"	74.079/74.092 2.9165/2.9170"	0.056/0.109 .0022/.0043"	2.096 .0825"	23.80 .937"
	4	66468 RA		69.812/69.825 2.7485/2.7490"	74.079/74.092 2.9165/2.9170"	0.056/0.109 .0022/.0043"	2.096 .0825"	39.17 1.542"



Perkins

1 (continuación)

3-152, A3, 152, B3.152, D3.152

3	Axial	2783 BF	STD-5-10				3.124	
91.44 x 127 2500 c.c. (152")	Levas						.01230"	
	1			47.473/47.498 1.8690/1.8700"	47.600/47.676 1.8740/1.8770"	0.100/0.200 .0039/.0079"	1.631	38.23
	2			47.219/47.244 1.8590/1.8600"	47.346/47.422 1.8640/1.8670"	0.100/0.200 .0039/.0079"		1.505"
	3			46.711/46.736 1.8390/1.8400"	46.838/46.914 1.8400/1.8470"	0.100/0.200 .0039/.0079"		
	Buje de Biela	2019 Y	SEMI	31.743/31.750 1.2497/1.2500"	34.919/34.995 1.3748/1.3778"	0.013/0.043 .0005/.0017"	1.830	26.85
							.076"	1.0571"

Perkins

2

4-203, A4.203D, D4.203, 4203 Potenciado, T-433

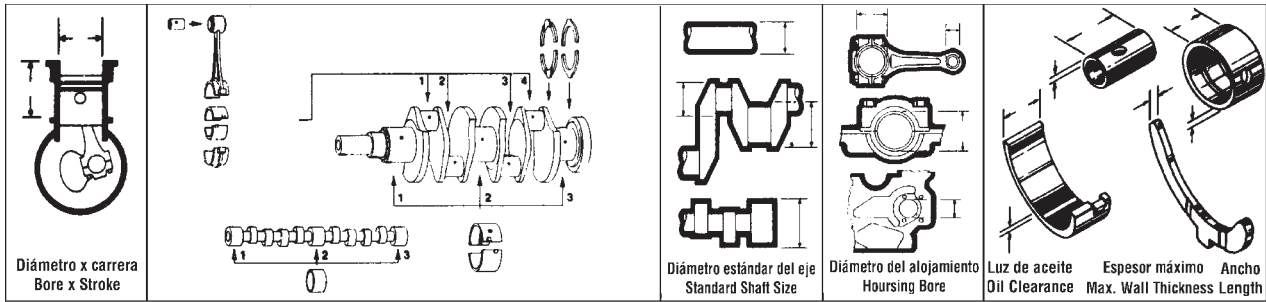
4	Bielas	4-2015 RAA	STD-10-20-30-40	57.112/57.125 2.2485/2.2490"	60.833/60.846 2.3950/2.3955"	0.046/0.084 .0018/.0033"	1.829	30.23
91.44 x 127 3300 c.c.	Bancada	6526 M 66466 RA	STD-10-20-30-40	69.812/69.825 2.7485/2.7490"	74.079/74.092 2.9165/2.9170"	0.080/0.130 .0031/.0051"	2.096	32.36
	1						.0825"	1.274"
	2-4	66467 RA		69.812/69.825 2.7485/2.7490"	74.079/74.092 2.9165/2.9170"	0.080/0.130 .0031/.0051"	2.096	23.80
	3-5	66468 RA		69.812/69.825 2.7485/2.7490"	74.079/74.092 2.9165/2.9170"	0.080/0.130 .0031/.0051"	2.096	39.17
							.0825"	1.542"
	Axial	2783 BF	STD-5-10				3.124	
							.1230"	
	Levas		STD-10					
	1 (*)			47.473/47.498 1.8690/1.8700"	47.549/47.600 1.8720/1.8740"	0.050/0.130 .0020/.0051"	1.631	38.23
	2			47.219/47.244 1.8590/1.8600"	47.346/47.422 1.8640/1.8670"	0.100/0.200 .0039/.0079"		1.505"
	3			46.711/46.736 1.8390/1.8400"	46.838/46.914 1.8400/1.8470"	0.100/0.200 .0039/.0079"		
	Buje de Biela	2019 Y	SEMI	31.743/31.750 1.2497/1.2500"	34.919/34.995 1.3748/1.3778"	0.013/0.043 .0005/.0017"	1.830	26.85
							.076"	1.0571"

Perkins

3

6-305 5004 c.c. - 1957 /

6	Bielas	6-2015 RAA	STD-10-20-30-40-50 STD/30-30/60	57.112/57.125 2.2485/2.2490"	60.833/60.846 2.3950/2.3955"	0.046/0.084 .0018/.0033"	1.829	30.23
91.44 x 127 5004 c.c. (305")	Bancada	6527 M	STD-10-20-30-40 STD/30-30/60	69.812/69.825 2.7485/2.7490"	74.079/74.092 2.9165/2.9170"	0.076/0.127 .0030/.0050"	2.096	32.36
	1	66466 RA					.0825"	1.274"
	2-3	66467 RA		69.812/69.825 2.7485/2.7490"	74.079/74.092 2.9165/2.9170"	0.076/0.127 .0030/.0050"	2.096	23.80
	4	66468 RA		69.812/69.825 2.7485/2.7490"	74.079/74.092 2.9165/2.9170"	0.076/0.127 .0030/.0050"	2.096	39.17
							.0825"	1.542"



Perkins



3 (continuación)

6-305 5004 c.c. - 1957 /

6	Axial	2783 BF	STD-5-10					3.124
91.44 x 127 5004 c.c. (305")	Levas							
	1			47.473/47.498 1.8690/1.8700"	47.600/47.676 1.8740/1.8770"	0.102/0.203 .0040/.0080"		
	2			47.219/47.244 1.8590/1.8600"	47.346/47.422 1.8640/1.8670"	0.102/0.203 .0040/.0080"		
	3			46.711/46.736 1.8390/1.8400"	46.838/46.914 1.8400/1.8470"	0.102/0.203 .0040/.0080"		
	Buje de Biela 4099Y							
	TORSION DE APRIETO: Bielas 5.6/ 6.22 mkg (cadmiadas), 8.99/9.7 mkg (sin cadmiar). Bancada 12.1/ 13.1 mkg (cuando no se utilicen las arandelas de suplemento debajo de la cabeza de los bulones de bancadas, el torque deberá elevarse a 15,2/ 15.9 mkg). Tapa de cilindros 7.6/ 8.3 mkg.							

Perkins



4 **4-236, 4-248, Q-20B4, Turbo TQ-20-B4**

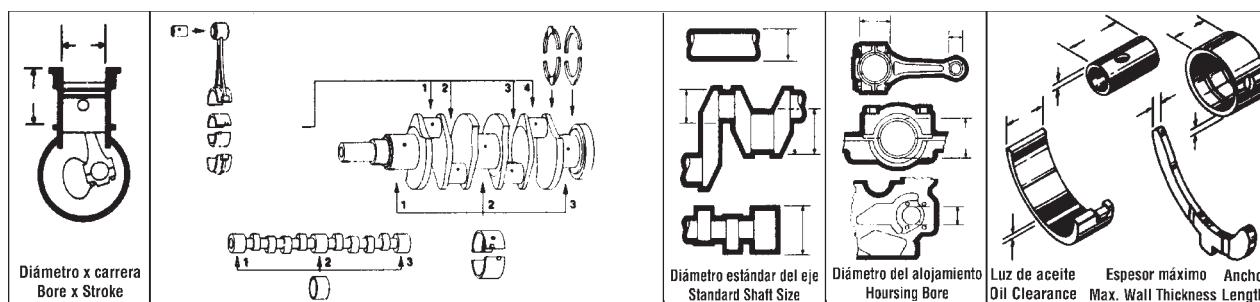
4	Bielas	4-3135 RA	STD-10-20-30-40	63.475/63.487 2.4990/2.4995"	67.208/67.221 2.6460/2.6465"	0.038/0.089 .0015/.0035"	1.844 .0726"	31.88 1.255"
98.43 x 127 3865 c.c.	Bancada	6477 M	STD-10-20-30-40					
4100.97 x 127 4067 c.c.	1-2-5-6	3136 RA		76.162/76.175 2.9985/2.9990"	80.416/80.442 3.1660/3.1670"	0.084/0.157 .0033/.0062"	2.091 .0823"	31.88 1.255"
	3	3137 RA		76.0162/76.0175 2.9985/2.9990"	80.416/80.442 3.1660/3.1670"	0.084/0.157 .0033/.0062"	2.091 .0823"	36.70 1.445"
	Axial	3137 BFA	STD-5-10				2.311 .0910"	
	Levas							
	1		STD-10	50.711/50.737 1.9965/1.9975"	50.800/50.825 2.0000/2.0010"	0.064/0.114 .0025/.0045"	2.390 .0941"	31.88 1.255"
	2			50.457/50.483 1.9865/1.9875"	50.546/50.597 1.9900/1.9920"	0.064/0.140 .0025/.0055"		
	3			50.203/50.229 1.9765/1.9775"	50.292/50.343 1.9800/1.9820"	0.064/0.140 .0025/.0055"		
	4			49.949/49.975 1.9665/1.9675"	50.038/50.089 1.9700/1.9720"	0.064/0.140 .0025/.0055"		
	Buje de Biela 3139 Y 15			SEMI				
				34.920/34.925 1.3748/1.3750"	38.895/38.920 1.5313/1.5323"	0.019/0.043 .0007/.0017"	2.350 .0925"	33.93 1.3358"

Perkins



5 **TQ-20-B4 Turbo**

4	Bielas	4-3135 CPB	STD-10-20-30-40	63.475/63.487 2.4990/2.4995"	67.208/67.221 2.6460/2.6465"	0.038/0.089 .0015/.0035"	1.844 .0726"	31.88 1.255"
98.43 x 127 3863 c.c. (236")								



Perkins

5 (continuación)

TQ-20-B4 Turbo

4 98.43 x 127 3863 c.c. (236")	Bancada	6477 M	STD-10-20-30-40	76.162/76.175 2.9985/2.9990"	80.416/80.442 3.1660/3.1670"	0.084/0.157 .0033/.0062"	2.091 .0823"	31.88 1.255"
	1-2-5-6	3136 RA						
	3	3137 RA		76.0162/76.0175 2.9985/2.9990"	80.416/80.442 3.1660/3.1670"	0.084/0.157 .0033/.0062"	2.091 .0823"	36.70 1.445"
	Axial	3137 BFA	STD-5-10				2.311 .0910"	
Levas								
	1		STD-10	50.711/50.737 1.9965/1.9975"	50.800/50.825 2.0000/2.0010"	0.064/0.114 .0025/.0045"	2.390 .0941"	31.88 1.255"
	2			50.457/50.483 1.9865/1.9875"	50.546/50.597 1.9900/1.9920"	0.064/0.140 .0025/.0055"		
	3			50.203/50.229 1.9765/1.9775"	50.292/50.343 1.9800/1.9820"	0.064/0.140 .0025/.0055"		
	4			49.949/49.975 1.9665/1.9675"	50.038/50.089 1.9700/1.9720"	0.064/0.140 .0025/.0055"		
	Buje de Biela	3139 Y 15	SEMI	34.920/34.925 1.3748/1.3750"	38.895/38.920 1.5313/1.5323"	0.019/0.043 .0007/.0017"	2.350 .0925"	33.93 1.3358"

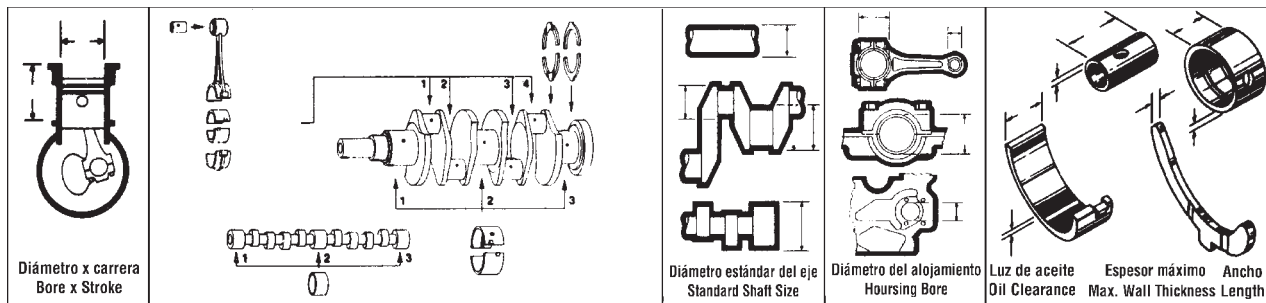
Perkins

6

6-354-4, Q-20-B6, 6-354, 6-354-2, 6-354 Fase 1 y 2

6 98.43 x 127 5798 c.c.	Bielas	6-3135 RA	STD-10-20-30-40	63.475/63.487 2.4990/2.4995"	67.208/67.221 2.6460/2.6465"	0.038/0.089 .0015/.0035"	1.844 .0726"	31.88 1.255"
	Bancada	6467 M	STD-10-20-30-40					
	1-2-3-5-6-7	3136 RA		76.162/76.175 2.9985/2.9990"	80.416/80.442 3.1660/3.1670"	0.094/0.157 .0037/.0062"	2.091 .0823"	31.88 1.255"
	4	3137 RA		76.0162/76.0175 2.9985/2.9990"	80.416/80.442 3.1660/3.1670"	0.094/0.157 .0037/.0062"	2.091 .0823"	36.70 1.445"
	Axial	3137 BFA	STD-5-10				2.311 .0910"	
Levas								
	1		STD-10	50.711/50.737 1.9965/1.9975"	50.800/50.825 2.0000/2.0010"	0.064/0.114 .0025/.0045"	2.390 .0941"	31.88 1.255"
	2			50.457/50.483 1.9865/1.9875"	50.546/50.597 1.9900/1.9920"	0.064/0.140 .0025/.0055"		
	3			50.203/50.229 1.9765/1.9775"	50.292/50.343 1.9800/1.9820"	0.064/0.140 .0025/.0055"		
	4			49.949/49.975 1.9665/1.9675"	50.038/50.089 1.9700/1.9720"	0.064/0.140 .0025/.0055"		
	Buje de Biela	3139 Y 15	SEMI	34.920/34.925 1.3748/1.3750"	38.895/38.920 1.5313/1.5323"	0.019/0.043 .0007/.0017"	2.350 .0925"	33.93 1.3358"

TORSION DE APRIETO: Bielas 9.0/ 9.7 mkg (Cadmiadas).
Bancada 19.3/ 20.7 mkg.
Tapa de cilindro 11.1/ 11.7 mkg (fosfatadas); 9.0/ 9.7 mkg (cadmiadas)



Perkins



7

T 5798 c.c., 6-354 Fase 2 Turbo - 1978 /

6 98.43 x 127 5798 c.c. (354")	Bielas	6-3135 CP/A	STD-10-20-30-40 STD/30-30/60	63.475/63.487 2.4990/2.4995"	67.208/67.221 2.6460/2.6465"	0.038/0.089 .0015/.0035"	1.844 .0726"	31.88 1.255"
	Bancada	6467 M	STD-10-20-30-40					
	1-2-3-5-6-7	3136 RA		76.162/76.175 2.9985/2.9990"	80.416/80.442 3.1660/3.1670"	0.094/0.157 .0037/.0062"	2.091 .0823"	31.88 1.255"
	4	3137 RA		76.0162/76.0175 2.9985/2.9990"	80.416/80.442 3.1660/3.1670"	0.094/0.157 .0037/.0062"	2.091 .0823"	36.70 1.445"
	Axial	3137 BFA	STD-5-10				2.311	
Levas								
	1			50.711/50.737 1.9965/1.9975"	50.800/50.825 2.0000/2.0010"	0.064/0.114 .0025/.0045"		
	2			50.457/50.483 1.9865/1.9875"	50.546/50.597 1.9900/1.9920"	0.064/0.140 .0025/.0055"		
	3			50.203/50.229 1.9765/1.9775"	50.292/50.343 1.9800/1.9820"	0.064/0.140 .0025/.0055"		
	4			49.949/49.975 1.9665/1.9675"	50.038/50.089 1.9700/1.9720"	0.064/0.140 .0025/.0055"		

Buje de Biela 4241Y

TORSION DE APRIETO: Bielas 9.0/ 9.7 mkg (Cadmiadas).
Bancada 19.3/ 20.7 mkg.
Tapa de cilindro 11.1/ 11.7 mkg (fosfatadas); 9.0/ 9.7 mkg (cadmiadas)

Perkins



8

TQ-20-B6, Turbo 6-354 Fase 4, Fase 4 Turbo - 1982 /

6 98.43 x 127 5798 c.c. (354")	Bielas	6-3135 CPB	STD-10-20-30-40	63.475/63.487 2.4990/2.4995"	67.208/67.221 2.6460/2.6465"	0.038/0.089 .0015/.0035"	1.844 .0726"	31.88 1.255"
	Bancada	7050 MRA	STD-10-20-30-40					
	1-2-3-5-6-7	66786 RA		76.162/76.175 2.9985/2.9990"	80.416/80.442 3.1660/3.1670"	0.094/0.157 .0037/.0062"	2.091 .0823"	31.88 1.255"
	4	66787 RA		76.0162/76.0175 2.9985/2.9990"	80.416/80.442 3.1660/3.1670"	0.094/0.157 .0037/.0062"	2.091 .0823"	36.70 1.445"
	Axial	3137 BFA	STD-5-10				2.311 .0910"	
Buje de Biela 3139 Y 15				SEMI				
				34.920/34.925 1.3748/1.3750"	38.895/38.920 1.5313/1.5323"	0.019/0.043 .0007/.0017"	2.350 .0925"	33.93 1.3358"

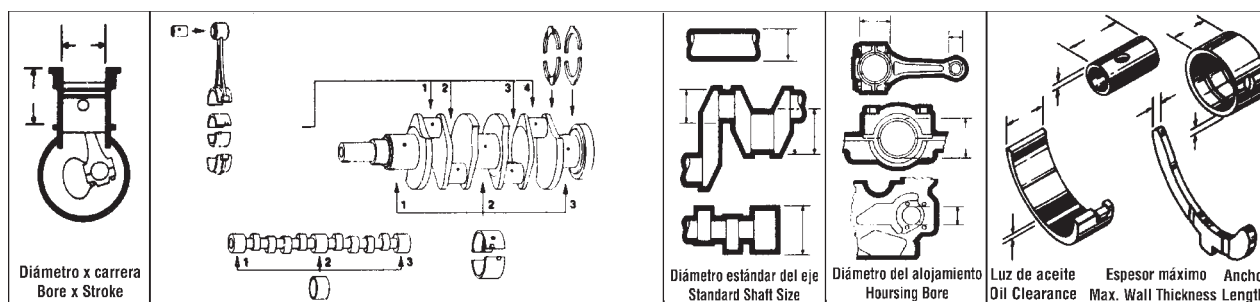
Peugeot



1

106, 205, 306, 405, TU3F, TU5F, TU5JP, TU5JPH

4 75 x 77 1360 c.c. 4 78.5 x 82 1587 c.c.	Bielas	4-62045 RA	STD-0.30-0.50	44.975/44.991 1.7707/1.7713"	48.655/48.671 1.9155/1.9162"	0.024/0.081 .0024/0.081"	1.820 .0717"	19.75 .7776"
	Bancadas	7196 M	STD-0.30					
	2-3-4	4151 RA		49.962/49.981 1.9670/1.9678"	53.712/53.731 2.1146/2.1154"	0.023/0.086 .0023/0.086"	1.854 .0730"	17.29 .6807"
	1-5	4152 RA		49.962/49.981 1.9670/1.9678"	53.712/53.731 2.1146/2.1154"	0.023/0.086 .0023/0.086"	1.854 .0730"	17.29 .6807"
	Axial	62047 BFA	STD-0.10-0.15-0.20	55.250/55.500 2.1752/2.1850"	71.450/71.700 2.8130/2.8228"		2.450 .0965"	



Peugeot

2 205 D/DT, 309D, 405 TD, Partner, XUD7, 7A, 9A, 7TE, 7TEY

4 88 x 88 1796 c.c.	Bielas	71-3730/4	STD-0.30	49.984/50.000 1.9679/1.9685"	53.695/53.708 2.1140/2.1145"	0.020/0.075	1.838 .0724"	20.00 .7874"
	71-3730 = Eng./Mot XUD 7 TE / XUD 9 TE / XUD 9 TF							
	Bancadas	6610 M	STD-0.30	59.981/60.000 2.3615/2.3622"	63.731/63.750 2.5091/2.5098"	0.041/0.104	1.845 .0726"	20.29 .7988"
	Axial	4141 AF	STD	67.050/67.300 2.6398/2.6496"	83.050/83.300 3.2697/3.2795"		1.880 .0740"	
4141 AF = Eng./Mot. XUD 7 TE / XUD 9 TE / XUD 9 TF								

Peugeot

3 205, 405 XU5-1.6

4 83 x 73 1580 c.c.	Biela	4-66720 RA	STD-0.30	44.975/44.991 1.7701/1.7713"	48.655/48.671 1.9156/1.9162"	0.024/0.080	1.820 .0717"	20.25 .7972"
	Bancada	7044 M	STD-0.30	59.990/60.000 2.3618/2.3622"	63.718/63.727 2.5086/2.5089"	0.028/0.071	1.845 .0726"	20.29 .7988"
	Traba derecha							
	Axial	66721 BF	STD-0.10-0.15	67.050/67.300 2.6398/2.6496"	83.050/83.300 3.2697/3.2795"		2.330 .0917"	

Peugeot

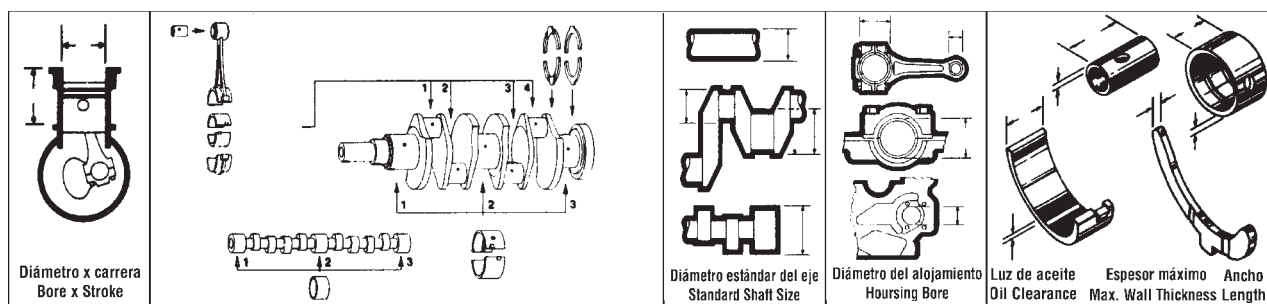
4 205, 306, 405, 806, Boxer Partner, 1.9 - DWB

4 83 x 88 1905 c.c.	Biela	4-66725 RA	STD-0.30	49.984/50.000 1.9679/1.9685"	53.695/53.708 2.1140/2.1145"	0.035/0.088	1.830 .0720"	20.50 .7894"
	Bancada	6610 M	STD-0.30	59.981/60.000 2.3615/2.3622"	63.371/63.750 2.5091/2.5098"	0.041/0.103	1.845 .0726"	20.29 .7988"
	Traba izquierda							
	Axial	66721 BF	STD-0.10-0.15	67.050/67.300 2.6398/2.6496"	83.050/83.300 3.2697/3.2795"		2.330 .0917"	

Peugeot

5 306, 405, 406, 806, Partner, L8, XU7JP

4 83 x 81.4 1762 c.c.	Biela	4-4300 RA	STD-0.30	44.975/44.991 1.7707/1.7713"	48.655/48.671 1.9156/1.9162"	0.020/0.070	1.825 .0719"	22.75 .8957"
	Bancada	7044 M	STD-0.30	59.990/60.000 2.3618/2.3622"	63.718/63.727 2.5086/2.5089"	0.028/0.071	1.845 .0726"	20.29 .7988"
	Traba derecha							
	Axial	66721 BF	STD-0.10-0.15	67.050/67.300 2.6398/2.6496"	83.050/83.300 3.2697/3.2795"		2.330 .0917"	



Peugeot



6 205, 405, 1.9 XU9

4 83 x 88 1905 c.c.	Biela	4-1305 RA	STD-0.30	49.984/50.000 1.9679/1.9685"	53.695/53.708 2.1140/2.1145"	0.023/0.076	1.836 .0723"	20.25 .7972"
	Bancada	7022 M	STD-0.30	59.990/60.000 2.3618/2.3622"	63.718/63.727 2.5086/2.5089"	0.028/0.071	1.845 .0726"	20.29 .7988"
	Traba derecha							
	Axial	66721 BF	STD-0.10-0.15	67.050/67.300 2.6398/2.6496"	83.050/83.300 3.2697/3.2795"		2.330 .0917"	

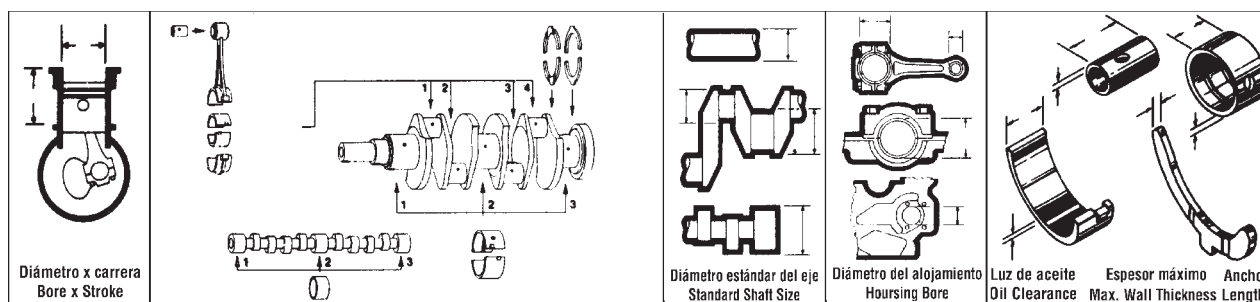
Peugeot



7 404, 504, 505, XC7, XC6-A, XMA-7, XM7, XN1, XN1-TN - 1965 /

4 84 x 73 1618 c.c.	Bielas	4-2865 RA	STD-0.30-0.50-0.80-1.00 1.30	49.981/50.000	53.655/53.674	0.017/0.073	1.819	23.40
	Bancadas	4432 M	STD-0.30-0.50-0.80-1.00					
85 x 73 1657 c.c.	1	2866 RA		59.401/59.419	63.227/63.246	0.033/0.085	1.889	29.50
	2	3077 RAA		58.548/58.573	63.384/62.403	0.033/0.085	1.889	21.50
	3	3078 RAA		57.174/57.189	61.000/61.019	0.033/0.085	1.889	29.50
	4	3079 RAA		56.140/56.165	59.976/59.995	0.033/0.085	1.889	21.50
84 x 81 1796 c.c.	5	2869 RA		51.166/51.181	54.992/55.011	0.033/0.085	1.889	29.35
	Axial (2 MEDIOS)	2869 BF	STD-5-10				2.330	
MODELOS 404/504 A PARTIR DE 1975 (BANCADA Nº 5 MODIFICADA)								
	Bancadas	6354 M	STD-0.30-0.50-0.80					
	1	2866 RA		59.401/59.419	63.227/63.246	0.033/0.085	1.889	29.50
	2	3077 RAA		58.548/58.573	62.384/62.403	0.033/0.085	1.889	21.50
	3	3078 RAA		57.174/57.189	61.000/61.019	0.033/0.085	1.889	29.50
	4	3079 RAA		56.140/56.165	59.976/59.995	0.033/0.085	1.889	21.50
	5	62099 RAA		54.905/54.920	58.731/58.750	0.033/0.085	1.889	29.35
	Axial (4 MEDIOS)	62099 BF	STD-5-10				2.300	
APARTIR DEL MOTOR Nº 132000								
	Levas							
	1			47.925/47.950	48.000/48.039	0.050/0.114		
	2			45.925/45.950	46.000/46.039	0.050/0.114		
	3			43.925/43.950	44.000/44.039	0.050/0.114		
	Levas							
	1			47.915/47.940	48.000/48.039	0.060/0.124		
	2			47.915/47.940	48.000/48.039	0.060/0.124		
	3			47.915/47.940	48.000/48.039	0.060/0.124		
Buje de Biela 4275 X								

TORSION DE APRIETO: Bielas (hasta motor Nº 97558) 4/ 4.5 mkg (desde motor Nº 97559) 3.5/ 4 mkg. Bancada 7/8 mkg.
Tapa de cilindros 7/8 mkg.
Motor 1796 c.c. Bielas 3.5/4 mkg. Bancada 7.5 mkg.
Tapa de cilindros preaprieto 4 mkg, final 7 mkg.
Motor 1971 c.c. Bielas 3.5/4 mkg. Bancada 7.5 mkg.
Tapa de cilindros preaprieto 5 mkg, final aflojar bulones y reapretar a 2 mkg + 90°



Peugeot

8

306, 406 2.0 TD, DWIO TD/ATED

4 85 x 88 1997 c.c.	Bielas	71-3952/4	STD	49.984/50.000 1.9679/1.9685"	53.695/53.708 2.1140/2.1145"	0.020/0.073	1.836 .0724"	20.00 .7874"
	Medio cojinete marcado "SPUTTER" = medio cojinete superior							
	Bancadas	6610 M	STD-0.30	59.981/60.000 2.3615/2.3622"	63.731/63.750 2.5091/2.5098"	0.041/0.104	1.845 .0726"	20.29 .7988"
	Axial	66721 BF	STD-0.10-0.15	67.050/67.300 2.6398/2.6496"	83.050/83.300 3.2697/3.2795"		2.330 .0917"	

Peugeot

9

306, 405, 806, Boxer-2.0 XU10

4 86 x 86 1998 c.c.	Biela	4-4150 RA	STD-0.30	49.984/50.000 1.9679/1.9685"	63.695/63.708 2.1140/2.1145"	0.027/0.080	1.834 .0722"	21.950 .8642"
		4-4305 RA	STD-0.30	49.984/50.000 1.9679/1.9685"	63.695/63.708 2.1140/2.1145"	0.027/0.080	1.834 .0722"	20.250 .7972"
	Bancada	6610 M	STD-0.30	59.981/60.000 2.3618/2.3622"	63.731/63.750 2.5091/2.5098"	0.041/0.103	1.845 .0726"	20.29 .7988"
	Traba izquierda							
	Axial	4141 AF	STD	67.050/67.300 2.6398/2.6496"	83.050/83.300 3.2697/3.2795"		1.880 .0740"	

Peugeot

10

604 3.0 - V6 - Z7X - 1978 / 1980

6 88 x 73 2664 c.c.	Biela	6-66310 RA	STD-0.30-0.50	52.267/52.286	56.000/56.019		1.848	17.65
	Bancada	6443 M	STD-0.30	70.043/70.062	74.000/74.019		1.967	23.47
	Axial	66311 BF	STD-0.10-0.15				2.350	

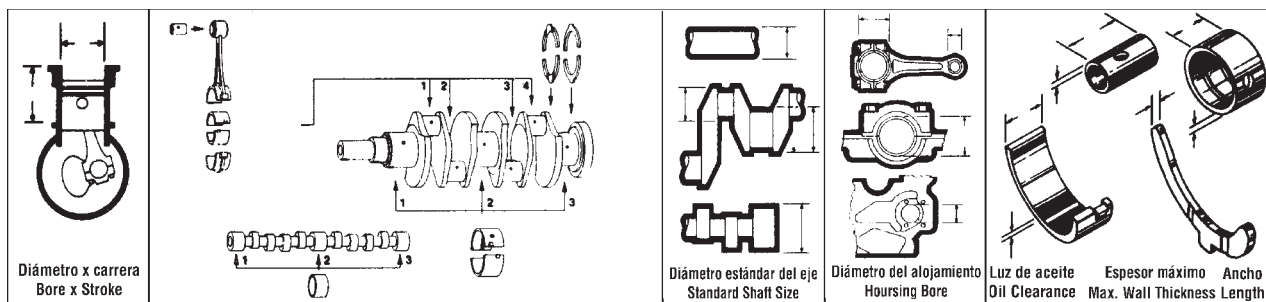
Rastrojero

1

42 HP, NP 6, D4M 1.8 - 1952 / 1966

4 78 x 92 1758 c.c.	Bielas	4-60275 RA	STD-10-20-30-40-50	49.984/50.000	54.000/54.019	0.030/0.060	1.990	25.00
	Bancadas	4509 M	STD-10-20-30-40-50					
	1	60276 CP		54.981/55.000	59.000/59.019	0.040/0.070	1.986	38.90
	2-3	60277 RA		54.981/55.000	59.000/59.019	0.040/0.070	1.986	28.10
	Levas	1301 M	STD-5-10-15-20					
	1-2-3	60271 DR		40.950/40.975	45.000/45.025	0.025/0.075	2.105	28.00
	Buje de Biela 4059 Y							

TORSION DE APRIETO: Bielas 4,5 mkg. Bancada 10 mkg.
Tapa de cilindros 6 y 10 mkg (en frío). Primeros 500 mkg 10 mkg (en caliente)



Rastrojero



2

NC62, NP66, P63, P64, C66, P67, D301 E1 - 1965 / 1971

4 78 x 94 1797 c.c.	Bielas	4-61910 RA	STD-0.25-0.50-0.75-1.00	49.984/50.000	54.000/54.019	0.040/0.060	1.990	24.10
	Bancadas	6288 M	STD-0.25-0.50-0.75-1.00					
	1	61911 CP		63.981/64.000	68.000/68.019	0.050/0.070	1.990	34.90
	2-3	61912 RA		63.981/64.000	68.000/68.019	0.050/0.070	1.990	25.10
	Levas	1345 M	STD-0.25-0.50-0.75					
	1-2-3	61911 DR		40.950/40.975	45.000/45.025	0.025/0.075	2.105	24.10

TORSION DE APRIETO: Bielas 4.5 mkg. Bancada 12 mkg.
Tapa de cilindros 6 y 10 mkg (en frío). Primeros 500 mkg 10 mkg
(en caliente)

Renault



1

Dauphine, Gordini, 4L, 4F, 845 c.c. - 1960 / 1969

4 58 x 80 845 c.c.	Bielas	4-2390 RA	STD-2-10-20-30-40-50	37.959/37.975	41.128/41.139	0.015/0.054	1.569	21.18
	Bancadas	4072 M	STD-2-10-20-30-40-50					
	1	2391 RA		39.975/39.991	44.000/44.013	0.026/0.059	2.000	27.13
	2	2392 RA		39.975/39.991	44.000/44.013	0.026/0.059	2.000	30.12
	3	2393 RA		39.975/39.991	44.000/44.013	0.026/0.059	2.000	32.87
	Axial	2392 BF	STD-5-10				2.000	
	Levas							
	1			37.888/37.950	38.000/38.025	0.050/0.137		18.00
	2			37.888/37.950	38.000/38.025	0.050/0.137		14.50
	3			37.888/37.950	38.000/38.025	0.050/0.137		18.00
	4			37.888/37.950	38.000/38.025	0.050/0.137		12.00

Buje de Biela 4054 Y

TORSION DE APRIETO: Bielas 3/ 3.5 mkg. Bancada 5.5/ 7 mkg
Tapa de cilindros 6 mkg

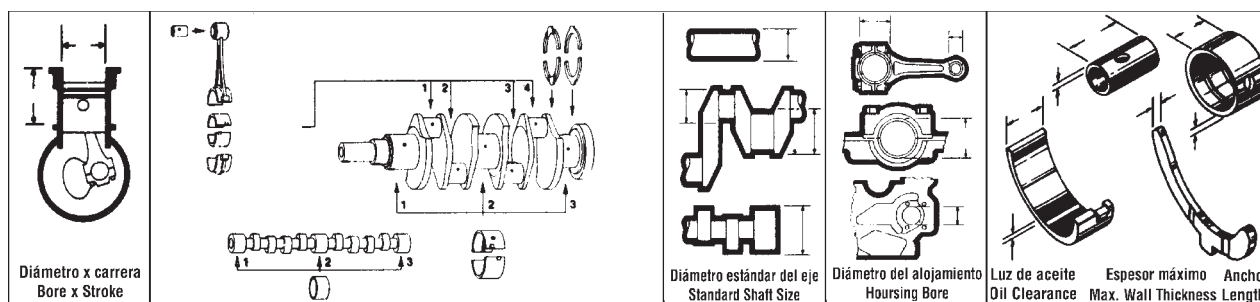
Renault



2

4L, 4S, 4F, R6, R12, M1000, M1100, M1300, M1400 -1970 / 1982

4 65 x 77 1020 c.c.	Bielas	4-2950 RAA	STD-0.25-0.50-0.75-1.00	43.960/43.980	47.614/47.625	0.022/0.061	1.806	17.75
	Bancadas	6286 M	STD-0.25-0.50-0.75-1.00					
	1-3	2951 RAA		45.980/46.000	49.874/49.890	0.020/0.058	1.932	24.50
	2-4-5	2952 RAA		45.980/46.000	49.874/49.890	0.020/0.058	1.932	19.50
69 x 77 1118 c.c.	Axial	2952 BF	STD-5-140				2.410	
73 x 77 1289 c.c.	Levas (*)	1399 M	RENAULT 4, 6 STD-0.25-0.50-0.75					
76 x 77 1397 c.c.	1	2954 DR		34.484/34.500	38.000/38.025	0.050/0.100	1.722	19.25
	2-3-4			37.925/37.950	38.000/38.025	0.050/0.100		



Renault



2 (continuación)

4L, 4S, 4F, R6, R12, M1000, M1100, M1300, M1400 -1970 / 1982

4	RENAULT 12			
65 x 77 1020 c.c.	Levas ^(?) 1-2-3-4	37.925/37.950	38.000/38.025	0.050/0.100
69 x 77 1118 c.c.				
73 x 77 1289 c.c.				
76 x 77 1397 c.c.				

TORSION DE APRIETO: Bielas 4.5 mkg. Bancada 6 mkg.
Tapa de cilindros 6 mkg.
Las posiciones 2-3-4 no usan originalmente cojinetes de levas.
(¹) No usa cojinete de levas en ninguna de las posiciones.
(²) El juego 1399 M está compuesto por un sólo cojinete (Nº 1)

Renault



3

Clio, Twingo, Express, Kangoo 1.2 - D7F

4	Biela	4-4405 RA	STD	39.984/40.000	43.639/43.650	0.020/0.065	1.813	16.25
69 x 76.8 1149 c.c.	Bancada	7264 M	STD					
		4407 RAH		43.990/44.010	47.612/47.625	0.020/0.064	1.798	23.93
		4406 RA		43.990/44.010	47.612/47.625	0.020/0.064	1.798	18.80

Renault



4

9, 11, 12, 18, 1.4 - 1.6, Twingo 1.3, 19 RN, Re 1.6

4	Biela	4-2950 RA	STD-0.25-0.50-0.75-1.00	43.964/43.980 1.7309/1.7315"	47.614/47.625 1.8746/1.8750"	0.020/0.066	1.810	20.75
74 x 72 1239 c.c.								
76 x 77 1397 c.c.	Bancada	6444 M	STD-0.25-0.50-0.75	54.785/54.805 2.1569/2.1577"	58.731/58.750 2.3122/2.3130"	0.026/0.090	1.950	19.50
77 x 84 1565 c.c.								
	Axial	60476 BF	STD-0.10-0.15	60.500/60.750 2.3918/2.3917"	79.750/80.000 3.1398/3.1496"		2.850	.1122"

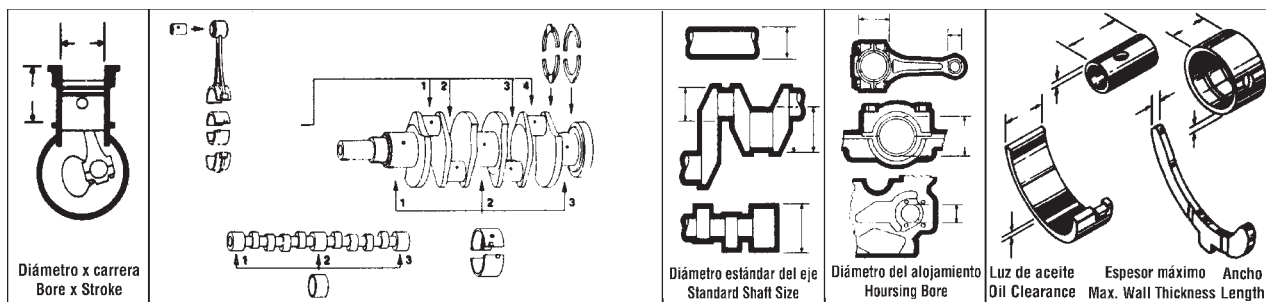
Renault



5

Clio, Kangoo, Express- C3J, E6J, E7J - 1.4

4	Biela	4-2950 RAA	STD-0.25-0.50-0.75	43.964/43.980 1.7309/1.7315"	47.614/47.625 1.8746/1.8750"	0.022/0.074	1.806	17.75
75.8 x 77 1367 c.c.								
	Bancada	AEM5544	STD-0.25-0.50-0.75	54.795/54.805 2.1573/2.1577"	58.731/58.750 2.3122/2.3130"	0.020/0.060 .0008/.0024"	1.950	19.50
	Axial	60478 BF	STD-0.05-0.10-0.15	60.500/60.750 2.3819/2.3917"	79.750/80.000 3.1398/3.1496"		2.850	.1122"



Renault



**Express, Megane, Clio, 1.9, Kangoo Diesel 1.9 - F8Q, F9Q,
Clio, 1.9, Megane, 1.8, 2.0 - F3P - F3R**

6

4 82.7 x 93 1794 c.c.	Biela	4-66740 RA	STD-0.25	48.000/48.020 1.8898/1.8906"	51.587/51.600 2.0310/2.0315"	0.0311/0.089	1.768 .0696"	20.50 .8071"
80 x 93 1870 c.c.	Bancada	7175 M	STD-0.25-0.50	54.785/54.795 2.1569/2.1573"	58.731/58.744 2.3122/2.3128"	0.045/0.094	1.945 .0766"	18.50 .7283"
	Axial	66741 AF	STD-0.10	60.750 2.3918"	80.000 3.1497"		2.350 .0926"	

Renault



**Express, Megane, Clio, 1.9, Kangoo Diesel 1.9 - F8Q, F9Q,
Clio, 1.9, Megane, 1.8, 2.0 - F3P - F3R**

7

4 82.7 x 93 1794 c.c.	Biela	4-66740 RA	STD-0.25	48.970/48.990 1.9280/1.9287"	51.587/51.600 2.0310/2.0315"	0.031/0.089	1.768 .0696"	20.50 .8071"
80 x 93 1870 c.c.	Bancada	7330 M	STD-0.25-0.50	54.785/54.795 2.1569/2.1573"	58.731/58.744 2.3122/2.3128"	0.045/0.094	1.945 .0766"	18.50 .7283"
	Axial	66741 AF	STD-0.10	60.750 2.3918"	80.000 3.1497"		2.350 .0926"	

Renault



**Megane, Clio, Williams 2.0 16V - F7R
Megane, Scenic, Laguna 1.9 TD - F7Q**

8

4 82 x 83.5 1764	Biela	4-4060 CP	STD-0.25-0.50	48.000/48.020 1.8898/1.8906"	51.587/51.600 2.0310/2.0315"	0.031/0.089	1.768 .0696"	20.50 .8071"
82.7 x 93 1998 c.c.	Bancada	7175 M	STD-0.25-0.50	54.785/54.795 2.1569/2.1573"	58.731/58.744 2.3122/2.3128"	0.045/0.094	1.945 .0766"	18.50 .7283"
	Axial	66741 AF	STD-0.10	60.750 2.3918"	80.000 3.1497"		2.350 .0926"	

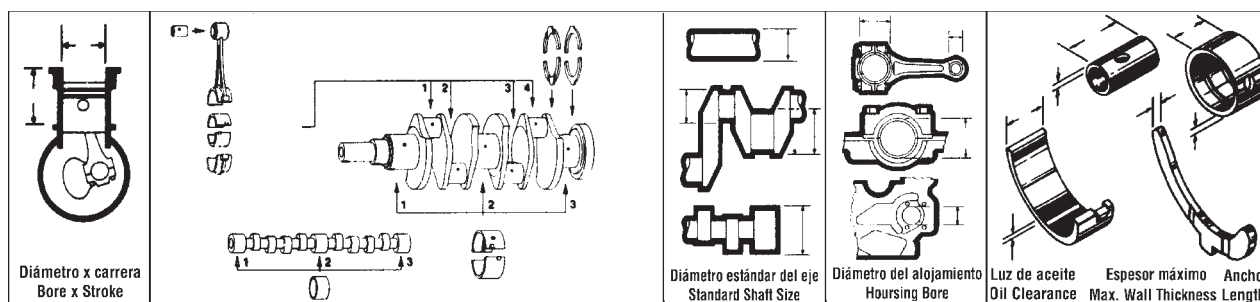
Renault



R18, Coupe Fuego, Trafic, M2000, 829 - 1981

9

4 88 x 82 1995 c.c.	Bielas	4-66600 RA	STD-0.25-0.50-0.75	52.267/52.286	56.000/56.019	0.035/0.080	1.852	18.75
	Bancadas	7016 M	STD-0.25-0.50-0.75					
	1-2-3-4-5	66646 RA		62.861/62.880	66.680/66.687	0.037/0.069	1.879	23.50
	Axial	66506 BF	STD-5-10				2.85	



Renault



10

Laguna RXE 16V, Volvo N7Q

4 83 x 90 1948 c.c.	Biela	71-3754/4	STD-0.25-0.50	48.984/50.000 1.9679/1.9685"	53.000/53.013 2.0866/2.0871"	0.020/0.063	1.496 .0589"	19.50 .7677"
	Bancada	H 046/5	STD-0.25-0.50					
	4	02-3755		64.987/65.000 2.5585/2.5591"	69.000/69.014 2.7165/2.7171"	0.020/0.064	1.994 .0785"	19.60 .7717"
	1	02-3756 H		64.987/65.000 2.5585/2.5591"	69.000/69.014 2.7165/2.7171"	0.020/0.056	1.998 .0787"	25.88 1.0189"

Renault



11

18, 21, Trafic Diesel, M2200, 851 - 1987 /

4 88 x 89 2165 c.c.	Bielas	4-66645 RAB	STD-0.25-0.50-0.75	56.267/56.286	60.005/60.019		1.852	20.75
	Bancada	7016 M	STD-0.25-0.50-0.75					
	1-2-3-4-5	66646 RA		62.861/62.880	66.680/66.687	0.037/0.069	1.879	23.50
	Axial	66506 BF	STD-5-10				2.85	

Rover



1

100, 114, 200, 214, 200, 214, 214 Cabrio, 400, 414 16V

4 75 x 79 1396 c.c.	Bielas	4-4130 RA	STD.010-.020	43.000/43.007 1.6929/1.6932"	46.666/46.687 1.8372/1.8381"	0.020/0.064	1.824 .0718"	21.61 .8508"
	Bancadas	7192 M	STD.010-.020					
	1-3-5	4131 RA		47.993/48.000 1.8895/1.8898"	51.579/51.600 2.0307/2.0315"	0.020/0.054	1.789 .0704"	17.45 .6870"
	2-4	4132 RA		47.993/48.000 1.8895/1.8898"	51.579/51.600 2.0307/2.0315"	0.020/0.054	1.789 .0704"	17.45 .6870"
	Axial	4131 AF	STD	56.800/57.050 2.2362/2.2461"	73.300/73.550 2.8858/2.8957"		2.400 .0945"	

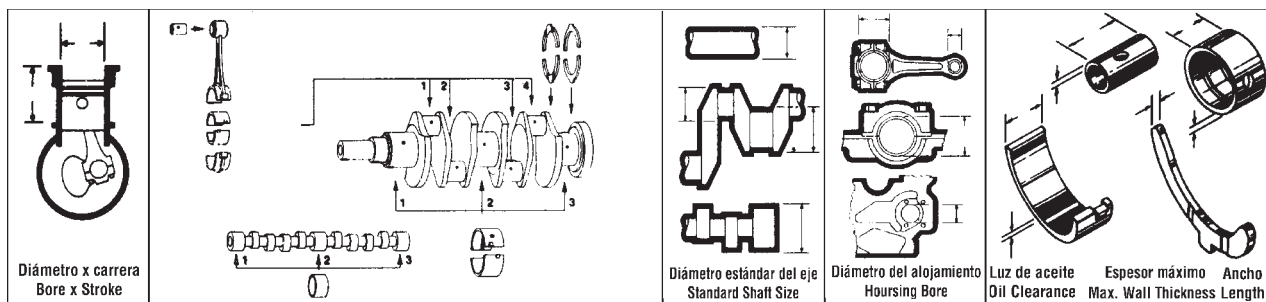
Rover



2

New Series 200, 220, 400, 420, 620, 800, 820 16V

4 84.5 x 88.9 1994 c.c.	Bielas	4-8180 RA	STD.010-.020-.030	47.648/47.660 1.8759/1.8764"	51.330/51.349 2.0209/2.0216"	0.020/0.066	1.830 .0720"	19.85 .7815"
	Bancadas	8151 M	STD.010-.020-.030					
	1-3-5	8181 RA		54.013/54.026 2.1265/2.1270"	57.683/57.696 2.2710/2.2715"	0.020/0.048	1.830 .0720"	28.70 1.1299"
	2-4	8182 RA		54.013/54.026 2.1265/2.1270"	57.683/57.696 2.2710/2.2715"	0.020/0.048	1.830 .0720"	19.56 .7701"
	Axial	65077 BF	STD	59.690 2.3500"	74.752 2.9430"		2.360 .0929"	



Scania



1	DS/DSC/DSI II						
6 127 x 145 11020 c.c.	Bielas	AEB 747	STD0.25-0.50-0.75-1.00	84.234/84.254 3.3163/3.3171"	88.483/88.503 3.4836/3.4844"	2.080 .0819"	47.75 1.8799"
	Bancadas	AEM 748	STD-0.25-0.50-0.75-1.00				
	1-3-5-7			101.597/101.620 3.9999/4.0008"	106.260/106.286 4.1835/4.1845"	2.290 .0902"	35.71 1.4059"
	4			101.597/101.620 3.9999/4.0008"	106.260/106.286 4.1835/4.1845"	2.290 .0902"	43.74 1.7220"
	Axial	AEW 95022	STD-.005-.010	112.649/113.030 4.4350/4.4500"	132.512/132.893 5.2170/5.2320"	3.430 .1350"	



Subaru

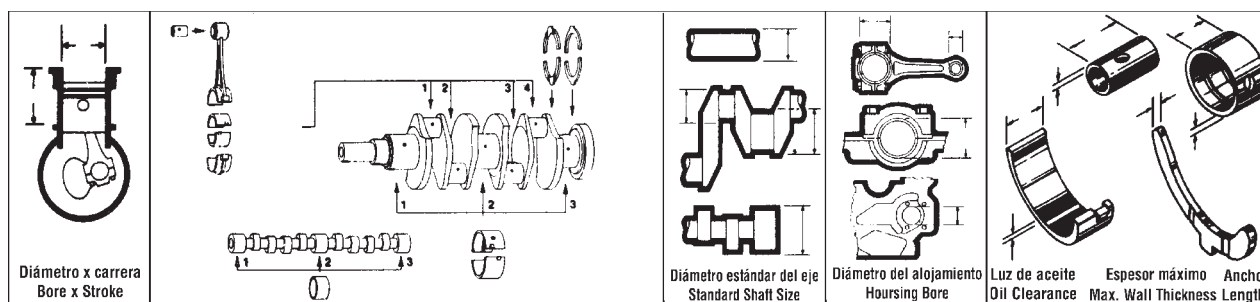


1	GL 1400/1600/1800, Elf, Leone 1200/1400/1600, EA61/62/63/64/71							
4 76 x 60 1088 c.c.	Biela	4-67365 RA	STD-0.25-0.50-0.75	44.988/45.001 1.7712/1.7717"	48.000/48.016 1.8898/1.8904"	0.0279/0.0813 .0011/.0032"	1.486 .0585"	13.26 .522"
79 x 60 1176 c.c.	Bancada 1-3	6791 M 67361 RA	STD-0.25-0.50-0.75	49.954/49.969 1.9667/1.9673"	54.000/54.018 2.1260/2.1267"	0.0127/0.0711 .0005/.0028"	2.006 .0790'	21.13 .832"
82 x 60 1267 c.c.	2	67367 RA		49.954/49.969 1.9667/1.9673"	54.000/54.018 2.1260/2.1267"	0.0127/0.0711 .0005/.0028"	2.006 .0790"	
85 x 60 1361 c.c.	Bancada 1-3	6788 M 67361 RA	STD-0.25-0.50-0.75	49.954/49.969 1.9667/1.9673"	54.000/54.018 2.1260/2.1267"	0.0127/0.0711 .0005/.0028"	2.006 .0790'	21.13 .832"
92 x 60 1595 c.c.	2	67367 RA		49.954/49.969 1.9667/1.9673"	54.000/54.018 2.1260/2.1267"	0.0127/0.0711 .0005/.0028"	2.006 .0790"	

Suzuki



1	ST90 Van, F8A, F10A						
4 62 x 66 797 c.c.	Biela	4-66755 RA	STD-0.25-0.50	37.975/37.988 1.4951/1.4956"	40.998/41.018 1.6141/1.6149"	0.028/0.078 .0586"	17.983 .708"
65.5 x 72 970 c.c.	Bancada	6660 M	STD-0.25-0.50	49.974/49.989 1.9675/1.9681"	53.997/54.018 2.1259/2.1267"	0.030/0.091 .0783"	19.989 .7870"
	Axial	66756 BF	STD				
		66756 BFL		57.000 2.2441"	73.000 2.8740"	2.519 .0992"	
		66756 BFU		57.000 2.2441"	73.000 2.8740"	2.519 .0992"	



Toyota

1

Hi-Lux Diesel, Crown Diesel, Hi-ACE, L, 2L

4 750x073 1290 c.c.	Biela	4-67270 CP	STD-0.25-0.50-0.75-1.00	41.976/41.998 1.6526/1.6535"	44.998/45.024 1.7716/1.7726"	0.020/0.086	1.493 .0588"	18.011 .7091"
	Bancada	6768 M	STD-0.25-0.50-0.75-1.00	49.974/49.999 1.9675/1.9685"	54.002/54.028 2.1261/2.1271"	0.020/0.091	1.993 .0785"	21.844 .8600"
	Axial	67271 BF	STD					
		67271 BFL		55.300 2.2441"	67.800 2.6693"		2.463 .0970"	
		67271 BFU		55.300 2.2441"	67.800 2.6693"		2.463 .0970"	

Toyota

2

Carina, Celica, Corolla, 1400, 1600, 1800 - 1978 / 1982

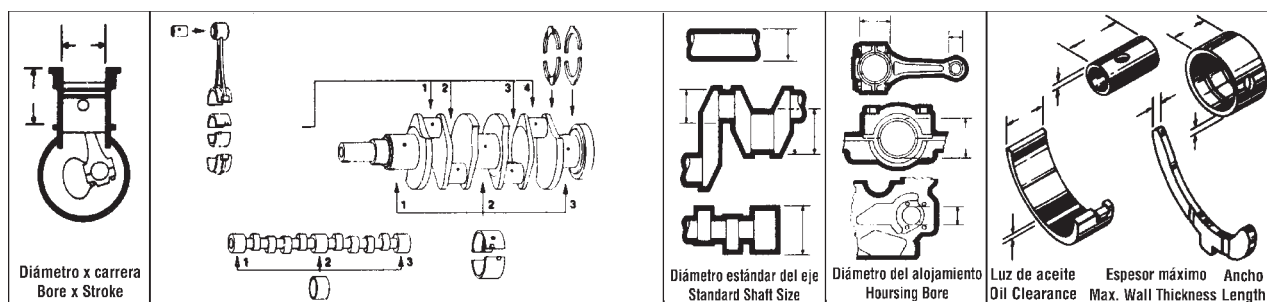
4 85 x 70 1588 c.c.	Biela	4-67275 CP	STD-0.25-0.50-0.75	47.976/47.991 1.8888/1.8894"	51.001/51.021 2.0079/2.0087"	0.023/0.084 .0009/0.0033"	1.494 .0588"	21.97 .865"
	Bancada	6786 M	STD-0.25-0.50-0.75					
		1-2-3-4-5 67276 CPA		57.976/58.001 2.2825/2.2835"	61.999/62.024 2.4409/2.4419"	0.013/0.086 .0005/0.0034"	1.994 .0785"	20.80 .819"
	Axial	67277 BF	STD-				2.489 .098"	
85 x 78 1770 c.c.	Levas	1708 M	STD-					
	1	67276 DR		46.459/46.474 1.8291/1.8297"	50.000/50.030 1.9685/1.9697"	0.036/0.097 .0014/0.0038"	1.737 .0684"	17.91 .705"
	2	67277 DR		46.210/46.225 1.8193/1.8199"	49.751/49.779 1.9587/1.9598"	0.036/0.094 .0014/0.0037"	1.737 .0684"	15.49 .610"
	3	67278 DR		45.959/45.974 1.8094/1.8100"	48.999/49.030 1.9291/1.9393"	0.038/0.099 .0015/0.0039"	1.494 .0588"	17.91 .705"
	4	67279 DR		45.710/45.725 1.7996/1.8002"	49.251/49.281 1.9390/1.9402"	0.036/0.097 .0014/0.0038"	1.745 .0687"	15.49 .610"
	5	67279 DRI		45.458/45.476 1.7897/1.7904"	49.500/49.530 1.9488/1.9500"	0.036/0.094 .0014/0.0037"	1.994 .0785"	17.91 .705"

Toyota

3

Celica 2000, 18R

4 88.5 x 80 1968 c.c.	Biela	4-3960 CPA	STD-0.25-0.50-0.75	52.974/52.989 2.0856/2.0862"	55.996/56.017 2.2046/2.2054"	0.025/0.086	1.491 .0587"	21.844 .8600"
	Bancada	7143 M	STD-0.25-0.50-0.75					
		1-3 67.331 CPA		59.974/59.990 2.3612/2.3618"	64.002/64.023 2.5198/2.506"	0.029/0.091	1.991 .0784"	22.500 .8858"
		2 67282 CPAH		59.979/59.994 2.3614/2.3620"	64.002/64.023 2.5198/2.5206"	0.025/0.086	1.991 .0784"	35.991 1.4170"



Toyota



								Hi-Lux 18R	
4		Biela	4-3960 CPA	STD-0.25-0.50-0.75	52.974/52.989 2.0856/2.0862"	55.996/56.017 2.2046/2.2054"	0.025/0.086	1.491 .0587"	21.844 .8600"
88.5 x 80 1968 c.c.		Bancada	6658 M	STD-0.25-0.50					
			67393 CP		59.969/59.984 2.3610/2.3616"	64.002/64.023 2.5198/2.5206"	0.035/0.096	1.991 .0784"	21.996 .8660"
			67394 CP		59.969/59.984 2.3610/2.3616"	64.002/64.023 2.5198/2.5206"	0.035/0.096	1.991 .0784"	27.996 1.1020"
		Axial	67332 AF	STD					
			67332 AFL		66.000 2.5984"	80.000 3.1496"		1.995 .0770"	
			67332 AFJ		66.000 2.5984"	80.000 3.1496"		1.995 .0770"	

Volkswagen



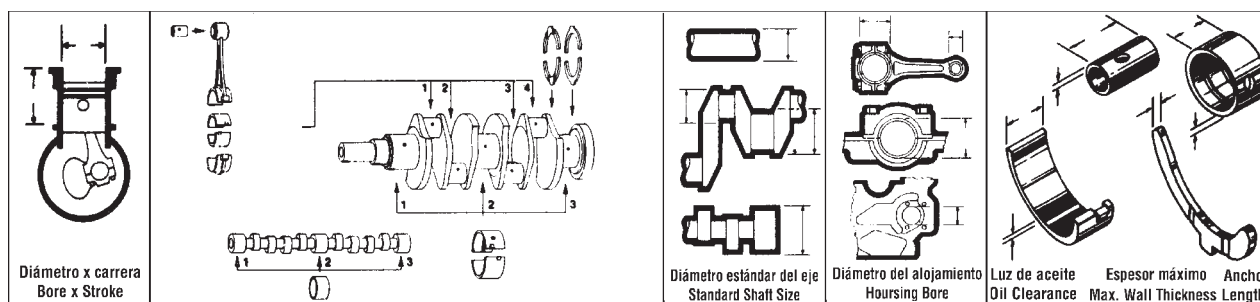
								Gol 1.000	
4		Biela	4-68105 RA	STD-0.25-0.50-0.75	41.695/41.980	45.000/45.012	0.010/0.049	1.505	18.00
70.0 x 64.20 1000 c.c.		Bancada	6838 M	STD-0.25-0.50-0.75					
			1-2-3-5 68096 RA		53.958/53.978 53.958/53.978	59.000/59.019 59.000/59.019	0.030/0.080 0.030/0.080	2.503 2.503	18.5 18.5
		Axial	68097 BF					1.975	

Volkswagen



								Fusca, Brasilia, Gol, Kombi, Variant, Karmann Guia, Furgoneta, 1.3, 1.5, 1.6	
4		Bielas	4-2930 RA	STD-0.25-0.50-0.75-1.00	54.971/54.990	57.800/57.819	0.010/0.056	1.397	17.6
77 x 69 1285 c.c.		Bancada	4213 M	STD-0.25-0.50-0.75-1.00					
		1▲	2931 ATA		54.971/54.990	65.000/65.019	0.040/0.100	4.987	26.94
83 x 69 1493 c.c.		2	2932 CP		54.971/54.990	65.000/65.019	0.040/0.100	5.035	22.00
		3	2934 AT		54.971/54.990	65.000/65.019	0.040/0.100	4.990	22.00
		4	2933 AT		54.971/54.990	65.000/65.019	0.040/0.100	4.983	15.00
85.50 x 69 1584 c.c.		Bancada	6820 M	STD-0.25-0.50-0.75-1.00	Igual al 4213 con diámetro ext. + 0.25-pestaña + 2 mm				
		Bancada	6821 M	STD-0.25-0.50-0.75-1.00	Igual al 4213 con diámetro ext. + 0.50-pestaña + 2 mm				
		Bancada	6822 M	STD-0.25-0.50-0.75-1.00	Igual al 4213 con diámetro ext. + 0.75-pestaña + 2 mm				
		Levas	1724 M	STD-0.25-0.50					
		1	2931 DRA		24.987/25.000 .9837/.9843"	27.500/27.521 1.0827/1.0835"	0.010/0.064 .0003/.0025"	1.245 .0490"	20.00 .787"
		2	2932 DRA		24.987/25.000 .9837/.9843"	27.500/27.521 1.0827/1.0835"	0.010/0.064 .0003/.0025"	1.245 .0490"	23.00 .906"
		3▲	2933 DRA		24.987/25.000 .9837/.9843"	27.500/27.521 1.0827/1.0835"	0.010/0.064 .0003/.0025"	1.245 .0490"	28.00 1.102"
		Buje de Biela	2934 Y		21.994/22.000 21.994/22.000	23.970/24.003 23.970/24.003	0.005/0.024 0.005/0.024	1.300 1.400	23.40 23.40

▲ Cojinete con pestaña - Flanged bearing



Volkswagen



3

Passat, Voyage, Parati,
Passat TS/LS, Gacel, 1.6, 1.5

4 79.5 x 80 1598 c.c.	Bielas	4-68095CP	STD-0.25-0.50-0.75-100 FLANGE +0.50	45.940/45.960	49.000/49.016	0.025/0.060	1.506	19.00
76.5 x 80 1471 c.c.	Bancada	6819 M	STD-0.25-0.50-0.75-1.00					
	1-2-4-5	68096 RA		53.940/53.960	59.000/59.019	0.025/0.060	2.503	18.50
	3 ▲	68097 CP		53.940/53.960	59.000/59.019	0.025/0.060	2.503	25.40
	Bancada	6834 M	STD-0.25-0.50-0.75					
	1-2-4-5	68096 RA		53.940/53.960	59.000/59.019	0.025/0.060	2.503	18.50
	3 ▲	68097 CPA		53.940/53.960	59.000/59.019	0.025/0.060	2.503	24.93
	Buje de Biela	1726 M	STD					
	1	68096 DR		42.959/42.975 1.6913/1.6919"	46.000/46.025 1.8110/1.8120"	0.025/0.066 .0009/.0026"	1.770	15.00 .590"
	2	68097 DR		41.959/41.975 1.6519/1.6526"	45.000/45.025 1.7717/1.7726"	0.025/0.066 .0009/.0026"	1.770	15.00 .590"
	Buje de Biela	4-68099 Y	SEMI	22.000/ .8661"/	25.000/25.032 .9843/.9855"	0.012/0.022 .0004/.0008"	1.770	25.00 .984"

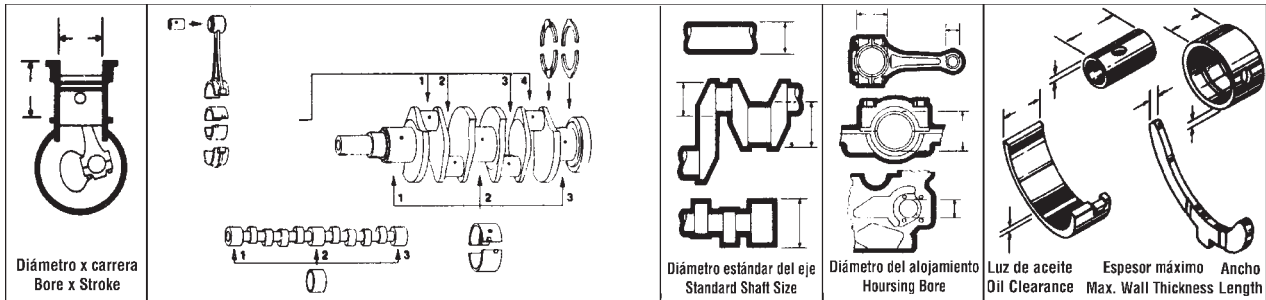
Volkswagen



Passat D, Pick-Up, Furgao, Gol, Parati, Voyage, Passat
Saveiro, Apollo, Santana, Gol GTI, Logus, Quantum, Gacel, GS, Carat, Senda
1.6, 1.8, 2.0, 2.0i, 1.6 Diesel - 1985 /

4

4 81 x 77.4 1584 c.c.	Bielas	4-68150 CP	STD-0.25-0.50-0.75	47.758/47.778	50.600/50.619	0.010/0.080	1.406	19.00
76.5 x 86.4 1588 c.c.	Bancada	6838 M	STD-0.25-0.50-0.75					
	1-2-3-5	68096 RA		53.958/53.978	59.000/59.019	0.030/0.080	2.503	18.50
	4			53.958/53.978	59.000/59.019	0.030/0.080	2.503	18.50
81 x 86.4 1781 c.c.			FLANGE + 0.50					
82.5 x 92.8 1984 c.c.	Bancada	6819 M	STD-0.25-0.50-0.75					
	1-2-4-5	68096 RA		53.940/53.960	59.000/59.019	0.025/0.060	2.503	18.50
	3 ▲	68097 CPA		53.940/53.960	59.000/59.019	0.025/0.060	2.503	24.50
	Bancada	6834 M	STD-0.25-0.50-0.75					
	1-2-4-5	68096 RA		53.958/53.978	59.000/59.019	0.025/0.060	2.503	18.50
	3 ▲	68097 CPA		53.958/53.978	59.000/59.019	0.025/0.060	2.503	24.93
	Axial ▲	68097 BF	STD-5-10				1.975	
	Buje de Biela	4-68099 YA	SEMI	24.000 .945"	27.000/27.021 1.062/1.064"	0.016/0.029 .0006/.0011"	1.780	25.00 .984"
	Buje de Biela	1726 M	STD					
	1	68096 DR		42.959/42.975 1.6913/1.6919"	46.000/46.025 1.8110/1.8120"	0.025/0.066 .0009/.0026"	1.770	15.00 .590"
	2	68097 DR		41.959/41.975 1.6519/1.6526"	45.000/45.025 1.7717/1.7726"	0.025/0.066 .0009/.0026"	1.770	15.00 .590"



Volkswagen

Golf III/IV, Golf Cabrio, Golf Variant, Polo Classic, Pasat, Sharan, Vento, Bora, Beetle, Caravelle, Transporter 1.9 TD/TDI Eng. AAZ/ABL/ADF/AFN/AHU/ARD/1Z Turbo Diesel - 1993 /

5	4 79.5 x 95.6 1896 c.c.	Bielas	71-3728/4	STD-0.25-0.50	47.758/47.778 1.8802/1.8810"	50.600/50.619 1.9921/1.9929"	0.020/0.074	1.406 .0554"	20.00 .7874"
		Bancadas	H 027/5 02-3085	STD-0.25-0.50-0.75	53.958/53.978 2.1243/2.1251"	59.000/59.019 2.3228/2.3236"	0.020/0.082	2.502 .0985"	18.50 .7283"
				02-3473	53.958/53.978	59.000/59.019	0.020/0.082	2.502	18.50
				Ranura de aceite de 360°					
		Axial	A 123/4	STD	59.500 2.3425"	73.200 2.8819"		1.975 .0778"	

Volkswagen

Golf III/IV, Pasat, Sharan, Vento, Caravelle, Bora VR6 (AAA/AES) - 1991 /

6	6 81 x 90.3 2792 c.c.	Bielas	01-3841/6	STD-	53.958/53.978 2.1243/2.1251"	56.800/56.819 2.2362/2.2370"	0.020/0.074	1.406 .0554"	17.00 .6693"
		Bancadas	H 035/7		53.958/59.978 2.3605/2.3613"	65.000/65.010 2.5591/2.5594"	0.020/0.075	2.501 .0985"	17.00 .6693"
		Axial	A 139/4 04-3717	STD	65.450 2.5768"	80.000 3.1496"		1.723 .0678"	
			04-3718		65.450 2.5768"	80.000 3.1496"		1.723 .0678"	